Master's Degree in Mathematics

Università di Pisa

M.Sc. Thesis

On the Construction of Elliptic Curves with a Prescribed Number of Points

Author:
Eugenio Paracucchi

Supervisor:
Davide Lombardo

Examiner:
Andrea Bandini

Academic Year 2021/2022

Abstract

This thesis describes a procedure (the 'CM method'), based on the theory of complex multiplication, to construct elliptic curves over finite fields having a prescribed number of rational points. A major ingredient in the method is the computation of certain Hilbert class polynomials, whose coefficients grow very quickly with the input data. Faster algorithms can be obtained by replacing $H_D(x)$ with other class polynomials, first introduced by Weber. Such polynomials come from class invariants, that is, modular functions related to the Hilbert class field. In this work, we take an elementary point of view and we focus on the γ-function, a specific determination of the cube root of the classical j-invariant. In particular, we show that γ is a class invariant for all quadratic discriminants relatively prime to 3, and explain how to compute the corresponding class polynomial.

The CM method has important applications in cryptography since it can be exploited to generate elliptic curves suitable for key exchange protocols, primality proving, and pairing-based cryptosystems. For this reason, we provide a working implementation of the algorithm using the SageMath language.

Key words: elliptic curves, complex multiplication, CM method, Hilbert class polynomial, class invariants, cryptography.

Contents

Abstract **iii**

Introduction **1**

1 Background **3**
1.1 Elliptic Curves 3
1.1.1 The Weierstrass equation and the group law 3
1.1.2 Morphisms of elliptic curves 7
1.1.3 The j-invariant 9
1.2 Cryptography 10
1.2.1 Key establishment 11
1.2.2 The discrete logarithm problem 12

2 Elliptic Curves over $\mathbb{C}$ **15**
2.1 Complex tori of dimension 1 16
2.1.1 Lattices 16
2.1.2 Complex tori 18
2.2 From Complex Tori to Elliptic Curves 21
2.2.1 Elliptic Functions 22
2.2.2 Weierstrass $\wp$-function 24
2.3 Uniformization 33
2.3.1 The j-invariant of a lattice 33
2.3.2 Modularity 35
2.3.3 The j-function 38

3 Complex Multiplication **43**
3.1 Introduction 43
3.1.1 Quadratic Fields 43
3.1.2 Fractional Ideals 46
3.1.3 Endomorphisms 50
3.2 Complex Multiplication 52

3.2.1 Elliptic curves with complex multiplication 52
3.2.2 The Hilbert Polynomial 55
3.2.3 Binary Quadratic Forms 57
3.2.4 Relation with the ideal class group 59
3.3 The integrality of j 62
3.3.1 Isogenies of degree n 63
3.3.2 Computing the Hilbert polynomial 66

4 Class Invariants **73**
4.1 Introduction 74
4.1.1 The Hilbert class field 74
4.1.2 The curve $X_{\mathrm{ns}}^{+}(3)$ 75
4.2 On the class invariant $\gamma(\tau) = j(\tau)^{1/3}$ 79
4.2.1 The Galois representation 79
4.2.2 3-torsion points 83
4.2.3 Computing the conjugates of $\gamma(\tau_0)$ 89

5 The CM method **93**
5.1 Preliminary 93
5.1.1 Elliptic curves over finite fields 93
5.1.2 Ramification of primes 94
5.1.3 Good reduction 95
5.1.4 The norm equation 99
5.2 The method 102
5.2.1 Cornacchia's algorithm 102
5.2.2 Roots modulo p 104
5.2.3 Recovering the correct curve 105
5.2.4 Analysis of the cost 105
5.2.5 Examples 107

A Implementations **109**
A.1 Class polynomials 109
A.1.1 Modular forms 109
A.1.2 Representatives 110
A.1.3 Hilbert and Weber class polynomials 111
A.2 The CM method 113
A.2.1 The Cornacchia algorithm 113
A.2.2 The CM method 114

Bibliography **120**

Introduction

Being able to construct elliptic curves over finite fields with a given number of rational points is certainly useful for cryptographic applications.

The security of classical *elliptic curve cryptography*, such as *key exchange protocols*, is based on the complexity of the *discrete logarithm problem* on the group of the $\mathbb{F}_q$-rational points of an elliptic curve $E/\mathbb{F}_q$. In the light of the simplification of Pohlig and Hellman, suitable elliptic curves to be used in practice should have order divisible by a huge prime factor. Also primality proving algorithms (such as the *Goldwasser-Kilian* algorithm) and pairing-based crypto-systems can be considerably improved by choosing an appropriate elliptic curve.

There exists more than one way to accomplish such tasks. In this thesis we are going to discuss the CM method to construct elliptic curves over a finite field, based on the theory of complex multiplication (CM) of elliptic curves over $\mathbb{C}$. The core idea of the method is to reduce an adequate CM elliptic curve E, defined over a number field F, modulo a prime ideal $\mathfrak{p} \subseteq \mathcal{O}_F$ in order to obtain an elliptic curve $E_\mathfrak{p}$ over $\mathcal{O}_F/\mathfrak{p} = \mathbb{F}_q$ with the desired properties.

In the first chapter we introduce the basics of elliptic curves and motivate their use in cryptography. After that, in the second chapter, elliptic curves over the complex numbers are presented. In particular, we establish the fundamental correspondence between complex tori, lattices and elliptic curves.

Chapters three and four deal with the theory of CM elliptic curves over $\mathbb{C}$, i.e., elliptic curves whose endomorphism ring is an order $\mathcal{O}$ in an imaginary quadratic field. We show that there exist only finitely many isomorphism classes of elliptic curves with CM by $\mathcal{O}$, and that their j-invariants are algebraic integers. The Hilbert class polynomial $H_D(x)$, whose roots are exactly the j-invariants of elliptic curves with CM by the order $\mathcal{O}_D$ of discriminant D, is introduced and an algorithm for its computation is given. The coefficients of this polynomial grow very quickly as D increases: they are, in fact, of the magnitude of $\exp(C \cdot |D|^{1/2} \cdot (\log |D|)^2)$, where C is a constant. Faster algorithms can be obtained by replacing $H_D(x)$ with other class polynomials,

first introduced by Weber. Such polynomials come from class invariants, that is, modular functions related to the Hilbert class field, and have much smaller coefficients than $H_D(x)$. In this work, we take an elementary point of view and we focus on the γ-function, a specific determination of the cube root of the classical j-invariant. In particular, we show that γ is a class invariant for all quadratic discriminants D relatively prime to 3, and explain how to compute the corresponding class polynomial.

In the final chapter we state the Deuring lifting theorem and we explain how CM elliptic curves can be reduced modulo a prime, yielding elliptic curves over finite fields. We then put together all the results established in the body of the thesis to obtain the desired algorithm. Given a positive integer N, together with its factorization, the algorithm computes a prime number p and an elliptic curve $E/\mathbb{F}_p$ such that $|E(\mathbb{F}_p)| = N$.

Finally, in the appendix, we provide a working implementation of the algorithms for the CM method and the computation of the class polynomials using the SageMath language.

Chapter 1

Background

To better understand what we will develop in the body of the thesis, in this introductory chapter we provide the main definitions and general results of the theory of elliptic curves.

We will also give a brief overview of classical elliptic curve cryptography in order to motivate our work.

1.1 Elliptic Curves

1.1.1 The Weierstrass equation and the group law

This thesis deals with elliptic curves, in particular with their cryptographic applications. There are several ways to introduce elliptic curves, for practical reasons we give the following definition.

Let K be a field and $\overline{K}$ an algebraic closure. To keep it simple, we will always assume that $\operatorname{Char} K \neq 2, 3$.

Definition 1.1.1. An **elliptic curve** E over K is a smooth projective variety in $\mathbb{P}^2(\overline{K})$ defined by the equation

$$Y^2Z = X^3 + AXZ^2 + BZ^3, \text{ with } A, B \in K. \tag{1.1}$$

We will write E/K to emphasize that E is defined over K.

Remark 1.1.1. If there is no restriction on the characteristic of the field K, the equation defining an elliptic curve should be replaced with the longer

$$Y^2Z + A_1XYZ + A_3YZ^2 = X^3 + A_2X^2Z + A_4XZ^2 + A_6Z^3, \tag{1.2}$$

with $A_i \in K$ for $i = 1, 2, 3, 4, 6$. Because of their sizes, equation (1.2) is called *long Weierstrass equation* and (1.1) *short Weierstrass equation*. If

$\operatorname{Char} K \neq 2, 3$ we can in fact reduce, completing the squares and the cubes, to a short equation. We will never deal with long equations, hence we will simply refer to (1.1) as **Weierstrass equation**.

We are interested in the solutions of the equation defining an elliptic curve. Let L be a field containing K, the set of L**-rational points** of E is

$$E(L) = \big\{[X_0 : Y_0 : Z_0] \in \mathbb{P}^2(L) \text{ satisfying equation (1.1)}\big\}.$$

Remark 1.1.2. We note that, independently of the equation and the field of definition, the point $[0 : 1 : 0]$ always belong to $E(L)$. This exceptional point is called the **point at infinity** and will be denoted by the symbol ∞.

All the other points of $E(L)$ lie in the affine chart $Z \neq 0$, therefore we can write

$$E(L) = \big\{(x, y) \in \mathbb{A}^2(K) \,\big|\, y^2 = x^3 + Ax + B\big\} \cup \big\{\infty\big\}$$

where $x = X/Z$, $y = Y/Z$ and the equation

$$y^2 = x^3 + Ax + B \tag{1.3}$$

is obtained from (1.1) by dehomogenizing with respect to Z. In the light of this, we will often define our elliptic curves using the affine equation (1.3), that we call Weierstrass equation as well, and work with that instead of (1.1).

Remark 1.1.3. The smoothness condition[1] is equivalent to require $V(I) = \emptyset$ where $I = (f(x), f'(x))$ and $f(x) = x^3 + Ax + B$. By the Hilbert Nullstellensatz, this is equivalent to the condition $(f(x), f'(x)) = 1$, and therefore that $f(x)$ has three distinct roots in $\overline{K}$, which is in turn is equivalent to the condition

$$\operatorname{disc}(f(x)) = -(4A^3 + 27B^2) \neq 0.$$

We define the **discriminat** of E as

$$\Delta(E) := 4A^3 + 27B^2.$$

Therefore $E : y^2 = x^3 + Ax + B$ is an elliptic curve if and only if $\Delta(E) \neq 0$.

The set of L-rational points $E(K)$ has, remarkably, a natural group structure for which ∞ is the neutral element.

Given $P, Q \in E(L)$ we define $P + Q$ as follows:

[1] A projective curve $C : G(X, Y, X) = 0$ is smooth if $\nabla G(X, Y, Z)$ is non-zero at any point $[X : Y : Z] \in C(\overline{K})$. In the case of the curve E/K defined by $Y^2Z - X^3 - AXZ^2 - BZ^3 = 0$ the point at infinity $[0 : 1 : 0]$ is always smooth, therefore the smoothness condition becomes $\nabla g(x, y) \neq 0$ for any $(x, y) \in E(\overline{K}) \setminus \{\infty\}$ where $g(x, y) = y^2 - x^3 - Ax - B$ is the dehomogenized.

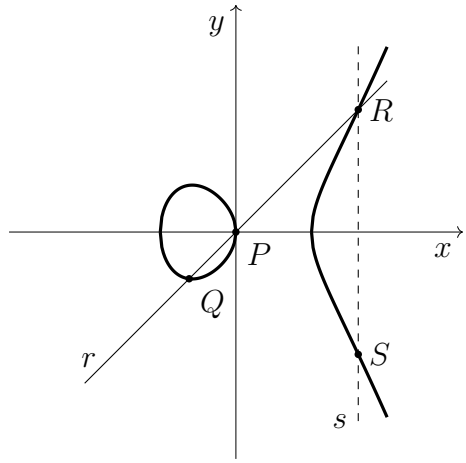


Figure 1.1: Adding points on the curve $y^2 = x^3 - x$.

(1) draw the line r between P and Q, if $P = Q$ draw the tangent line at P (which is well defined because E is smooth);

(2) compute $r \cap E(L) = \{P, Q, R\}$, to regard as a multi-set since it could be that $P = Q$;

(3) draw the line s through R and ∞ (if $R = \infty$ draw the tangent) and compute the multi-set $s \cap E(L) = \{R, \infty, S\}$. By definition

$$P + Q := S.$$

If $L = \mathbb{R}$ we can draw pictures, and we can easily visualize the group law (see figure 1.1).

Theorem 1.1.1. *Let E/K be an elliptic curve and L a field containing K. Then $(E(L), +)$ is an abelian group with neutral element ∞.*

Proof. The only non-trivial point is the proof of associativity. An elementary proof can be found in [Rei88] Chapter 2, Theorem 2.8. Another, that uses the Riemann-Roch theorem, can be found in [Sil09], Chapter 3, Proposition 3.4. □

Example 1. As an example, we compute the sum $S = P + Q$ for $P, Q \in E(L) \setminus \{\infty\}$.

Let $E : y^2 = x^3 + Ax + B$ and $P = (x_1, y_1),\ Q = (x_2, y_2)$. If $x_1 \neq x_2$ the line through P and Q has slope

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

and the equation is

$$r : y = m(x - x_1) + y_1.$$

To find the third intersection point R between E and r we substitute to get

$$(m(x - x_1) + y_1)^2 = x^3 + Ax + B.$$

Rearranging the terms we obtain the cubic equation

$$0 = x^3 - m^2x^2 + \dots$$

for which we know the roots x_1 and x_2. To recover the third solution x_R we use the relation [2] $m^2 = x_1 + x_2 + x_R$, therefore

$$x_R = m^2 - x_1 - x_2.$$

The y coordinate of $R = (x_R, y_R)$ can be computed plugging x_R in the equation of r:

$$y_R = m(x_R - x_1) + y_1.$$

The line passing through R and ∞ is vertical and, by symmetry, the point $S = (x_3, y_3)$ can be obtained from R by flipping the sign of y_R:

$$x_3 = m^2 - x_1 - x_2, \qquad y_3 = m(x_1 - x_3) - y_1.$$

If $x_1 = x_2$ but $y_1 \neq y_2$ then P and Q are vertically aligned and the line r through P and Q passes through ∞. The tangent at ∞ is the (projective) line $s : Z = 0$ and therefore $P + Q = \infty$.

For the remaining case $P = Q = (x_1, y_1)$ we take the tangent at P

$$r : y = m(x - x_1) - y_1, \qquad m = \frac{3x_1^2 + A}{2y_1}.$$

If $y_1 = 0$ then r is vertical and $P + Q = \infty$, otherwise we can proceed as in the first case and we obtain $S = (x_3, y_3)$ where

$$x_3 = m^2 - 2x_1, \qquad y_3 = m(x_1 - x_3) - y_1.$$

Remark 1.1.4. From the previous example, for any $P = (x, y) \in E(L) \setminus \{\infty\}$ we have $-P = (x, -y)$. In fact P and $-P$ must be vertically aligned and hence they must have the same x-coordinate and hence $y_P^2 = y_{-P}^2 \Rightarrow y_{-P} = -y_P$.

[2] It is a well know fact that for the cubic polynomial $x^3 + ax^2 + bx + c$ with roots r, s, t we have $r + s + t = -a$.

For any positive integer n and any $P \in E(\overline{K})$ we will denote by

$$[n]P := \underbrace{P + \cdots + P}_{n \text{ times}}.$$

If $n < 0$ we then define $[n]P := -[-n]P$ and we set $[0]P = \infty$. We will also denote by

$$E[n] = \big\{P \in E(\overline{K}) \,\big|\, [n]P = \infty\big\}$$

the n-**torsion subgroup** of $E(\overline{K})$.

Example 2. Let $P \in E[2] \setminus \{\infty\}$, $P = (x_P, y_P)$. By hypothesis

$$[2]P = P + P = \infty \implies P = -P,$$

therefore $(x_P, y_P) = (x_P, -y_P)$ and hence $y_P = 0$. It follows that

$$E[2] = \big\{(x, y) \in \mathbb{A}^2(\overline{K}) \,\big|\, y = 0,\ x^3 + Ax + B = 0\big\} \cup \big\{\infty\big\}.$$

1.1.2 Morphisms of elliptic curves

Elliptic curves are algebraic curves, hence it is natural to consider *rational maps* between them. Note that, since elliptic curves are smooth projective curves, any rational map between them is globally defined and is in fact a *morphism* of algebraic varieties.

On the other hand, we have seen that the set of rational points of an elliptic curve forms an abelian group. Therefore we can equally naturally consider *group homomorphisms* as maps between elliptic curves.

Quite strikingly, these two structures are closely related. Let E, E' be two elliptic curves defined over K.

Theorem 1.1.2 ([Sil09], Chapter 3, Theorem 4.8)**.** *Let $\varphi : E \longrightarrow E'$ be a rational map such that $\varphi(\infty_E) = \infty_{E'}$. Then $\varphi : E(\overline{K}) \longrightarrow E(\overline{K})$ is a group homomorphism.*

Therefore we give the following definition.

Definition 1.1.2. We say that $\varphi : E \longrightarrow E'$ is a **morphism** of elliptic curves if φ is a rational map and $\varphi(\infty_E) = \infty_{E'}$. We will say that φ is defined over K if it is given by rational functions with coefficients in K. We will call φ an **isomorphism** if there exists another morphism of elliptic curves $\psi : E' \longrightarrow E$ such that $\psi \circ \varphi = \mathrm{Id}_E$ and $\varphi \circ \psi = \mathrm{Id}_{E'}$. Moreover, we will say that E and E' are isomorphic (over K) if there exists an isomorphism $\varphi : E \longrightarrow E'$ (defined over K).

Theorem 1.1.3. *Let E/K be an elliptic curve, then the sum and the negation*

$$\begin{array}{rccc} + : & E \times E & \longrightarrow & E \\ & (P,Q) & \mapsto & P+Q \end{array} \quad \text{and} \quad \begin{array}{rccc} [-1] : & E & \longrightarrow & E \\ & P & \mapsto & -P \end{array}$$

are rational maps, in particular the negation is a morphism of elliptic curves.

Proof. Since $[-1](x,y) = (x,-y)$ and $-\infty = \infty$ the negation is a morphism of elliptic curves. For the sum we refer to [Sil09], Chapter 3, Theorem 3.6. □

Corollary 1.1.1. *The the multiplication by n map*

$$\begin{array}{rccc} [n] : & E & \longrightarrow & E \\ & P & \mapsto & [n]P \end{array}$$

is a morphism of elliptic curve for every $n \in \mathbb{Z}$.

Proof. By induction on n using theorem 1.1.3 □

It can be shown, see [Was08], Chapter 3, Theorem 3.6, that

$$[n](x,y) = \left(\frac{\phi_n(x,y)}{\psi_n(x,y)^2}, \frac{\omega_n(x,y)}{\psi_n(x,y)^3} \right),$$

where $\phi_n, \psi_n, \omega_n \in K[x,y]$. The polynomials $\psi_n(x,y)$ are called the **division polynomials** and, if $E : y^2 = x^3 + Ax + B$, they are defied recursively by

$$\begin{aligned}
&\psi_0 = 0 \\
&\psi_1 = 1 \\
&\psi_2 = 2y \\
&\psi_3 = 3x^4 + 6Ax^2 + 12Bx - A^2 \\
&\psi_4 = 4y(x^6 + 5Ax^4 + 20Bx^3 - 5Ax^2 - 4ABx - 8B^2 - A^3) \\
&\psi_{2k+1} = \psi_{k+2}\psi_k^3 - \psi_{k-1}\psi_{k+1}^3 && \text{for } m \geq 2 \\
&\psi_{2k} = (2y)^{-1}\psi_k(\psi_{k+2}\psi_{k-1}^2 - \psi_{k-2}\psi_{m+1}^2) && \text{for } m \geq 3.
\end{aligned}$$

It can also be shown that

$$\psi_n \in \begin{cases} \mathbb{Z}[x,A,B] & \text{if } n \text{ is odd} \\ 2y \cdot \mathbb{Z}[x,A,B] & \text{if } n \text{ is even.} \end{cases}$$

Remark 1.1.5. If n is odd then the roots of ψ_n (in $\overline{K}$) are the x-coordinates of the n-torsion points of $E[n] \setminus \{\infty\}$. Similarly, if n is even the roots of ψ_n/y are the x-coordinates of the points of the set $E[n] \setminus E[2]$.

1.1.3 The j-invariant

We are mostly interested in the study of elliptic curves up to isomorphism. Two different Weierstrass equations could define isomorphic elliptic curves. Consider, for example, the elliptic curves E, E' (over $\mathbb{Q}$) defined by

$$E : y^2 = x^3 + x + 1, \text{ and } E' : y^2 = x^3 + 16x + 64.$$

The equations defining E and E' are different, even though they are isomorphic over $\mathbb{Q}$: the map

$$\begin{array}{rccc} \varphi : & E & \longrightarrow & E' \\ & (x, y) & \mapsto & (4x, 8y) \\ & \infty & \mapsto & \infty \end{array}$$

is in fact an isomorphism. The shape of φ is not specific, indeed we have the following proposition.

Proposition 1.1.1 ([Sil09], Chapter 3, Proposition 3.1)**.** *Let $E : y^2 = x^3 + Ax + B$ and $E' : y^2 = x^3 + A'x + B$ be two elliptic curves defined over K. Every isomorphism of elliptic curves between E and E' has the form*

$$\varphi((x, y)) = (u^2x, u^3y)$$

for some $u \in \overline{K}^\times$. In addition, φ is defined over K if and only if $u \in K^\times$.

Example 3. The field of definition makes a difference. Consider, for example, $E : y^2 = x^3 + 2$ and $E' : y^2 = x^3 + 1$ over $\mathbb{Q}$. The curves E and E' are not isomorphic over $\mathbb{Q}$ (it can be shown that the torsion subgroups are, respectively, trivial and isomorphic to $\mathbb{Z}/6\mathbb{Z}$) while they are over $\mathbb{Q}(\sqrt[6]{2})$ and in fact $(x, y) \mapsto (u^{-2}x, u^{-3}x)$ where $u = \sqrt[6]{2}$ is an isomorphism.

Two different Weierstrass equations $y^2 = x^3+Ax+B$ and $y^2 = x^3+A'x+B$ over K define isomorphic elliptic curves (over K) if and only if $A' = v^4A$ and $B' = v^6B$ for some $v \in K^\times$. In particular, the point $J = [A^3 : B^2] \in \mathbb{P}^1(K)$ only depends on the isomorphism class of the elliptic curve. We would like to convert this point into a number.

Remark 1.1.6. If $P = [a : b] \in \mathbb{P}^1(K)$ then $\mathbb{P}^1(K) \setminus \{P\} \cong \mathbb{A}^1(K)$ via

$$[X : Y] \mapsto \frac{X}{aY - bX}.$$

Since the discriminant $\Delta(E) = 4A^3 + 27B^2 \neq 0$, we are led to the following definition.

Definition 1.1.3. The j**-invariant** of the elliptic curve $E : y^2 = x^3+Ax+B$ is the quantity

$$j(E) = 1728\frac{4A^3}{4A^3 + 27B^2}.$$

Remark 1.1.7. The j-invariant $j(E)$ is well defined since $\Delta(E) \neq 0$ and it only depends on the $\overline{K}$-isomorphism class of E.

Example 4. There are two special cases of j which appear quite often:

- $j = 0$, for the elliptic curves $E : y^2 = x^3 + B$;
- $j = 1728$, for the curves $E : y^2 = x^3 + Ax$.

Theorem 1.1.4. *Let $E : y^2 = x^3 + Ax + B$ and $E' : y^2 = x^3 + A'x + B$ be two elliptic curves defined over K.*

(*i*) *If $E \cong E'$ over $\overline{K}$ then $j(E) = j(E')$.*

(*ii*) *If $j(E) = j(E')$ then there exists an isomorphism $\varphi : E \longrightarrow E'$ defined over $\overline{K}$.*

Proof. The first assertion follows from what we have said above. For (*ii*) we refer to [Was08], Chapter 2, Theorem 2.19. □

Remark 1.1.8. The j-invariant determines isomorphism classes of elliptic curves over $\overline{K}$ but, in general, not over K: the elliptic curves of example 3 have the same j-invariant $j(E) = j(E') = 0$ and hence they are isomorphic over $\overline{\mathbb{Q}}$, but not over $\mathbb{Q}$.

Corollary 1.1.2. *Let K be a field. There is a one to one correspondence*

$$\frac{\{\text{Elliptic curves over } \overline{K}\}}{\text{isomorphism}} \longleftrightarrow \overline{K}.$$

Proof. The association is $E \mapsto j(E)$, which is injective by theorem 1.1.4. To show that is onto, for any $j_0 \in \overline{K}$ we must exhibit an elliptic curve $E/\overline{K}$ such that $j(E) = j_0$. If $j_0 = 0, 1728$ then we choose, respectively, $E : y^2 = x^3 + 1$ and $E : y^2 = x^3 + x$. Otherwise we can take $E : y^2 = x^3 + Ax + B$ where $A = 3j_0(1728 - j_0)$ and $B = 2j_0(1728 - j_0)^2$. □

1.2 Cryptography

In this section we will give a brief glimpse of how elliptic curves are used for cryptographic applications. In particular, we are going to present the classical elliptic curve *Diffie-Hellman* key establishment, and we also provide an overview of the *elliptic curve discrete logarithm problem*, pointing out some precautions.

1.2.1 Key establishment

Tancredi and Caterina want to establish a common secret key to be used to encrypt (with some *symmetric cipher*) their future messages. To do this, they are forced to use a virtual public (and hence insecure) channel, since they live in different towns and they are not able to meet in person.

They decide to do the following:

1. Global parameters. They agree on an elliptic curve $E/\mathbb{F}_p$ and on a randomly chosen point $P \in E(\mathbb{F}_p) \setminus \{\infty\}$.

2. Personal parameters. Caterina generates her own pair of parameters $K_C = (s_C, P_C)$, where

   - s_C is a (secret) natural number,
   - $P_C := [s_C]P$.

   Tancredi does the same yielding $K_T = (s_T, P_T)$.

3. Key establishment. Tancredi sends Caterina P_T and she computes $P_{CT} = [s_C]P_T = [s_C s_T]P$. Conversely, Tancredi will receive P_C from Caterina and will compute $P_{TC} = [s_T]P_C = [s_T s_C]P$. Since $P_{TC} = P_{CT} = S$ they now agree on the common secret key S.

This cryptographic scheme is known as the (elliptic curve) *Diffie-Hellman* key establishment protocol.

Remark 1.2.1. The security relies on the difficulty of the discrete logarithm problem (DLP) on the (finite) group of the $\mathbb{F}_p$-rational points of E. More generally let $(G, \cdot)$ be a finite cyclic group, by the discrete logarithm problem we mean to determine the least positive integer x (if it exists) which satisfies the equation

$$g^x = h$$

for two given elements $h, g \in G$. In our case $G = \langle P \rangle$ is the subgroup of $E(\mathbb{F}_p)$ generated by the point P and the problem is to determine the least $t \in \mathbb{N}$ such that

$$[t]P = Q, \text{ where } P, Q \in G.$$

In its generality, there is no efficient algorithm (for a classical computer) to recover x (resp. t) given g and h (resp. P and Q).

In the above situation, if the malicious Damiano wanted to interfere in the channel, in order for him to recover the secret key S he would have to obtain s_T and s_C from P_T and P_C, that is he would have to solve a DLP.

Now suppose that Omar, a friend of Tancredi and Caterina, wants to establish a tri-party secure conversation. They generate their global (E, P) and personal parameters K_T, K_C, K_O as before, and modify step 3 with the following.

3′. Tancredi (T), Caterina (C) and Omar (O) do as follows:

- T sends P_T to C who sends P_C to O who sends P_O to T;
- T computes $P_{TO} = [s_T]P_O$, C computes $P_{CT} = [s_C]P_T$ and O computes $P_{OC} = [s_O]P_C$;
- T sends P_{TO} to C who sends P_{CT} to O who sends P_{OC} to T;
- T (resp. C, O) computes $P_{TOC} = [s_T]P_{OC} = [s_T s_O s_C]P$ (resp. P_{CTO}, P_{OCT}). And they finally agree with the secret key $S = P_{TOC} = P_{CTO} = P_{OCT}$.

Remark 1.2.2. The tri-party key establishment can be, of course, generalized to a multi-party protocol involving N users. In the case $N = 3$, six transmission of data (2 rounds of 3) and 9 total computations (3 for each user) are necessary. Similarly, for N users, $N(N-1)$ transmissions and N^2 computations are necessary. A faster key establishment protocol, using only one round of communications, can be accomplished using pairings, see [BSS05], Chapter X.

1.2.2 The discrete logarithm problem

In this section we are going to present in more detail the DLP on the points of an elliptic curve. Let E be an elliptic curve defined over the finite field $\mathbb{F}_q$, and let P be a point of $E(\mathbb{F}_q) \setminus \{\infty\}$. The elliptic curve discrete logarithm problem (ECDLP) is, given $Q \in \langle P \rangle = G$, to find the least positive integer m such that

$$[m]P = Q. \tag{1.4}$$

A first simplification, due to Pohlig and Hellman [BSS00], consists in noting that solving the DLP in a group G is equivalent to solving the DLP in each p-Sylow subgroup and then using the Chinese remainder theorem. Which is in turn equivalent to lift the solution to the DLP for a cyclic subgroup of order p.

The *baby-step giant-step* algorithm can efficiently solve the DLP for cyclic subgroups of a small prime order. Therefore one should, for the security, work with elliptic curves whose order is divisible by a large prime factor.

Elliptic curves to be used for the applications should be *ordinary* in order to avoid the MOV attack (see [BSS00]). Let $n = |\langle P \rangle|$ and let l be the

least positive integer such that $\mathbb{F}_{q^l}$ contains the set μ_n of the n-th root of unity, that is, the smallest l such that $q^l \equiv 1 \pmod{n}$. The ECDLP can be reduced to a DLP in the cyclic multiplicative group $\mathbb{F}_{q^l}^{\times}$, where there exist sub-exponential algorithms for the solution to DLP.

Remark 1.2.3. If $E/\mathbb{F}_p$ is supersingular, then $|E(\mathbb{F}_p)| = p + 1$ and hence, since $n \mid |E(\mathbb{F}_p)|$, we have

$$p \equiv -1 \pmod{n} \implies p^2 \equiv 1 \pmod{p} \implies l = 2.$$

More generally if $E/\mathbb{F}_q$ is supersingular, then it can be shown, [BSS00], Chapter IX, Corollary IX.21, that

$$l \in \{1, 2, 3, 4, 5, 6\}.$$

By the remark, supersingular elliptic curves have small values of l, and hence they should be avoided.

The CM method, that we are going to present, enables us to produce elliptic curves with a prescribed number of points and it will return ordinary elliptic curves.

Chapter 2

Elliptic Curves over $\mathbb{C}$

In this chapter we will delve into elliptic curves defined over the complex numbers, exploring their geometry.

An elliptic curve $E/\mathbb{C}$, in addition to its algebraic structure (algebraic variety and abelian group), is a Riemann surface (i.e. a complex manifold of dimension 1). Because E is a closed subset of $\mathbb{P}^2(\mathbb{C})$, E is compact, hence[1] it is homeomorphic to a g-holed torus where g is the genus of E. We will prove that an **elliptic curve** $E/\mathbb{C}$ is in fact (isomorphic to) a **complex torus** $\mathbb{C}/\Lambda$ defined by a **lattice** Λ.

The aim of this chapter is indeed to establish an explicit correspondence between these algebro-geometric objects: elliptic curves, complex tori and lattices.

$$\left\{\begin{matrix}\textit{Elliptic curves over } \mathbb{C} \\ \textit{up to isomorphism}\end{matrix}\right\} \longleftrightarrow \left\{\begin{matrix}1-\textit{dimensional} \\ \textit{complex tori} \\ \textit{up to isomorphism}\end{matrix}\right\} \longleftrightarrow \left\{\begin{matrix}\textit{Lattices in } \mathbb{C} \\ \textit{up to homothety}\end{matrix}\right\}$$

In section 2.1 we will handle complex tori and lattices showing that they are, essentially, the same thing. Then we will show that every torus $\mathbb{C}/\Lambda$ defines an elliptic curve $E_\Lambda/\mathbb{C}$ to which it is isomorphic[2] : we will construct an explicit map parameterized by elliptic functions, in particular the Weierstrass $\wp$-function (and its derivative). To complete the correspondence, in section 4, we will introduce the j-function and use its properties to show that every elliptic curve over $\mathbb{C}$ comes from a torus.

Concretely, the equivalence between elliptic curves is presented in theorem 2.2.4 and corollary 2.3.2, and that between lattices and (1-dim.) complex tori in remarks 2.1.3 and 2.1.5

[1] As a consequence of the classification theorem of compact Riemann surfaces

[2] Both algebraically (as groups) and analytically (as Riemann surfaces).

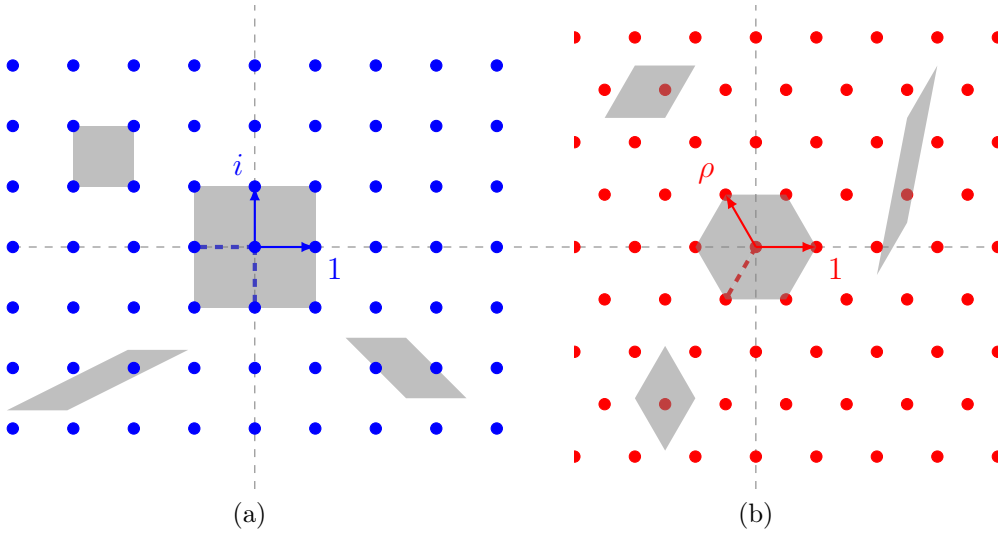


Figure 2.1: Some points of Λ_1 and Λ_2 are displayed (respectively in blue and red). The gray shaded regions are fundamental parallelograms.

2.1 Complex tori of dimension 1

2.1.1 Lattices

Definition 2.1.1. A **lattice** Λ is an additive discrete subgroup of $\mathbb{C}$ isomorphic, as a $\mathbb{Z}$-module, to $\mathbb{Z}^2$. We say that $\{\lambda_1, \lambda_2\}$ is a **basis** for Λ if $\Lambda = \langle \lambda_1, \lambda_2 \rangle_{\mathbb{Z}}$, and in this case we write

$$\Lambda = \mathbb{Z}\lambda_1 \oplus \mathbb{Z}\lambda_2.$$

A **fundamental parallelogram** for Λ is any set of the form

$$\mathcal{P} = \left\{\alpha + t_1\lambda_1 + t_2\lambda_2 \mid 0 \leq t_1, t_2 < 1\right\}$$

where $\alpha \in \mathbb{C}$ and $\{\lambda_1, \lambda_2\}$ is a basis of the lattice.

Example 5. $\Lambda_1 = \mathbb{Z}[i] = \mathbb{Z} \oplus \mathbb{Z}i$, $\Lambda_2 = \mathbb{Z}[\rho] = \mathbb{Z} \oplus \mathbb{Z}\rho$, where $\rho = \zeta_3$ is a primitive cube root of unity, are examples of lattices. (See figure 2.1)

Remark 2.1.1. Any two fundamental parallelograms $\mathcal{P}_1$ and $\mathcal{P}_2$ of a lattice Λ have the same area. Since area is invariant under translation we can suppose

$$\mathcal{P}_1 = \left\{t_1\lambda_1 + t_2\lambda_2 \mid 0 \leq t_1, t_2 < 1\right\}$$

and

$$\mathcal{P}_2 = \left\{t_1\mu_1 + t_2\mu_2 \mid 0 \leq t_1, t_2 < 1\right\}$$

where $\{\lambda_1, \lambda_2\}$ and $\{\mu_1, \mu_2\}$ are two bases for Λ, then

$$\mathrm{Vol}(\mathcal{P}_1) = \int_{\mathcal{P}_1} 1 \; dxdy = \int_{\mathcal{P}_2} |\det(B)| dxdy = |\det(B)| \, \mathrm{Vol}(\mathcal{P}_2)$$

where $B \in \mathrm{GL}(2, \mathbb{Z})$ is the change of basis matrix, then $\det(B) = \pm 1$ and $\mathrm{Vol}(\mathcal{P}_1) = \mathrm{Vol}(\mathcal{P}_2)$. Hence the quantity

$$\mathrm{Vol}(\Lambda) := \mathrm{Vol}(\mathcal{P}),$$

where $\mathcal{P}$ is any fundamental parallelogram, is well defined.

We now state and prove a lemma that we will use in Section 3.

Lemma 2.1.1. *Let Λ be a lattice. There exists a constant $c = c(\Lambda)$ such that for all $N \geq 1$*

$$\#\{\lambda \in \Lambda \mid N \leq |\lambda| < N + 1\} < cN. \tag{2.1}$$

That is, the number of lattice points in the annulus $A_N = \{z \in \mathbb{C} \mid N \leq |z| < N + 1\}$ is approximately linear.

Proof. Let $\mathcal{P} = \left\{t_1\lambda_1 + t_2\lambda_2 \mid 0 \leq t_1, t_2 < 1\right\}$ be a fundamental parallelogram for Λ, as we have just shown above $\mathrm{Vol}(\mathcal{P}) = \mathrm{Vol}(\Lambda)$. Let $d = \sup\{|x - y| \mid x, y \in \mathcal{P}\}$ and, for $R > 0$, let k_R be the number of translates of $\mathcal{P}$ that intersect $B_R = \{z \in \mathbb{C} \mid |z| < R|\}$ (we consider only translates by elements of the lattice), then

$$k_{R-d} \leq \#\{\lambda \in \Lambda \mid |\lambda| < R\} \leq k_R$$

and these fundamental parallelograms are contained in B_{R+d}. On the other hand, we have

$$\mathrm{Vol}(B_R) \leq k_R \cdot \mathrm{Vol}(\Lambda) \leq \mathrm{Vol}(B_{R+d}).$$

Thanks to these inequalities, follow that

$$\#\{\lambda \in \Lambda \mid |\lambda| < R\} = \frac{\pi R^2}{\mathrm{Vol}(\Lambda)} + O(R), \text{ as } R \to +\infty.$$

Now we can easily conclude that

$$\begin{aligned}\#\{\lambda \in \Lambda \mid R \leq |\lambda| < R + 1\} &= \#\{\lambda \in \Lambda \mid |\lambda| < R + 1\} - \#\{\lambda \in \Lambda \mid |\lambda| < R\} \\ &= \frac{2\pi R}{\mathrm{Vol}(\Lambda)} + O(R) < c(\Lambda)R.\end{aligned}$$

□

Definition 2.1.2. We say that two lattices Λ, Λ' are **homothetic** if there exists $\alpha \in \mathbb{C}^\times$ such that $\alpha\Lambda' = \Lambda$. In this case we will write

$$\Lambda \sim \Lambda'.$$

Remark 2.1.2. The relation of homothethy $\sim$ is an equivalence relation.

Let $\Lambda = \mathbb{Z}\lambda_1 \oplus \mathbb{Z}\lambda_2$. Up to swapping λ_1, λ_2 we can suppose $\mathrm{Im}(\lambda_2/\lambda_1) > 0$. Setting $\alpha = 1/\lambda_1$ and $\tau = \lambda_2/\lambda_1$, we have

$$\Lambda \sim \alpha\Lambda = \mathbb{Z} \oplus \mathbb{Z}\tau.$$

Every lattice Λ is thus homothetic to $\Lambda_\tau := \mathbb{Z} \oplus \mathbb{Z}\tau$ for some $\tau \in \mathcal{H}$.

Example 6. The lattices Λ_1 and Λ_2 of Example 1 are not homothetic.

We define

$$\mathrm{Hom}(\Lambda, \Lambda') = \{\alpha \in \mathbb{C} \mid \alpha\Lambda \subseteq \Lambda'\}.$$

It is easy to check that it is an additive subgroup of $\mathbb{C}$; if $\Lambda = \Lambda'$ it is also a ring (commutative with 1) and we will denote it with $\mathrm{End}(\Lambda)$.

2.1.2 Complex tori

Given a lattice $\Lambda = \mathbb{Z}\lambda_1 \oplus \mathbb{Z}\lambda_2$ we can consider the quotient of additive groups

$$X = \mathbb{C}/\Lambda$$

together with the quotient map

$$\begin{array}{rccc} \pi : & \mathbb{C} & \longrightarrow & X \\ & z & \mapsto & z + \Lambda \end{array}.$$

We put on X the quotient topology induced by π. X is then homeomorphic to the torus $T = \mathbb{R}^2/\mathbb{Z}^2$. Because the action of Λ is free and properly discontinuous, π is its universal cover. More is true, in fact we can give X a complex structure that makes it a Riemann surface: for $x \in X$ pick $z \in \pi^{-1}(x)$ and set $U = B(z, \delta)$ where $0 < \delta < \frac{1}{2}\min\{|\lambda| \mid \lambda \in \Lambda \setminus \{0\}\}$, $V = \pi(U)$

$$\pi^{-1} : V \xrightarrow{\sim} U \cong \mathbb{C}$$

is a chart around x, and of course transition maps are holomorphic (they are all the identity map).

We will call X a **complex torus** (of dimension 1).

Remark 2.1.3. On the other hand every Riemann surface X homeomorphic to a torus is the quotient $\mathbb{C}/\Lambda$ for an appropriate lattice Λ. We only give a rough idea of how this can be shown. It is well known that

$$H_1(X,\mathbb{Z}) \cong \pi_1^{ab}(X,x_0) = \pi_1(X,x_0) \cong \mathbb{Z}^2.$$

Take two generators γ_1, γ_2 of $H_1(X,\mathbb{Z})$ (that can be regarded as paths on X) and ω any holomorphic 1-form on X (on a torus we can take $\omega = dz$), consider $\lambda_1, \lambda_2 \in \mathbb{C}$ defined as

$$\lambda_1 = \int_{\gamma_1} \omega, \quad \lambda_2 = \int_{\gamma_2} \omega$$

and set $\Lambda = \mathbb{Z}\lambda_1 \oplus \mathbb{Z}\lambda_2$. Then Λ is a lattice and $X \cong \mathbb{C}/\Lambda$.

Now let's focus on *morphisms* between complex tori. Let $X = \mathbb{C}/\Lambda$, $Y = \mathbb{C}/\Lambda'$ and let $f : X \longrightarrow Y$ be a holomorphic map. Examples are translations

$$\begin{array}{rccc} \tau_{x_0}: & X & \longrightarrow & X \\ & x & \mapsto & x + x_0 \end{array}$$

for any $x_0 \in X$; we can suppose, after composing by $\tau_{-f(0)}$, that $f(0) = 0$.

Proposition 2.1.1. *Let $f : X \longrightarrow Y$ as above, $f(0) = 0$. Then there exists a unique holomorphic map $F : \mathbb{C} \longrightarrow \mathbb{C}$ that induces f, i. e. the diagram below*

$$\begin{array}{ccc} \mathbb{C} & \xrightarrow{F} & \mathbb{C} \\ {\scriptstyle\pi}\downarrow & & \downarrow{\scriptstyle\pi} \\ X & \xrightarrow{f} & Y \end{array}$$

commutes, and $F(0) = 0$; in particular $F(\Lambda) \subseteq \Lambda'$. More precisely $F(z) = \alpha z$, for some $\alpha \in \mathbb{C}$.

Proof. Because π is the universal cover, F can be taken to be the unique holomorphic lift with $F(0) = 0$. It follows, by commutativity, that $F(\Lambda) \subseteq \Lambda'$. We note that

$$F(z+\lambda) \equiv F(z) \pmod{\Lambda'} \ \forall z \in \mathbb{C} \ \ \forall \lambda \in \Lambda.$$

Since Λ' is discrete and F is continuous, $\forall \lambda \in \Lambda \ \exists \lambda' \in \Lambda'$ such that $F(z+\lambda) = F(z) + \lambda' \ \forall z \in \mathbb{C}$, and differentiating leads to

$$F'(z+\lambda) = F'(z) \ \forall z \in \mathbb{C}, \ \forall \lambda \in \Lambda$$

that is F' is periodic with respect to Λ. Than if $\mathcal{P}$ is any fundamental parallelogram for Λ we have:

$$\sup_{z\in\mathbb{C}} |F'(z)| = \sup_{z\in\overline{\mathcal{P}}} |F'(z)| = \max_{z\in\overline{\mathcal{P}}} |F'(z)| = L < +\infty$$

by compactness of $\overline{\mathcal{P}}$, therefore $F'(z)$ is bounded hence, by Liouville's theorem, constant: $F'(z) = \alpha$. It follows that $F(z) = \alpha z + \beta$, but $F(0) = 0 \implies \beta = 0$ and $F(z) = \alpha z$. $\square$

Corollary 2.1.1. *Let $f : X \longrightarrow Y$ be a morphism of complex tori, $f(0) = 0$, then f is a group homomorphism.*

Definition 2.1.3. Let X, Y be complex tori. We define

$$\mathrm{Hom}(X, Y) = \{f : X \longrightarrow Y \mid f \text{ holomorphic}, \ f(0) = 0\}.$$

If $X = Y$ then we write $\mathrm{End}(X)$ for $\mathrm{Hom}(X, X)$.

Remark 2.1.4. $(\mathrm{Hom}(X, Y), +)$, where $+$ is the pointwise sum of functions, is an abelian group and $(\mathrm{End}(X), +, \circ)$, where $\circ$ is the composition of functions, is a ring (commutative with 1).

Remark 2.1.5. If $X = \mathbb{C}/\Lambda$ and $Y = \mathbb{C}/\Lambda'$, then, by proposition 2.1.1, the association

$$\begin{array}{ccc} \mathrm{Hom}(\Lambda, \Lambda') & \longrightarrow & \mathrm{Hom}(X, Y) \\ \alpha & \mapsto & \phi_\alpha(z) \end{array}$$

where $\phi_\alpha(z) = \alpha z \pmod{\Lambda}$, is a bijection. This, together with remark 2.1.3, establishes an equivalence between (the category of) lattices and (the category of) complex tori. This allows us to interchange these two objects.

We conclude this section with a discussion about the isomorphism class of a complex torus. Given a lattice Λ, there exists $\alpha \in \mathbb{C}^\times$ and $\tau \in \mathcal{H}$ such that $\alpha\Lambda = \Lambda_\tau$. The map ϕ_α, defined above, induces an isomorphism of complex tori

$$\phi_\alpha : \mathbb{C}/\Lambda \xrightarrow{\sim} \mathbb{C}/\Lambda_\tau.$$

Therefore it is not restrictive to consider lattices of the form $\Lambda_\tau = \mathbb{Z} \oplus \tau\mathbb{Z}$, $\tau \in \mathcal{H}$.

Let $f : \mathbb{C}/\Lambda_\tau \longrightarrow \mathbb{C}/\Lambda_{\tau'}$ be a non-constant map of complex tori, then there exists $\alpha \in \mathbb{C}^\times$ such that $\alpha\cdot 1$, $\alpha\cdot\tau \in \Lambda_{\tau'}$ therefore there exist $a, b, c, d \in \mathbb{Z}$ such that $\alpha\tau = a\tau' + b$ and $\alpha = c\tau' + d$, and

$$\tau = \frac{a\tau' + b}{c\tau' + d}.$$

If, in addition, f is an isomorphism, then there also exist $a', b', c', d' \in \mathbb{Z}$ such that

$$\tau' = \frac{a'\tau + b'}{c'\tau + d'}$$

and

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot \begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix} = \begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix} \cdot \begin{pmatrix} a & b \\ c & d \end{pmatrix} = I.$$

Therefore the matrices live in $\mathrm{GL}(2, \mathbb{Z})$. Moreover we have

$$\tau = \frac{a\tau' + b}{c\tau' + d} \cdot \frac{c\bar{\tau}' + d}{c\bar{\tau}' + d} = \frac{bd + ac\tau\tau' + ad\tau' + bc\bar{\tau}'}{|c\tau' + d|^2}$$

and taking the imaginary part

$$\operatorname{Im} \tau = \frac{(ad - bc) \operatorname{Im} \tau'}{|c\tau' + d|^2}$$

$\implies ad - bc > 0$ since $\operatorname{Im} \tau,\ \operatorname{Im} \tau' > 0 \implies ad - bc = 1$.

We have just proved the following

Theorem 2.1.1. *Let $X = \mathbb{C}/\Lambda_\tau$ $Y = \mathbb{C}/\Lambda_{\tau'}$ be complex tori. Then $X \cong Y$ if and only if there exists $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$ such that*

$$\tau = \frac{a\tau' + b}{c\tau' + d}.$$

In terms of lattices: $\Lambda_\tau \sim \Lambda_{\tau'} \iff \exists \gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$ such that $\tau = \frac{a\tau'+b}{c\tau'+d}$.

2.2 From Complex Tori to Elliptic Curves

So far we have noticed some analogies between elliptic curves $E/\mathbb{C}$ and complex tori X: they are both abelian groups together with a complex structure and corollary 2.1.1 is analogous to theorem 1.1.2.

In this section we are going to prove that every complex torus $X = \mathbb{C}/\Lambda$ is isomorphic to an elliptic curve $E_\Lambda/\mathbb{C}$. We will write an affine equation for X and we will discover that it is (almost) a Weierstrass equation! In order to do this we are interested in maps

$$\psi : \ \mathbb{C}/\Lambda \ \longrightarrow \ \mathbb{P}^2(\mathbb{C})$$

given by meromorphic functions $x : \mathbb{C}/\Lambda \longrightarrow \mathbb{C}$. If we look at the following diagram

$$\begin{tikzcd} \mathbb{C} \arrow[d, "\pi"'] \arrow[dr, "f"] & \\ \mathbb{C}/\Lambda \arrow[r, "\psi"] & \mathbb{P}^2(\mathbb{C}) \end{tikzcd}$$

ψ is determined by $f(z) = [x_0(z) : x_1(z) : x_2(z)]$ for $x_i : \mathbb{C} \longrightarrow \mathbb{C}$ meromorphic and periodic with respect to the lattice Λ. We call such functions *elliptic functions*: they will be the main character of this section.

2.2.1 Elliptic Functions

Definition 2.2.1. Let Λ a lattice, a meromorphic function $f : \mathbb{C} \longrightarrow \mathbb{C}$ periodic with respect to Λ, that is

$$f(z + \lambda) = f(z) \ \ \forall z \in \mathbb{C}, \ \forall \lambda \in \Lambda$$

is called an **elliptic function** for Λ. We denote with $\mathbb{C}(\Lambda)$ the set of these functions.

Remark 2.2.1. $\mathbb{C}(\Lambda)$ is a field that contains $\mathbb{C}$, identified with the constant functions.

We will use the standard convention for

$$\operatorname{ord}_z f = \text{vanishing order of } f \text{ at } z$$

and

$$\operatorname{Res}_z f = \text{residue of } f \text{ at } z$$

of a meromorphic function $f : \mathbb{C} \longrightarrow \mathbb{C}$.

Lemma 2.2.1. *$f \in \mathbb{C}(\Lambda)$ with no zeros (or poles) is constant.*

Proof. Suppose f has no poles, hence $f : \mathbb{C} \longrightarrow \mathbb{C}$ is holomorphic. By periodicity we have the equality

$$\sup_{z \in \mathbb{C}} |f(z)| = \sup_{z \in \overline{\mathcal{P}}} |f(z)|$$

where $\mathcal{P}$ is any fundamental parallelogram. By compactness of $\overline{\mathcal{P}}$ and by continuity of f, this value is finite then f is bounded hence constant by Liouville's Theorem.

If f has no zeros take $1/f$ and use the argument above. $\square$

Remark 2.2.2. If $a_x(f)$ is a quantity depending on $f \in \mathbb{C}(\Lambda)$ and $x \in \mathbb{C}/\Lambda$, we define the sum

$$\sum_{x \in \mathbb{C}/\Lambda} a_x := \sum_{z \in \mathcal{P}} a_z,$$

that is independent, by periodicity of f, of the choice of the fundamental parallelogram $\mathcal{P}$.

Proposition 2.2.1 (Vanishing formulas)**.** *Let $f \in \mathbb{C}(\Lambda)$, then*

(i) $\sum_{x \in \mathbb{C}/\Lambda} \operatorname{Res}_x(f) = 0$*;*

(ii) $\sum_{x \in \mathbb{C}/\Lambda} \operatorname{ord}_x(f) = 0$*;*

(iii) $\sum_{x \in \mathbb{C}/\Lambda} \operatorname{ord}_x(f) x \in \Lambda$.

Proof. These formulas are consequences of the *residue theorem* (see, for example, [Fre11] III.6).

(i) Let $\mathcal{P}$ be a fundamental parallelogram with no zeros and poles on $\partial\mathcal{P}$. By the residue theorem we have

$$2\pi i \sum_{x \in \mathbb{C}/\Lambda} \operatorname{Res}_x(f) = \int_{\partial\mathcal{P}} f(z) dz.$$

By periodicity of f the integrals along the opposite sides of the parallelogram cancel each other out, hence the integral on the right side is zero.

(ii) Since $f(z)$ is periodic with respect to Λ, also $f'(z)$ is. Now consider the elliptic function $g(z) = f'(z)/f(z)$ and observe that

$$\operatorname{Res}_x(g) = \operatorname{ord}_x(f),$$

and apply *(i)* to the function $g(z)$.

(iii) Let $\mathcal{P}$ be a fundamental parallelogram as in *(i)*. Consider the function $h(z) = zf'(z)/f(z)$ and observe that

$$\operatorname{Res}_x(g) = x \operatorname{ord}_x(f).$$

By the residue theorem we have

$$2\pi i \sum_{x \in \mathbb{C}/\Lambda} x \operatorname{ord}_x(f) = \int_{\partial\mathcal{P}} z \frac{f'(z)}{f(z)} dz = \left(\int_a^{a+\lambda_1} - \int_{a+\lambda_2}^{a+\lambda_1+\lambda_2} \right) \left(z \frac{f'(z)}{f(z)} dz \right) +$$
$$+ \left(\int_{a+\lambda_1}^{a+\lambda_1+\lambda_2} - \int_a^{a+\lambda_2} \right) \left(z \frac{f'(z)}{f(z)} dz \right);$$

by periodicity of $f'(z)/f(z)$ we get

$$\int_{a+\lambda_2}^{a+\lambda_1+\lambda_2} z\frac{f'(z)}{f(z)}dz = \int_a^{a+\lambda_1} (z-\lambda_2)\frac{f'(z)}{f(z)}dz$$

therefore

$$\left(\int_a^{a+\lambda_1} - \int_{a+\lambda_2}^{a+\lambda_1+\lambda_2}\right)\left(z\frac{f'(z)}{f(z)}dz\right) = \lambda_2 \int_a^{a+\lambda_1} \frac{f'(z)}{f(z)}dz = 2\pi i\lambda_2 m$$

for an integer m, the winding number around 0 of the path

$$\gamma : [0,1] \longrightarrow \mathbb{C},\ \gamma(t) = f((1-t)a + t(a+\lambda_1))$$

that is an integer because $f(a) = f(a+\lambda_1)$. The same argument holds for the other integral, hence $\sum_{x\in\mathbb{C}/\Lambda} x\,\mathrm{ord}_x(f) \in \Lambda$.

□

Definition 2.2.2. The order of $f \in \mathbb{C}(\Lambda)$ is the number of poles (counted with multiplicity) of f in any fundamental parallelogram.

Remark 2.2.3. From (ii) of proposition 2.2.1 it follows that the order of f is also equal to the number of zeros (counted with multiplicity) of f in any fundamental parallelogram.

Corollary 2.2.1. *A non-constant elliptic function has order at least 2.*

Proof. If f has a single simple pole at w, then $\mathrm{Res}_w(f) \neq 0$ that is in contradiction with (i) of proposition 2.2.1. □

2.2.2 Weierstrass $\wp$-function

So far we haven't exhibited any example of elliptic function, except for the constants; according to corollary 2.2.1 any other elliptic function has order at least 2, so a possible candidate is a function with a pole of order two at each point of Λ, something like

$$f(z) = \sum_{\lambda\in\Lambda} g_\lambda(z).$$

where g_λ is a meromorphic function with a double pole at λ.

Definition 2.2.3. Given a lattice Λ, the **Weierstrass $\wp$-function** relative to Λ is

$$\wp(z,\Lambda) = \frac{1}{z^2} + \sum_{\lambda\in\Lambda\setminus\{0\}} \left(\frac{1}{(z-\lambda)^2} - \frac{1}{\lambda^2} \right).$$

We also define the **Eisenstein series of weight** $2k$ relative to Λ as

$$G_{2k}(\Lambda) = \sum_{\lambda\in\Lambda\setminus\{0\}} \frac{1}{\lambda^{2k}}.$$

When the lattice Λ is fixed we will simply write $\wp(z)$ and G_{2k}.

We fix a lattice Λ.

Theorem 2.2.1. *1. G_{2k} is absolutely convergent for every $k > 1$;*

2. the series defining $\wp$ is absolutely convergent and converges uniformly on every compact subset $K \subseteq \mathbb{C} \setminus \Lambda$;

3. $\wp$ is an even elliptic function with a pole of order two at each $\lambda \in \Lambda$.

Proof. 1. By lemma 2.1.1, there exists a constant $c = c(\Lambda) > 0$ such that for all $N \geq 1$

$$\#\{\lambda \in \Lambda \mid N \leq |\lambda| < N+1\} < cN.$$

Using this inequality we get

$$\sum_{\substack{\lambda\in\Lambda,\\ |\lambda|\geq 1}} \frac{1}{|\lambda|^{2k}} \leq \sum_{N=1}^{+\infty} \frac{\#\{\lambda \in \Lambda \mid N \leq |\lambda| < N+1\}}{N^{2k}} < \sum_{N=1}^{+\infty} \frac{c}{N^{2k-1}} < +\infty.$$

2. Let $K \subset \mathbb{C} \setminus \Lambda$ be a compact set, and let $M = \max\{|z| \mid z \in K\}$. If $z \in K$ and $|\lambda| > 2M$, then we have $|z-\lambda| \geq |\lambda|/2$ and $|2\lambda - z| \leq 5|\lambda|/2$. Therefore

$$\left| \frac{1}{(z-\lambda)^2} - \frac{1}{\lambda^2} \right| = \left| \frac{z(2\lambda - z)}{\lambda^2(z-\lambda)^2} \right| \leq \frac{10|z|}{|\lambda|^3} \leq \frac{10M}{|\lambda|^3}.$$

By 1. we see that the series defining $\wp(z)$ converges absolutely for every $z \in \mathbb{C} \setminus \Lambda$ and uniformly on every compact set $K \subset \mathbb{C} \setminus \Lambda$.

3. $\wp$ is a meromorphic function by 2. and it is clear, by its defining series, that it has a pole of order two at each $\lambda \in \Lambda$. It is also clear that

$$\wp(-z) = \wp(z) \ \forall z. \tag{2.2}$$

In order to show that $\wp$ is in fact an elliptic function, let's compute its derivative:

$$\wp'(z, \Lambda) = -2 \sum_{\lambda \in \Lambda} \frac{1}{(z-\lambda)^3}.$$

It is clear that $\wp'$ is an elliptic function:

$$\wp'(z+\lambda) = \wp'(z) \ \ \forall \lambda \in \Lambda \tag{2.3}$$

so integrating (2.3) yields

$$\wp(z+\lambda) = \wp(z) + \alpha(\lambda) \ \ \forall \lambda \in \Lambda.$$

Evaluating at $z = -\lambda/2$ and using (2.2) we obtain $\alpha(\lambda) = 0 \ \forall \lambda$, thus $\wp$ is periodic with respect Λ, hence an elliptic function.

□

Lemma 2.2.2. *Let Λ be a lattice and $z \in \mathbb{C}$. Then:*

(i) if $2z \notin \Lambda$ we have

$$\mathrm{ord}_w(\wp - \wp(z)) = \begin{cases} 1 & w \in \pm z + \Lambda \\ -2 & w \in \Lambda \\ 0 & \textit{otherwise} \end{cases}$$

(ii) if $2z \in \Lambda$ and $z \notin \Lambda$ we have

$$\mathrm{ord}_w(\wp - \wp(z)) = \begin{cases} 2 & w \in z + \Lambda \\ -2 & w \in \Lambda \\ 0 & \textit{otherwise} \end{cases}$$

Proof. It is enough to consider the function $\wp - \wp(z)$ and then apply *(ii)* of proposition 2.2.1. □

Theorem 2.2.2. *Let $\mathbb{C}^{even}(\Lambda) \subseteq \mathbb{C}(\Lambda)$ be the subfield of the even elliptic functions. Then*

$$\mathbb{C}^{even}(\Lambda) = \mathbb{C}(\wp(z)).$$

Proof. Let f be an even elliptic function. The idea of the proof is to modify f in order to obtain an even elliptic function with no zeroes or poles in $\mathbb{C}\setminus\Lambda$. Fix a fundamental parallelogram $\mathcal{P}$. Since $\overline{\mathcal{P}}$ is compact, f has only a finite number of zeroes or poles in $\mathcal{P}$. Let $w \in \mathcal{P},\ w \notin \Lambda$ such that $\operatorname{ord}_w(f) \neq 0$ (that is w is a zero or a pole). We distinguish two cases:

- $2w \notin \Lambda$. Because f is even, $\operatorname{ord}_w f = \operatorname{ord}_{-w} f$. Consider the new function
$$f' = f \cdot (\wp - \wp(w))^{-\operatorname{ord}_w f}.$$
By lemma 2.2.2, f' has the same zeroes and poles as f (with multiplicity) except for those $x \in \pm w + \Lambda$ where $\operatorname{ord}_x f' = 0$ and for $x \in \Lambda$.

- $2w \in \Lambda$. In this case $m = \operatorname{ord}_w f$ is even: writing $w = \lambda/2$ for $\lambda \in \Lambda$, for z sufficiently near to w we have
$$f(\lambda/2 + z) = a_m z^m + o(z^m)$$
and by periodicity and evenness
$$f(\lambda/2 - z) = f(-\lambda/2 + z) = f(\lambda/2 + z)$$
hence $a_m(-z)^m + \cdots = a_m z^m + \ldots$ and so m must be even. Now consider
$$f' = f \cdot (\wp - \wp(w))^{-\frac{1}{2}\operatorname{ord}_w f}$$
and, as before, by lemma 2.2.2, this new function has the same zeroes and poles (with multiplicity) as f except for $x \in w+\Lambda$, where $\operatorname{ord}_x f' = 0$ and for $x \in \Lambda$.

We replace f with f' and reiterate this procedure. Eventually we obtain an equality:
$$f \cdot \frac{r(\wp)}{s(\wp)} = g$$
for two polynomials r, s. Now $g \in \mathbb{C}^{even}(\Lambda)$ has no zeroes or poles on $\mathbb{C}\setminus\Lambda$, by remark 2.2.3 g can't have poles (or zeroes) at any $\lambda \in \Lambda$ hence it is constant by lemma 2.2.1 and then $f \in \mathbb{C}(\wp)$ as desired. □

Corollary 2.2.2. *Every elliptic function is a rational function of $\wp$ and $\wp'$, that is*
$$\mathbb{C}(\Lambda) = \mathbb{C}(\wp(z), \wp'(z)).$$

Proof. Given $f \in \mathbb{C}(\Lambda)$, we can write

$$f(z) = \frac{f(z) + f(-z)}{2} + \frac{f(z) - f(-z)}{2} = f_1(z) + f_2(z)$$

where f_1 is even and f_2 is odd. Thanks to theorem 2.2.2 we have $f_1 \in \mathbb{C}(\wp)$ and $f_2 = \frac{f_2\wp'}{\wp'} \in \frac{1}{\wp'}\mathbb{C}(\wp)$. □

The next step is to find an algebraic relation between $\wp$ and $\wp'$. Since $\wp'$ is odd then $(\wp')^2$ is even, therefore

$$(\wp')^2 = \frac{a(\wp)}{b(\wp)}$$

for two polynomials $a(x), b(x)$. On the left hand side we have a function with a pole of order 6 at $z = 0$, on the right the order of pole at 0 is $2\cdot(\deg a - \deg b)$, thus

$$\deg a = 3 + \deg b.$$

Comparing the zeroes on a fundamental parallelogram, on the left this number, counted with multiplicity, is 6, on the right is $2 \cdot \deg a$ so $\deg a = 3$. It follows that

$$(\wp')^2 = a(\wp)$$

for a cubic polynomial $a(x)$.

In order to find this polynomial we will use the following lemma.

Lemma 2.2.3. *The Laurent series for $\wp(z)$ around $z = 0$ is*

$$\wp(z) = \frac{1}{z^2} + \sum_{k=1}^{\infty}(2k+1)G_{2k+2}z^{2k}$$

Proof. Let $\delta = \min_{\lambda\in\Lambda\setminus\{0\}} |\lambda|$ (it is well defined since Λ is discrete). Then for $z \in \mathcal{B}(0, \delta)$ and $\lambda \in \Lambda \setminus \{0\}$ we have

$$\begin{aligned}\frac{1}{(z-\lambda)^2} - \frac{1}{\lambda} &= \lambda^{-2}\left(\left(\frac{1}{1 - z/\lambda}\right)^2 - 1\right)\\ &= \sum_{n=1}^{\infty}(n+1)\frac{z^n}{\lambda^{n+2}}\end{aligned}$$

where the second equality follows by the well known equality $1/(1-z/\lambda) = \sum_{n=0}^{\infty}(z/\lambda)^n$. Thus, for $z \in \mathcal{B}(0,\delta)$:

$$\begin{aligned}
\wp(z) &= \frac{1}{z^{-2}} + \sum_{\lambda\in\Lambda\setminus\{0\}} \frac{1}{(z-\lambda)^2} - \frac{1}{\lambda} \\
&= \frac{1}{z^{-2}} + \sum_{\lambda\in\Lambda\setminus\{0\}} \sum_{n=1}^{\infty}(n+1)\frac{z^n}{\lambda^{n+2}} \\
&= \frac{1}{z^{-2}} + \sum_{n=1}^{\infty}(n+1)z^n \sum_{\lambda\in\Lambda\setminus\{0\}} \frac{1}{\lambda^{n+2}} \\
&= \frac{1}{z^2} + \sum_{k=1}^{\infty}(2k+1)G_{2k+2}z^{2k}
\end{aligned}$$

because $G_{2k+1} = 0\ \forall k$. □

Thanks to the lemma we can compute the expansion around $z = 0$ of $\wp'$, thus

$$\begin{aligned}
\wp(z) &= z^{-2} + 3G_4z^2 + 5G_6z^4 + \dots \\
\wp(z)' &= -2z^{-3} + 6G_4z + 20G_6z^3 + \dots
\end{aligned}$$

Cubing and squaring the above formulas yields

$$\begin{aligned}
\wp(z)^3 &= z^{-6} + 9G_4z^{-2} + 15G_6 + \dots \\
\wp'(z)^2 &= -4z^{-6} - 24G_4z^{-2} - 80G_6 + \dots
\end{aligned}$$

Comparing these equations, we note that

$$f(z) = \wp'(z)^2 - 4\wp(z)^3 + 60G_4\wp(z) + 140G_6 = \alpha z + \beta z^2 + \dots$$

Thus f is an elliptic function holomorphic in a sufficiently small neighborhood of $z = 0$, that vanishes at $z = 0$. By periodicity, f is also holomorphic at each point of Λ. Since $\wp$ and $\wp'$ are holomorphic on $\mathbb{C}\setminus\Lambda$, $f : \mathbb{C} \longrightarrow \mathbb{C}$ is a holomorphic elliptic function, $f(0) = 0$, by lemma 2.2.1 $f \equiv 0$.

Theorem 2.2.3. *Let Λ be a lattice, set $g_2 = 60G_4(\Lambda)$ and $g_3 = 140G_6(\Lambda)$, then*

$$\wp'(z)^2 = 4\wp(z)^3 - g_2\wp(z) - g_3 \quad \forall z \in \mathbb{C}\setminus\Lambda. \tag{2.4}$$

Equation (2.4) is very similar to a Weierstrass equation !! Given a lattice Λ, define $E_\Lambda/\mathbb{C}$ to be the curve given by (the projective closure of)

$$E_\Lambda : y^2 = 4x^3 - g_2x - g_3.$$

Proposition 2.2.2. *E_Λ is non-singular, then it is isomorphic to an elliptic curve. In particular the discriminant*

$$\Delta(\Lambda) = g_2^2 - 27g_3^3$$

is nonzero.

Proof. We prove that the polynomial $f(x) = 4x^3 - g_2x - g_3$ has three distinct roots. Let $\{\lambda_1, \lambda_2\}$ be a basis for Λ, define $\lambda_3 = \lambda_1 + \lambda_2$. Because $\wp'(z)$ is an odd function we have

$$\wp'(\lambda_i/2) = -\wp'(-\lambda_i/2) = -\wp'(\lambda_i/2)$$

so $\wp'(\lambda_i/2) = 0$ for $i = 1, 2, 3$. By (4) it follows that $f(x)$ vanishes at each $\wp(\lambda_i/2)$. It is enough to show that these three numbers are distinct. The function $\wp(z) - \wp(\lambda_i/2)$ is even, therefore has a zero of order at least two at $z = \lambda_i/2$. On the other hand this function has order two thus $\lambda_i/2$ is the only zero (in the fundamental parallelogram $\mathcal{P} = \{\alpha\lambda_1 + \beta\lambda 2 \mid 0 \leq \alpha, \beta < 1\}$) hence $\wp(\lambda_i/2) \neq \wp(\lambda_j/2)$ for $i \neq 0$. □

Now let's consider the map $\Phi : \mathbb{C} \longrightarrow \mathbb{P}^2(\mathbb{C})$ given by

$$\Phi(z) = \begin{cases} [0:1:0] = \infty & \text{if } z \in \Lambda \\ [\wp(z) : \wp'(z) : 1] & \text{otherwise} \end{cases}$$

since $\wp, \wp' \in \mathbb{C}(\Lambda)$, the function Φ induces a map on $\mathbb{C}/\Lambda$ that we still denote by Φ:

$$\begin{tikzcd} \mathbb{C} \arrow[d, "\pi"'] \arrow[dr, "\Phi"] & \\ \mathbb{C}/\Lambda \arrow[r, "\Phi"] & \mathbb{P}^2(\mathbb{C}) \end{tikzcd}$$

Theorem 2.2.4. *The map $\Phi : \mathbb{C}/\Lambda \longrightarrow E_\Lambda$ is both an isomorphism of Riemann surfaces and a group homomorphism.*

Proof. Surjectivity. Let $(x, y) \in E_\Lambda$ (in the affine part) and consider the function $\wp(z) - x$ that is a non-constant elliptic function, hence, by lemma 2.2.1, it has a zero, say $z = a$. Using relation (2.4) we obtain $\wp'(a)^2 = y^2$, so $y = \pm\wp'(a)$ thus, up to replacing a with $-a$, we have $\Phi(a) = (x, y)$. To conclude, note that $\Phi(0) = (0 : 1 : 0)$.

Injectivity. Suppose $\Phi(z_1) = \Phi(z_2)$ and $2z_1 \notin \Lambda$ (the other case is analogous), then the order two elliptic function $\wp(z) - \wp(z_1)$ vanishes on $z_1, -z_1$ and z_2. It follows that $z_1 \equiv \pm z_2 \pmod{\Lambda}$, but

$$\wp'(z_2) = \wp'(z_1) = \wp'(\pm z_2) = \pm\wp'(z_2)$$

and $\wp'(z_2) \neq 0$, so $z_1 \equiv z_2 \pmod{\Lambda}$.

Holomorphy. It is clear that Φ is holomorphic on $\mathbb{C} \setminus \Lambda$ because $\wp$ and $\wp'$ are. Around $\lambda \in \Lambda$ we use the following representation for Φ:

$$z \mapsto \left[\frac{\wp(z)}{\wp'(z)} : 1 : \frac{1}{\wp'(z)}\right].$$

Since $\mathbb{C}/\Lambda$ and E_Λ are compact Riemann surfaces it follows that Φ is an isomorphism.

Group homomorphism. Consider the map

$$\begin{aligned} s : \mathbb{C}/\Lambda \times \mathbb{C}/\Lambda &\longrightarrow E_\Lambda \\ ([z_1],[z_2]) &\longmapsto \Phi([z_1]+[z_2]) - \Phi([z_1]) - \Phi([z_1]) \end{aligned}$$

which is holomorphic because Φ and the sum on E_Λ they are. We have

$$s|_{\mathbb{C}/\Lambda \times \{[0]\}} \equiv 0 \equiv s|_{\{[0]\} \times \mathbb{C}/\Lambda}$$

hence, by holomorphy, $s \equiv 0$ and therefore $\Phi([z_1]+[z_2]) = \Phi([z_1]) + \Phi([z_1])$. □

Remark 2.2.4. The definition of the Weierstrass $\wp$-function and the Eisenstein series depends on the lattice Λ. If we take an homothetic lattice $\Lambda' \sim \Lambda$, $\Lambda' = \alpha\Lambda,\ \alpha \in \mathbb{C}^\times$, then

$$\wp(z;\Lambda) = \alpha^2 \wp(\alpha z;\Lambda'),$$

and

$$G_{2k}(\Lambda) = \alpha^{2k} G_{2k}(\Lambda').$$

Homothetic lattices define different curves, but, as we are going to show, these curves are isomorphic. Let Λ and Λ' be homothetic lattices: $\Lambda' = \alpha\Lambda$ for $\alpha \in \mathbb{C}^\times$, then

$$E_\Lambda :\ y^2 = 4x^2 - g_2(\Lambda)x - g_3(\Lambda)$$

and

$$E_{\Lambda'} :\ y^2 = 4x^2 - g_2(\Lambda')x - g_3(\Lambda')$$

with $g_2(\Lambda) = 60G_4(\Lambda) = \alpha^4 60G(\Lambda') = \alpha^4 g_2(\Lambda')$ and $g_3(\Lambda) = \alpha^6 g_3(\Lambda')$. The change of variables

$$(x,y) \mapsto (x',y') = (x, y/2)$$

turns E_Λ in Weierstrass form: $y^2 = x^3 - \frac{g_2(\Lambda)}{4}x - \frac{g_3(\Lambda)}{4}$; similarly for $E_{\Lambda'}$.

Since $\mathbb{C}$ is algebraically closed, by theorem 1.1.4 the isomorphism class of an elliptic curve $E/\mathbb{C}$ defined by a Weierstrass equation

$$y^2 = x^3 + Ax + B$$

is determined by its j-invariant:

$$j(E) = 1728\frac{4A^3}{4A^3 + 27B^2}.$$

In our case we have:

$$\begin{aligned} j(E_\Lambda) &= 1728\frac{g_2(\Lambda)^3}{g_2(\Lambda)^3 - 27g_3(\Lambda)^2} = 1728\frac{\alpha^{12}g_2(\Lambda')^3}{\alpha^{12}g_2(\Lambda')^3 - 27\alpha^{12}g_3(\Lambda')^2} \\ &= 1728\frac{g_2(\Lambda')^3}{g_2(\Lambda')^3 - 27g_3(\Lambda')^2} = j(E_{\Lambda'}) \end{aligned}$$

therefore the elliptic curves E_Λ and $E_{\Lambda'}$ are isomorphic.

Example 7.

1. Let $\Lambda = \mathbb{Z}[i] = \mathbb{Z}\oplus\mathbb{Z}i$. We note that $i\Lambda = i\mathbb{Z}[i] = \mathbb{Z}[i] = \Lambda$, hence
$$g_3(\Lambda) = (i)^6 g_3(i\Lambda) = -g_3(\Lambda)$$
it follows that $g_3(\Lambda) = 0$. The elliptic curve E_Λ has, thus, equation
$$y^2 = 4x^3 - g_2(\Lambda)x$$
and $j(E_\Lambda) = 1728$. It follows that E_Λ is isomorphic, for example, to the curve $y^2 = x^3 + x$.

2. We do the same for $\Lambda = \mathbb{Z} \oplus \mathbb{Z}\rho = \mathbb{Z}[\rho]$, where $\rho = (\sqrt{-3} - 1)/2$. In this case, since $\rho^2 + \rho + 1 = 0$, we have $\rho\mathbb{Z}[\rho] = \mathbb{Z}[\rho]$, thus
$$g_2(\Lambda) = (\rho)^4 g_2(\rho\Lambda) = \rho g_2(\Lambda)$$
it follows that $g_2(\Lambda) = 0$. The elliptic curve E_Λ has equation
$$y^2 = 4x^3 - g_3(\Lambda)$$
and $j(E_\Lambda) = 0$, it is isomorphic to the curve $y^2 = x^3 + 1$.

We will denote the set of lattices modulo homothety by

$$\mathcal{L} = \frac{\{\text{Lattices in } \mathbb{C}\}}{\text{homothety}}.$$

We will write $[\Lambda] = \{\Lambda' \mid \Lambda' \sim \Lambda\}$ for the homothety class of Λ, and

$$\mathcal{ELL} = \frac{\{\text{Elliptic curves over } \mathbb{C}\}}{\text{isomorphism}}$$

for the set of all isomorphism classes of elliptic curves defined over $\mathbb{C}$. According to Remark 2.2.4 we have a well defined map

$$\begin{array}{ccc} \mathcal{L} & \longrightarrow & \mathcal{ELL} \\ [\Lambda] & \mapsto & [E_\Lambda] \end{array}.$$

Our goal it to show that this correspondence is one-to-one: in order to do this we must go in the opposite direction, that is, start with an elliptic curve $E/\mathbb{C}$ and find a lattice Λ_E such that $E = E_{\Lambda_E}$. This is known as the *Uniformization theorem* and will be the topic of the next section.

2.3 Uniformization

In chapter 1 we have seen that two elliptic curves E/K and E'/K are isomorphic over $\overline{K}$ if and only if $j(E) = j(E')$. Moreover if

$$E : y^2 = x^3 + Ax + B \text{ and } E' : y^2 = x^3 + A'x + B'$$

are defined by Weierstrass equations, then E and E' are isomorphic (over K) if and only if $A' = \mu^4 A$ and $B' = \mu^6 B$ for some $\mu \in K^\times$, and the isomorphism is given by the map

$$(x, y) \mapsto (\mu^2 x, \mu^3 y).$$

In our case $K = \mathbb{C}$ is algebraically closed, hence the j-invariant of an elliptic curve over $\mathbb{C}$ determine its isomorphism class. In this section we are going to define the analogue of the j-invariant for lattices.

2.3.1 The j-invariant of a lattice

Definition 2.3.1. Let Λ be a lattice. The **discriminant** of Λ is

$$\Delta(\Lambda) = g_2(\Lambda)^3 - 27g_3(\Lambda)^2$$

and its **j-invariant** is

$$j(\Lambda) = 1728\frac{g_2(\Lambda)^3}{\Delta(\Lambda)} = 1728\frac{g_2(\Lambda)^3}{g_2(\Lambda)^3 - 27g_3(\Lambda)^2}.$$

Remark 2.3.1. The j-invariant is well defined since, by proposition 2.2.2, $\Delta(\Lambda) \neq 0$ for every lattice Λ.

Remark 2.3.2. It is easy to check that $j(\Lambda) = j(E_\Lambda)$.

The j-invariant of a lattice characterizes lattices up to homothety.

Theorem 2.3.1. *Two lattices Λ and Λ' are homothetic if and only if $j(\Lambda) = j(\Lambda')$.*

Proof. If $\Lambda \sim \Lambda'$ then $\Lambda' = \alpha\Lambda$ for some $\alpha \in \mathbb{C}^\times$, and by Remark 3.2.1 we have $g_2(\Lambda') = \alpha^{-4}g_2(\Lambda)$ and $g_3(\Lambda') = \alpha^{-6}g_3(\Lambda)$. Therefore

$$\begin{aligned} j(\Lambda') &= 1728\frac{g_2(\Lambda')^3}{g_2(\Lambda')^3 - 27g_3(\Lambda')^2} = 1728\frac{\alpha^{-12}g_2(\Lambda)^3}{\alpha^{-12}g_2(\Lambda)^3 - 27\alpha^{-12}g_3(\Lambda)^2} \\ &= 1728\frac{g_2(\Lambda)^3}{g_2(\Lambda)^3 - 27g_3(\Lambda)^2} = j(\Lambda). \end{aligned}$$

Conversely, assume $j(\Lambda) = j(\Lambda')$. Let E_Λ and $E_{\Lambda'}$ be the corresponding elliptic curves: then $j(E_\Lambda) = j(E_{\Lambda'})$, hence they are isomorphic.

We may assume they are in short Weierstrass form:

$$E_\Lambda : y^2 = x^3 + Ax + B$$

with $-4A = g_2(\Lambda)$, $-4B = g_3(\Lambda)$ and similarly for $E_{\Lambda'}$ with $-4A' = g_2(\Lambda')$, $-4B' = g_3(\Lambda')$. Therefore there exists $\mu \in \mathbb{C}^\times$ such that $A' = \mu^4 A$ and $B' = \mu^6 B$, hence, if we let $\alpha = \mu^{-1}$, we have $g_2(\Lambda') = \alpha^{-4}g_2(\Lambda) = g_2(\alpha\Lambda)$ and $g_3(\Lambda') = \alpha^{-6}g_3(\Lambda) = g_3(\alpha\Lambda)$. We claim that $\Lambda' = \alpha\Lambda$.

Recall from Section 2 that the Weierstrass $\wp$-function satisfies the differential equation

$$\wp'(z)^2 = 4\wp(z)^3 - g_2\wp(z) - g_3$$

differentiating both sides yields

$$\wp''(z) = 6\wp(z)^2 - g_2/2. \tag{2.5}$$

Also recall that the Laurent expansion of $\wp$ around $w = 0$ is

$$\wp(z) = \frac{1}{z^2} + \sum_{n=1}^{\infty}(2n+1)G_{2n+2}z^{2n} = \frac{1}{z^2} + \sum_{n=1}^{\infty} a_n z^{2n}$$

where $a_1 = g_2/20$, $a_2 = g_3/28$. Comparing the coefficient of z^{2n} in (2.5) we obtain

$$(2n+2)(2n+1)a_{n+1} = 6\left(2a_{n+1} + \sum_{k=1}^{n-1} a_k a_{n-k}\right), \quad \forall n \geq 2$$

which can be rewritten as

$$a_{n+1} = \frac{6}{(2n+2)(2n+1) - 12}\sum_{k=1}^{n-1} a_k a_{n-k}, \quad \forall n \geq 2.$$

This allows us to compute a_{n+1} from $a_1, \ldots, a_{n-1}$. It follows that the series expansion of $\wp(z;\Lambda)$ at $w = 0$, hence $\wp(z;\Lambda)$ itself, is uniquely determined by $a_1 = g_2(\Lambda)$ and $a_2 = g_3(\Lambda)$.

In our case we have $g_2(\Lambda') = g_2(\alpha\Lambda)$ and $g_3(\Lambda') = g_3(\alpha\Lambda)$, therefore $\wp(z,\Lambda) = \wp(z,\alpha\Lambda)$ and in particular they have the same set of poles, that is $\Lambda' = \alpha\Lambda$. □

Corollary 2.3.1. *Two lattices are homothetic if and only if the corresponding elliptic curves are isomorphic.*

2.3.2 Modularity

The next thing we are going to prove is that for all $j_0 \in \mathbb{C}$ there is a lattice Λ such that $j(\Lambda) = j_0$.

In order to show this we need some basic facts on the theory of Modular Forms. The **modular group** is

$$\Gamma(1) = \mathrm{SL}_2(\mathbb{Z}) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \middle| \, a,b,c,d \in \mathbb{Z},\ ad - bc = 1 \right\}$$

and acts[3] on $\mathcal{H} = \{\tau \in \mathbb{C} \mid \operatorname{Im}\tau > 0\}$ via Möbius transformations

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot \tau = \frac{a\tau + b}{c\tau + d}.$$

We will denote $\mathcal{H}/\Gamma(1) = \{\Gamma(1)\cdot\tau \mid \tau \in \mathcal{H}\}$ the set of the orbits and with $\pi : \mathcal{H} \longrightarrow \mathcal{H}/\Gamma(1)$ the quotient map.

Remark 2.3.3. Consider $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $S = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, it is clear that $T, S \in \Gamma(1)$. We have, $\forall \tau \in \mathcal{H}$,

$$T \cdot \tau = \tau + 1,\ S \cdot \tau = -1/\tau.$$

They correspond, respectively, to a translation and an inversion.

Remark 2.3.4. The matrix I and $-I$ both act trivially on $\mathcal{H}$. Thus, if we define

$$\overline{\Gamma(1)} = \mathrm{SL}_2(\mathbb{Z})/\{\pm I\},$$

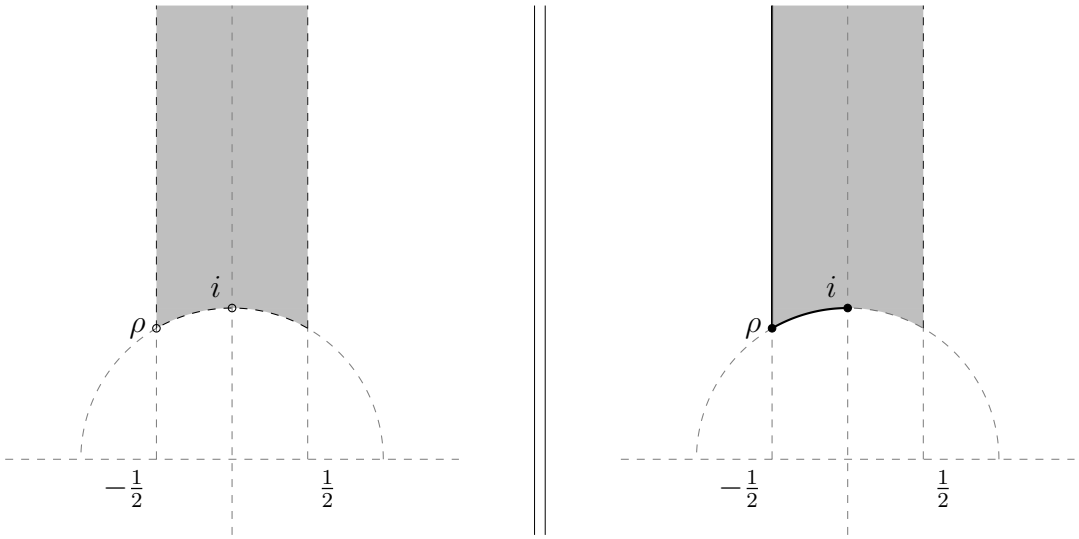
the actions of $\Gamma(1)$ and $\overline{\Gamma(1)}$ have the same orbits.

We are now looking for a fundamental domain for this action. Define

$$\mathcal{F} = \left\{ \tau \in \mathcal{H} \middle| \, |\tau| > 1 \wedge -\frac{1}{2} < \operatorname{Re}\tau < \frac{1}{2} \right\},$$

see figure 2.2.

[3] It is easy to check that this is in fact an action.



(a) Fundamental domain $\mathcal{F}$ (b) The set $\mathcal{F}_0$

Figure 2.2

Lemma 2.3.1. *$\mathcal{F}$ is a fundamental domain for the action of Γ on $\mathcal{H}$. That is*

1. *$\forall z \in \mathcal{H}\ \exists \gamma \in \Gamma$ such that $\gamma \cdot z \in \overline{\mathcal{F}}$;*
2. *$\forall z_1, z_2 \in \overline{\mathcal{F}}$ such that $z_1 \in \Gamma(1) \cdot z_2$ then:*
   - *(a) $\mathrm{Re}(z_1) = \pm\frac{1}{2}$ and $z_2 = z_1 \mp 1$, or*
   - *(b) $|z_1| = 1$ and $z_2 = S \cdot z_1$.*

In particular, $\mathcal{F}_0 = \big\{z \in \mathcal{H} \mid \big(|z| \geq 1 \ \wedge\ -\frac{1}{2} \leq \mathrm{Re}\, z \leq 0\big) \ \vee\ \big(|z| > 1 \ \wedge\ 0 < \mathrm{Re}\, z < \frac{1}{2}\big)\big\}$ is a set of representatives for the orbits of the action.

Proof. Let $\tau \in \mathcal{H}$, it is not difficult to prove that the set

$$\big\{\, \mathrm{Im}(\gamma \cdot \tau) \mid \gamma \in \Gamma(1)\big\} \subseteq \mathbb{R}_{>0}$$

has a maximum. Let $\gamma \in \Gamma(1)$ such that $\mathrm{Im}(\gamma \cdot \tau)$ assumes the maximum value. Applying T an appropriate number of times we have

$$-1/2 \leq \mathrm{Re}(T^k \gamma \cdot \tau) \leq 1/2$$

and this does not alter the imaginary part. We claim that $\tau' = T^k \gamma \cdot \tau \in \overline{\mathcal{F}}$. In fact we have

$$\frac{\mathrm{Im}(\tau')}{|\tau'|^2} = \mathrm{Im}(S \cdot \tau') \leq \mathrm{Im}(\tau') \implies |\tau'| \geq 1$$

because $\operatorname{Im}(\tau')$ is the maximum. This proves 1.

Let $\tau,\ \gamma\cdot\tau \in \overline{\mathcal{F}}$, $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. We distinguish two cases:

- $c = 0$. In this case
$$\gamma = \pm\begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}$$
but $-I$ acts trivially on $\mathcal{H}$ hence we can suppose γ is with positive sign. Therefore $\gamma\cdot\tau = \tau + b \implies \operatorname{Re}\tau = \pm 1/2$ and $b = \mp 1$.

- $c \neq 0$. Up to swapping γ with γ^{-1} we can suppose
$$\operatorname{Im}(\gamma\cdot\tau) \geq \operatorname{Im}\tau.$$
Letting $\tau = x + iy$, we then have
$$\begin{aligned}\operatorname{Im}(\gamma\cdot\tau) &= \frac{\operatorname{Im}\tau}{|c\tau + d|^2} = \frac{y}{c^2y^2 + (cx+d)^2} \\ &\leq \frac{y}{c^2y^2} = \frac{1}{c^2y}.\end{aligned}$$
Now we note that if $w \in \overline{\mathcal{F}}$ then $\operatorname{Im} w \geq \sqrt{3}/2$. Thus
$$\frac{\sqrt{3}}{2} \leq y \leq \frac{2\sqrt{3}}{3c^2}$$
$\implies c^2 = 1 \implies c = \pm 1$. We can assume, as above, $c = 1$. Now let's consider the inequalities
$$\operatorname{Im}(\gamma\cdot\tau) = \frac{\operatorname{Im}\tau}{|\tau + d|^2} \leq \frac{\operatorname{Im}\tau}{|\tau|^2} \leq \operatorname{Im}\tau \leq \operatorname{Im}(\gamma\cdot\tau)$$
it follows that they are all equalities, hence $|\tau| = 1$ and $|\tau + d| = |\tau|$. We have two cases:
  - $d = 0$, thus $\gamma = \begin{pmatrix} a & -1 \\ 1 & 0 \end{pmatrix}$ and $\gamma\cdot\tau = a - 1/\tau$. Since both τ and $\gamma\cdot\tau$ belongs to $\overline{\mathcal{F}}$, it follows that $a = 0$ or $a = \pm 1$ and $\tau = i$ or $\tau = \rho, -\overline{\rho}$ respectively (in the first case we are in (b) otherwise in (a)).
  - $d = \pm 1$, in this case we get $\tau = \rho$ or $\tau = -\overline{\rho}$, case (a).

$\square$

2.3.3 The j-function

Recall, from Section 2.1, that we denote $\Lambda_\tau = \mathbb{Z} \oplus \mathbb{Z}\tau$, $\tau \in \mathcal{H}$. The j-function is

$$\begin{array}{rccc} j: & \mathcal{H} & \longrightarrow & \mathbb{C} \\ & \tau & \mapsto & j(\Lambda_\tau). \end{array}$$

We also write $g_i(\tau) = g_i(\Lambda_\tau)$ for $i = 2, 3$ and $\Delta(\tau) = \Delta(\Lambda_\tau)$. With this notation

$$j(\tau) = 1728 \frac{g_2(\tau)^3}{\Delta(\tau)} = 1728 \frac{g_2(\tau)^3}{g_2(\tau)^3 - 27 g_3(\tau)^2}.$$

The j-function is an example of a *modular function.*

Lemma 2.3.2. *For any matrices*

$$\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma(1)$$

we have

$$j(\gamma \cdot \tau) = j\left(\frac{a\tau + b}{c\tau + d}\right) = j(\tau).$$

Also, $j(\tau)$ is periodic of period one, and its Fourier q-expansion is

$$j(\tau) = \frac{1}{q} + 744 + \sum_{n \geq 1} c_n q^n,$$

where $q = e^{2\pi i \tau}$, and c_n are positive integers.

Proof. The $\Gamma(1)$-invariance is a consequence of theorem 2.1.1 and theorem 2.3.1. Note that for $\gamma = T$ we get

$$j(\tau + 1) = j(\tau),$$

which says that j is periodic of period one. By this our function admits, in fact, a Fourier q-expansion. It is shown in [Sil94], Chapter 1, Proposition 7.4, that $g_2(\tau)$ and $\Delta(\tau)$ have q-expansions given by

$$\begin{aligned} g_2(\tau) &= \frac{(2\pi)^4}{12}(1 + 240q + O(q^2)), \\ \Delta(\tau) &= (2\pi)^{12} q (1 - 24q + O(q^2)). \end{aligned}$$

Therefore

$$\begin{aligned} j(\tau) = 1728\frac{g_2(\tau)^3}{\Delta(\tau)} &= \frac{(1+240q+O(q^2))^3}{q(1-24q+O(q^2))} \\ &= (1+720q+O(q^2))(1+24q+O(q^2)) \\ &= \frac{1}{q} + 744 + \sum_{n\geq 1} c_n q^n. \end{aligned}$$

Because $g_2(\tau)$ and $\Delta(\tau)$ have q-expansions with integer coefficients, it follows that also $j(\tau)$ admits a q-expansion in $\mathbb{Z}[[q]]$. □

More is true, in fact we have the following important result.

Theorem 2.3.2. *The j-function is holomorphic, $\Gamma(1)$-invariant, and surjective. It induces a bijection*

$$\bar{j} : \mathcal{H}/\Gamma(1) \longrightarrow \mathbb{C}$$

Proof. The holomorphy of j follows from the holomorphy of g_2 and g_3. The series

$$g_2(\tau) = 60 \sum_{\substack{c,d\in\mathbb{Z} \\ (c,d)\neq(0,0)}} \frac{1}{(c\tau+d)^4}$$

and

$$g_3(\tau) = 140 \sum_{\substack{c,d\in\mathbb{Z} \\ (c,d)\neq(0,0)}} \frac{1}{(c\tau+d)^6}$$

are absolutely convergent and converge uniformly over any compact subset of $\mathbb{C}$ as shown in theorem 2.2.1. It follows that g_2 and g_3 define holomorphic functions on $\mathcal{H}$. Since $\Delta(\tau) \neq 0\ \forall\tau \in \mathcal{H}$, it follows that j is holomorphic on $\mathcal{H}$. $\Gamma(1)$-invariance has been proved in the previous lemma. Note that for $\gamma = T, S$ we obtain

$$j(\tau+1) = j(\tau) \quad \forall\tau \in \mathcal{H}, \tag{2.6}$$

and

$$j(-1/\tau) = j(\tau) \quad \forall\tau \in \mathcal{H}. \tag{2.7}$$

For the surjectivity, consider, for $H > 2$, the region $\mathcal{F}(H)$ bounded by the curves

$$\operatorname{Re}\tau = \pm 1/2,\ |\tau| = 1,\ \operatorname{Im}\tau = H.$$

Let $\partial\mathcal{F}(H)$ be the boundary of $\mathcal{F}(H)$, which we orient anti-clockwise (see figure 2.3). Let $j_0 \in \mathbb{C}$, if there exists $\tau_0 \in \partial\mathcal{F}(H)$ such that $j(\tau_0) = j_0$ we are

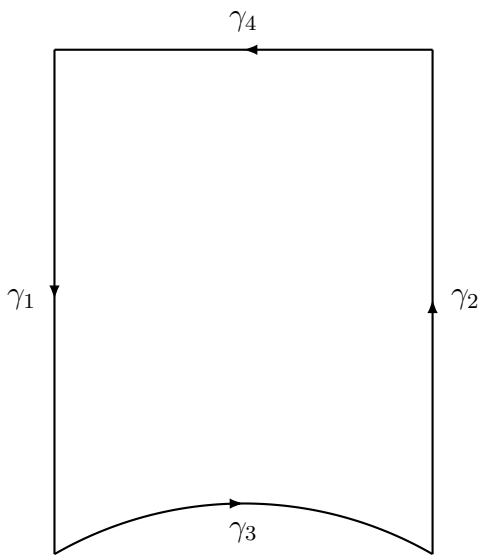


Figure 2.3: The boundary $\partial\mathcal{F}(H)$ and the integration path.

done. Suppose then that $j(\tau) \neq j_0$ $\forall \tau \in \partial\mathcal{F}(H)$, hence $j(\tau) - j_0 \neq 0$ $\forall \tau \in \partial\mathcal{F}(H)$. By the Residue theorem we have

$$\frac{1}{2\pi i}\int_{\partial\mathcal{F}(H)} \frac{j'(\tau)}{j(\tau)-j_0} d\tau = \#\{\tau \in \mathcal{F}(H) \mid j(\tau) = j_0\}$$

where the points are counted with multiplicity. We claim that

$$\lim_{H\to+\infty} \frac{1}{2\pi i}\int_{\partial\mathcal{F}(H)} \frac{j'(\tau)}{j(\tau)-j_0} d\tau = 1. \tag{2.8}$$

By equations (2.6) and (2.7) the integral vanishes on the vertical lines $\gamma_1 \cup \gamma_2$ and on the circular path γ_3. It remains to compute the contribution of the upper path γ_4. By lemma 2.3.2 we can write

$$j(\tau) = \widetilde{j}(q) = \frac{1}{q} + 744 + \sum_{n\geq 1} c_n q^n,$$

where $q = e^{2\pi i\tau}$. The change of variable $\tau \to q$ in the integral (2.8) leads to

$$\lim_{H\to+\infty} \frac{1}{2\pi i}\int_{\gamma_4} \frac{j'(\tau)}{j(\tau)-j_0} d\tau = -\lim_{r\to 0} \frac{1}{2\pi i}\int_{\partial D_r} \frac{\widetilde{j}'(q)}{\widetilde{j}(q)-j_0} dq$$

where $r = r(H)$, $D_r = \{z \in \mathbb{C} \mid |z| \leq r\}$ and the minus sign arises from the change of orientation of the curve. We can evaluate the integral on the right

with the Residue theorem using the series expansion: for a small enough r (i.e. for $H \gg 0$)

$$\frac{1}{2\pi i}\int_{\partial D_r} \frac{\widetilde{j}'(q)}{\widetilde{j}(q)-j_0}dq = \mathrm{Res}_0\left(\frac{\widetilde{j}'(q)}{\widetilde{j}(q)-j_0}\right) = -1.$$

This proves the claim. Therefore $j(\overline{\mathcal{F}}) = \mathbb{C}$, hence j is surjective, and, in addition, j is injective on $\mathcal{F}$. By lemma 2.3.1 and theorem 2.1.1 it follows that j is also injective on $\mathcal{F}_0$, hence it induces a bijection

$$\overline{j} : \mathcal{H}/\Gamma(1) \longrightarrow \mathbb{C}.$$

□

Remark 2.3.5. Actually we can say a little bit more about $\overline{j}$: we can give $\mathcal{H}/\Gamma(1)$ an holomorphic structure such that $\overline{j}$ is a bi-holomorphism.

Corollary 2.3.2 (Uniformization Theorem)**.** *For every elliptic curve $E/\mathbb{C}$ there exists a lattice Λ such that $E \cong E_\Lambda$.*

Proof. Let $E/\mathbb{C}$ be an elliptic curve, thanks to the theorem above there exists $\tau \in \mathcal{H}$ such that $j(\tau) = j(E)$. Therefore

$$j(E) = j(\tau) = j(\Lambda_\tau) = j(E_{\Lambda_\tau})$$

hence by theorem 1.1.4 the curves E and E_{Λ_τ} are isomorphic. □

Remark 2.3.6. We can actually say a little bit more. If E is defined over $\mathbb{C}$ by the equation $y^2 = x^3 + Ax + B$, then we can find a lattice Λ such that $g_2(\Lambda) = -4A$ and $g_3(\Lambda) = -4B$. The curve E_Λ is then given by

$$E_\Lambda : y^2 = 4x^3 + 4Ax + 4B$$

and, up the rescaling, $E = E_\Lambda$. For a proof of this we refer to [Sil94], Chapter 1, Corollary 4.3.

Corollary 2.3.3. *There is a one to one correspondence between lattices up to homothety and elliptic curves up to isomorphism:*

$$\mathcal{L} \longleftrightarrow \mathcal{ELL}$$

given by $[\Lambda] \mapsto [E_\Lambda]$.

Proof. Injectivity is corollary 2.3.1, while surjectivity is a consequence of the Uniformization theorem. □

Chapter 3

Complex Multiplication

This chapter deals with elliptic curves with **complex multiplication**, i.e. elliptic curves with endomorphism ring strictly bigger than $\mathbb{Z}$. In particular we will focus on CM elliptic curves defined over $\mathbb{C}$, for which the endomorphism ring is an **order** $\mathcal{O}$ in an imaginary quadratic field K. Note that these (isomorphism classes of) elliptic curves are very few. Indeed, we will see that there exist only finitely many isomorphism classes of elliptic curves with complex multiplication by $\mathcal{O}$, which are parameterized by the elements of $\mathrm{Cl}(\mathcal{O})$, the **ideal class group**.

We will also introduce the Hilbert class polynomial $H_D(x)$ and we will provide a possible algorithm for its computation. To do so, we will make use of **binary quadratic forms** and formulas for the j-function involving the Dedekind eta function.

3.1 Introduction

This introductory part aims to clarify the concepts that we will present later on in this chapter.

Some basic facts about quadratic fields, orders, ideal class group and other number theoretical results are recalled. Further, some general results about the endomorphism ring of an elliptic curve will be exposed.

3.1.1 Quadratic Fields

Definition 3.1.1. A quadratic field is a number field $K/\mathbb{Q}$ of degree $[K : \mathbb{Q}] = 2$. If $K \cap \mathbb{R} \neq K$ we say that K is **imaginary**, otherwise it is called *real*.

It is easy to show that every quadratic field is of the form $K = \mathbb{Q}(\sqrt{d})$ for some square-free integer d, and of course K is imaginary if and only if $d < 0$. We will denote by $\mathcal{O}_K$ the ring of integers of K.

Proposition 3.1.1 ([Mar18], Chapter 2, Corollary 2)**.** *Let $K = \mathbb{Q}(\sqrt{d})$, $d \in \mathbb{Z}$ squarefree. Then we have*

$$\mathcal{O}_K = \begin{cases} \mathbb{Z}[\sqrt{d}] & d \equiv 2, 3 \pmod 4 \\ \mathbb{Z}\left[\frac{1+\sqrt{d}}{2}\right] & d \equiv 1 \pmod 4 \end{cases}$$

Remark 3.1.1. We note that in general $\mathcal{O}_K \neq \mathbb{Z}[\alpha]$ if $K = \mathbb{Q}(\alpha)$.

Definition 3.1.2. Let $K/\mathbb{Q}$ be a number field, $\mathcal{O} \subseteq K$ is an **order** if

(*i*) $\mathcal{O}$ is a subring (with 1) of K;

(*ii*) $\mathcal{O}$ is a finitely generated $\mathbb{Z}$-module;

(*iii*) $\mathcal{O}$ contains a $\mathbb{Q}$-basis of K.

Remark 3.1.2. Condition (*iii*) is equivalent to $\mathcal{O} \otimes_{\mathbb{Z}} \mathbb{Q} = K$, which is equivalent to $\mathrm{Frac}(\mathcal{O}) = K$. Since $\mathcal{O}$ is torsion free (and $\mathbb{Z}$ is PID) then $\mathcal{O}$ is a free $\mathbb{Z}$-module of rank $n = [K : \mathbb{Q}]$.

Example 8. If $K = \mathbb{Q}(\sqrt{-3})$ then $\mathcal{O}_1 = \mathbb{Z}[\sqrt{-3}]$ and $\mathcal{O}_2 = \mathcal{O}_K = \mathbb{Z}\left[\frac{1+\sqrt{-3}}{2}\right]$ are two different orders in K.

Remark 3.1.3. $\mathcal{O} \subseteq \mathcal{O}_K$ for every order $\mathcal{O}$ of K, hence $\mathcal{O}_K$ is the maximal order. To see this, notice that if $\alpha \in \mathcal{O}$ then $\mathbb{Z}[\alpha]$ is a $\mathbb{Z}$-submodule of $\mathcal{O}$ and hence it is finitely generated, therefore α is integral over $\mathbb{Z}$ and $\alpha \in \mathcal{O}_K$.

Let $K/\mathbb{Q}$ be a number field and $\mathcal{O} \subseteq K$ an order. The **discriminant** of $\mathcal{O}$ is defined as

$$\mathrm{disc}(\mathcal{O}) := (\det(\sigma_i(\alpha_j)))^2$$

where $\{\alpha_1, \ldots, \alpha_n\}$ is a $\mathbb{Z}$-basis of $\mathcal{O}$ and $\sigma_1, \ldots, \sigma_n : K \longrightarrow \mathbb{C}$ are the field immersions. It can be shown that this definition is independent of the choice of the bases and hence the discriminant is well defined. We define $d_K = \mathrm{disc}(K) := \mathrm{disc}(\mathcal{O}_K)$, the discriminant of the maximal order.

The following equality holds:

Proposition 3.1.2. *Let K be a number field and let $\mathcal{O}_1 \subseteq \mathcal{O}_2 \subseteq K$ be two orders, then*

$$\mathrm{disc}(\mathcal{O}_1) = [\mathcal{O}_2 : \mathcal{O}_1]^2 \,\mathrm{disc}(\mathcal{O}_2).$$

Proof. We set $f = [\mathcal{O}_2 : \mathcal{O}_1]$ and take $\{\alpha_1, \dots, \alpha_n\}$, $\{\beta_1, \dots, \beta_n\}$ two $\mathbb{Z}$-bases of $\mathcal{O}_1$ and $\mathcal{O}_2$ respectively. Because $\mathcal{O}_1 \subseteq \mathcal{O}_2$ we have $\forall j$

$$\alpha_j = \sum_{k=1}^{n} b_{k,j}\beta_k$$

for some $b_{k,j} \in \mathbb{Z}$ such that $f = \det(b_{k,j})$. Therefore

$$\begin{aligned}\operatorname{disc}(\mathcal{O}_1) &= (\det(\sigma_i(\alpha_j)))^2 = (\det(\sum_{k=1}^{n} b_{k,j}\sigma_i(\beta_k)))^2 \\ &= (\det(b_{k,j}))^2 \cdot \operatorname{disc}(\mathcal{O}_2) = f^2 \operatorname{disc}(\mathcal{O}_2).\end{aligned}$$

□

For all $\alpha \in K$ define the **norm** of α as

$$\mathrm{N}_{K/\mathbb{Q}}(\alpha) = \prod_{i=1}^{n} \sigma_i(\alpha).$$

It is clear from the definition that:

Proposition 3.1.3. *Let $K/\mathbb{Q}$ be a number field. Then the following properties of the norm hold:*

1. *$\forall \alpha \in K$ (resp. in $\mathcal{O}_K$) $\mathrm{N}_{K/\mathbb{Q}}(\alpha) \in \mathbb{Q}$ (resp. $\in \mathbb{Z}$);*
2. *$\mathrm{N}_{K/\mathbb{Q}}(\alpha\beta) = \mathrm{N}_{K/\mathbb{Q}}(\alpha)\,\mathrm{N}_{K/\mathbb{Q}}(\beta)$ $\forall \alpha, \beta \in K$;*
3. *$|\,\mathrm{N}_{K/\mathbb{Q}}(u)| = 1$ $\forall u \in \mathcal{O}_K^\times$.*

In the case of a quadratic field K we can easily compute the discriminant d_K and the norm of any element:

Proposition 3.1.4. *Let $K = \mathbb{Q}(\sqrt{d})$, $d \in \mathbb{Z}$ square-free. Then*

$$\operatorname{disc}(K) = \begin{cases} 4d & d \equiv 2, 3 \pmod 4 \\ d & d \equiv 1 \pmod 4 \end{cases}$$

and for all $a + b\sqrt{d} = \alpha \in K$ we have

$$\mathrm{N}_{K/\mathbb{Q}}(\alpha) = a^2 - db^2.$$

Remark 3.1.4. Let K be a quadratic field and let $\mathcal{O} \subseteq \mathcal{O}_K$ be an order. By proposition 3.1.2 we have

$$\operatorname{disc}(\mathcal{O}) = f^2 \operatorname{disc}(K)$$

where $f = [\mathcal{O}_K : \mathcal{O}]$ is called **conductor**. If $\mathcal{O}_K = \mathbb{Z}[\alpha]$, it follows that $\mathcal{O} = \mathbb{Z}[f\alpha]$ and hence an order in a quadratic field is uniquely determined by its discriminant. We will denote by $\mathcal{O}_D$ the order of discriminant D in K. On the other hand if $D \equiv 0, 1 \pmod 4$ is a non-square then there exists a quadratic field K and an order $\mathcal{O} \subseteq K$ with discriminant D. For this we can take $K = \mathbb{Q}(\sqrt{D})$ and write

$$D = e^2 d$$

where d square-free, thus $K = \mathbb{Q}(\sqrt{d})$. We distinguish two cases:

1. $d \equiv 2, 3 \pmod 4 \implies 2 \mid e$ and $D = f^2 4d = f^2 d_K$. $\mathcal{O}_K = \mathbb{Z}[\sqrt{d}]$ and $\mathcal{O}_D = \mathbb{Z}[f\sqrt{d}]$ is an order with discriminant D.

2. $d \equiv 1 \pmod 4 \implies D = e^2 d_K$. $\mathcal{O}_K = \mathbb{Z}\left[\frac{1+\sqrt{d}}{2}\right]$ and $\mathcal{O}_D = \mathbb{Z}\left[e\frac{1+\sqrt{d}}{2}\right]$ is the required order.

Remark 3.1.5. In the rest of the chapter we will use the word *discriminant* to denote a non-square integer $D \equiv 0, 1 \pmod 4$.

3.1.2 Fractional Ideals

Let A be an integral domain and $K = \operatorname{Frac}(A)$ be its field of fractions. An A-module $I \subseteq K$ is a **fractional ideal** if there exists $d \in A$, $d \neq 0$ such that $dI \subseteq A$. A fractional ideal I is **invertible** if there exists a fractional ideal J such that $IJ = A$.

We denote by $\mathcal{F}(A)$ the set of all non-zero fractional ideals of A and with $\mathcal{F}(A)^\times$ the set of all invertible fractional ideals. We also define $\mathcal{P}(A)$ to be the collection of all non zero principal fractional ideals. Of course we have

$$\mathcal{P}(A) \subseteq \mathcal{F}(A)^\times \subseteq \mathcal{F}(A).$$

Example 9. In $A = \mathbb{Z}[\sqrt{-7}]$ the ideal $I = (1+\sqrt{-7}, 1-\sqrt{-7})$ is not invertible. By contradiction, suppose that there exists J such that $IJ = A$. Then $\forall \beta \in K = Q(A)$ such that $\beta I \subseteq I$ we have $\beta \in A$ indeed

$$\beta A = \beta IJ = (\beta I)J \subseteq IJ = A,$$

therefore $\beta \in A$. In our case $\beta = \frac{1+\sqrt{-7}}{2} \in K \setminus A$ but $\beta I \subseteq I$, a contradiction. This example shows that in general $\mathcal{F}(A)^\times \subsetneq \mathcal{F}(A)$.

Remark 3.1.6. If I, J are fractional ideals then also the product IJ is a fractional ideal itself. Hence $\mathcal{F}(A)$ is a monoid with product and $\mathcal{F}(A)^\times$ is a group. Principal ideals are invertible and a product of principal ideals is principal, therefore $\mathcal{P} \subseteq \mathcal{F}(A)^\times$ is a subgroup.

Definition 3.1.3. Let A be an integral domain. The **ideal class group** of A is the quotient

$$\mathrm{Cl}(A) = \mathcal{F}(A)^\times / \mathcal{P}(A).$$

An element of $\mathrm{Cl}(A)$ will be denoted by $[I]$ and, by definition, $I \sim J$ if and only if $\exists \alpha \in K^\times$ such that $IJ^{-1} = (\alpha)$.

Remark 3.1.7. We have seen that in general $\mathcal{F}(A)^\times \subsetneq \mathcal{F}(A)$. It can be shown (see [Mar18] Corollary 1 pag. 41) that if A is a Dedekind domain then every nonzero fractional ideal is invertible and hence $\mathcal{F}(A)^\times = \mathcal{F}(A)$. This is the case of $A = \mathcal{O}_K$, the ring of integers of a number field. An order $\mathcal{O} \subsetneq \mathcal{O}_K$ is not integrally closed and thus is not Dedekind.

We will focus on the case of $A = \mathcal{O}$ an order in a quadratic field $K = \mathbb{Q}(\sqrt{d})$. Let I be a fractional ideal of $\mathcal{O}$. We note that

$$\mathcal{O} \subseteq \{\beta \in K \mid \beta I \subseteq I\},$$

and the inclusion can be strict (as we have already noticed in example 9). We give the following definition

Definition 3.1.4. A fractional ideal I of $\mathcal{O}$ is **proper** if

$$\mathcal{O} = \{\beta \in K \mid \beta I \subseteq I\}.$$

We will see that for orders in a quadratic field the notion of proper and invertible ideals coincide. A fractional ideal $I \subseteq \mathcal{O}$ is called an **$\mathcal{O}$-ideal**.

Remark 3.1.8. If I is a nonzero $\mathcal{O}$-ideal then the quotient $\mathcal{O}/I$ is finite. For any $x \in I \setminus \{0\}$ we have $0 \neq m = \mathrm{N}_{K/\mathbb{Q}}(x) \in \mathbb{Z} \cap I$ and therefore

$$m\mathcal{O} \subseteq I \subseteq \mathcal{O}$$

and $[\mathcal{O} : I] < +\infty$ because $\mathcal{O}$ is f.g. over $\mathbb{Z}$. It follows that any fractional ideal I is a free $\mathbb{Z}$-module of rank 2 and we can write $I = \mathbb{Z}\omega_1 \oplus \mathbb{Z}\omega_2$ for some $\omega_1, \omega_2 \in K$.

Definition 3.1.5. Let $I \neq 0$ be an $\mathcal{O}$-ideal. We define the norm of I as

$$\mathrm{N}(I) = [\mathcal{O} : I] = \#\mathcal{O}/I.$$

Remark 3.1.9. By the previous remark $\mathrm{N}(I)$ is well defined.

Proposition 3.1.5. *The norm is multiplicative for proper ideals: if I, J are proper $\mathcal{O}$-ideals then*

$$\mathrm{N}(IJ) = \mathrm{N}(I)\,\mathrm{N}(J).$$

Moreover, if $\alpha \in \mathcal{O}$ then

$$\mathrm{N}((\alpha)) = |\,\mathrm{N}_{K/\mathbb{Q}}(\alpha)|.$$

Proof. A proof of this proposition (in the case $\mathcal{O} = \mathcal{O}_K$ the maximal order) can be found in [Mar18], Chapter 3, Theorem 22. □

Remark 3.1.10. Every fractional ideal I of $\mathcal{O}$ can be written in the form $I = \alpha J$ with $\alpha \in K^\times$ and J an $\mathcal{O}$-ideal. We can thus extend the norm for fractional ideals defining

$$\mathrm{N}(I) = |\,\mathrm{N}_{K/\mathbb{Q}}(\alpha)|\,\mathrm{N}(J).$$

This is well defined: if $I = \alpha J = \beta J'$ with $\alpha, \beta \in K^\times$ and J, J' $\mathcal{O}$-ideals, then, writing $\alpha = \alpha_1/m$ and $\beta = \beta_1/n$ with $\alpha_1, \beta_1 \in \mathcal{O}$ and $n, m \in \mathbb{Z}$ (we can do this since $K = \mathcal{O} \otimes_{\mathbb{Z}} \mathbb{Q}$), we have

$$\alpha J = \beta J' \iff n\alpha_1 J = m\beta_1 J'.$$

Taking the norms on both sides (now they are $\mathcal{O}$-ideals) we get

$$\begin{aligned} \mathrm{N}(n\alpha_1 J) = \mathrm{N}(m\beta_1 J') &\iff \mathrm{N}_{K/\mathbb{Q}}(n\alpha_1)\,\mathrm{N}(J) = \mathrm{N}_{K/\mathbb{Q}}(m\beta_1)\,\mathrm{N}(J') \\ &\iff n^2|\,\mathrm{N}_{K/\mathbb{Q}}(\alpha_1)|\,\mathrm{N}(J) = m^2|\,\mathrm{N}_{K/\mathbb{Q}}(\beta_1)|\,\mathrm{N}(J'). \end{aligned}$$

On the other hand $\mathrm{N}_{K/\mathbb{Q}}(\alpha) = \mathrm{N}_{K/\mathbb{Q}}(\alpha_1)/m^2$ and $\mathrm{N}_{K/\mathbb{Q}}(\beta) = \mathrm{N}_{K/\mathbb{Q}}(\beta_1)/n^2$, therefore $\mathrm{N}(I)$ is well defined.

Theorem 3.1.1. *Let $\mathcal{O} \subseteq K$ be an order in a quadratic field and let I be a fractional ideal of $\mathcal{O}$. Then I is proper if and only if I is invertible.*

Proof. We denote by σ the nontrivial automorphism of K. If I is invertible there exists J such that $IJ = \mathcal{O}$. Let $\beta \in K$ such that $\beta I \subseteq I$: we have

$$\beta\mathcal{O} = \beta IJ = (\beta I)J \subseteq IJ = \mathcal{O}$$

hence $\beta \in \mathcal{O}$ and $\mathcal{O} = \{\beta \in K \mid \beta I \subseteq I\}$, that is I is proper.

Conversely suppose I is proper, $I = \mathbb{Z}\omega_1 \oplus \mathbb{Z}\omega_2 = \omega_1(\mathbb{Z} \oplus \mathbb{Z}\tau)$ where $\tau = \omega_2/\omega_1 \in K$. Since I is proper, also $I' = \mathbb{Z} \oplus \mathbb{Z}\tau$ is. There exist minimal $a, b, c \in \mathbb{Z},\ \gcd(a, b, c) = 1$ such that

$$a\tau^2 + b\tau + c = 0.$$

Now we prove that $\mathbb{Z} \oplus \mathbb{Z}a\tau = \mathcal{O}$. Since I' is a fractional ideal and $1 \in I'$, we have $\mathcal{O} \subseteq \mathcal{O}I' \subseteq I'$ therefore, if $\mathcal{O} = \mathbb{Z} \oplus \mathbb{Z}\omega$, we have $\omega = m + n\tau$ for some $m, n \in \mathbb{Z}$ and, after replacing ω by $\omega - m$, we can suppose $\omega = n\tau$. We also have $\omega\tau = n\tau^2 \in I'$ which implies $a \mid n$, because, by assumption, $a\tau^2 = -b\tau - c$ with a, b, c relatively prime. It follows that $\mathcal{O} \subseteq \mathbb{Z} \oplus \mathbb{Z}a\tau$. On the other hand we have $a\tau I' \subseteq I'$ which implies, since I' is proper, that $a\tau \in \mathcal{O}$, hence that $\mathbb{Z} \oplus \mathbb{Z}a\tau = \mathcal{O}$.

We claim that

$$I\sigma(I) = \frac{\mathrm{N}_{K/\mathbb{Q}}(\omega_1)}{a}\mathcal{O}$$

(where $\sigma(I) = \sigma(\omega_1)(\mathbb{Z} \oplus \mathbb{Z}\sigma(\tau))$ is the conjugate of I), which implies that I is invertible. We have

$$\begin{aligned} aI\sigma(I) &= a\omega_1\sigma(\omega_1)\langle 1, \tau, \sigma(\tau), \tau\sigma(\tau)\rangle_{\mathbb{Z}} = \mathrm{N}_{K/\mathbb{Q}}(\omega_1)\langle a, a\tau, a\sigma(\tau), a\tau\sigma(\tau)\rangle_{\mathbb{Z}} \\ &= \mathrm{N}_{K/\mathbb{Q}}(\alpha)\langle a, a\tau, -b, c\rangle_{\mathbb{Z}} = \mathrm{N}_{K/\mathbb{Q}}(\alpha)\langle 1, a\tau\rangle_{\mathbb{Z}} \\ &= \mathrm{N}_{K/\mathbb{Q}}(\alpha)(\mathbb{Z} \oplus \mathbb{Z}a\tau) = \mathrm{N}_{K/\mathbb{Q}}(\alpha)\mathcal{O}. \end{aligned}$$

where we used that $\tau + \sigma(\tau) = -b/a$, $\tau\sigma(\tau) = c/a$, $\gcd(a, b, c) = 1$ and $\mathbb{Z} \oplus \mathbb{Z}a\tau = \mathcal{O}$. □

Remark 3.1.11. The proof of the above theorem gives an explicit formula for the inverse of a proper fractional ideal. For example, if K is an imaginary quadratic field, then σ is the complex conjugation, therefore, if I is a proper fractional ideal of $\mathcal{O}$, we have

$$I^{-1} = \frac{a}{\mathrm{N}_{K/\mathbb{Q}}(\alpha)}\overline{I}$$

where $\overline{I} = \sigma(I)$ is the conjugate of I.

We conclude with the following important result:

Theorem 3.1.2. *Let $\mathcal{O} \subseteq K$ be an order in a number field K, then* $\mathrm{Cl}(\mathcal{O})$ *is a finite group.*

Proof. We will prove this theorem for the special case of imaginary quadratic fields $K = \mathbb{Q}(\sqrt{D})$ in the next section: it will be a consequence of theorem 3.2.2. For the general case we refer to [Neu99], Chapter 1, Section 12. □

If $\mathcal{O} = \mathcal{O}_D$, the order of discriminant D in the quadratic field $K = \mathbb{Q}(\sqrt{D})$, the **class number** of $\mathcal{O}_D$ is

$$h(D) := \#\,\mathrm{Cl}(\mathcal{O}_D).$$

3.1.3 Endomorphisms

We now turn back to elliptic curves. In this section we will provide some basic facts about the endomorphism ring of an elliptic curve.

Let K be a field and E/K be an elliptic curve. A morphism of elliptic curves $\varphi : E \longrightarrow E$ is called an **endomorphism**. Examples are the *zero map* $[0](P) = \infty \ \forall P \in E(\overline{K})$ and the *identity* id_E. A non-constant morphism of elliptic curves is called an **isogeny**. We define

$$\mathrm{End}(E) = \{\varphi : E \longrightarrow E \text{ isogeny}\} \cup \{[0]\}$$

to be the set of all the endomorphisms of E.

Example 10. If $f, g \in \mathrm{End}(E)$ then the sum $(f+g)(P) = f(P) + g(P) \ \forall P \in E(\overline{K})$ and the composition $f \circ g$ are endomorphisms since the addition between points of E has rational formulas and composition of rational maps is rational.

It follows easily that $(\mathrm{End}(E), +, \circ, [0], id)$ is a ring.

Theorem 3.1.3. *Every rational map between elliptic curves $\varphi : E \longrightarrow E'$ is either constant or surjective.*

Proof (Sketch). E is a projective curve hence $f(E(\overline{K}))$ is a Zariski closed subset of $E'(\overline{K})$. Further, E is irreducible and hence also $f(E(\overline{K}))$ is. It follows that

$$f(E(\overline{K})) = \{P\} \ \vee \ f(E(\overline{K})) = E'(\overline{K})$$

where $P \in E(\overline{K})$. □

As a consequence we have

Corollary 3.1.1. $\mathrm{End}(E)$ *is an integral domain.*

Proof. If $f, g \neq 0$ then f and g are surjective and also the composition is, hence $f \circ g \neq 0$. □

Remark 3.1.12. If E, E' are isomorphic elliptic curves then $\mathrm{End}(E) \cong \mathrm{End}(E')$.

Example 11. For all $n \in \mathbb{Z}$ the multiplication by n, $[n] : E \longrightarrow E$, is a first nontrivial example of endomorphism. We will identify $\mathbb{Z} \subseteq \mathrm{End}(E)$ through the injection

$$\begin{array}{ccc} \mathbb{Z} & \hookrightarrow & \mathrm{End}(E) \\ n & \mapsto & [n] \end{array}$$

Is $\mathrm{End}(E)$ strictly bigger than $\mathbb{Z}$? Let us look at some examples.

Example 12. 1. Consider the curve $E : y^2 = x^3 + x + 1$ over $\mathbb{C}$. We will show that if $E/\mathbb{C}$ is an elliptic curve with $\mathrm{End}(E) \neq \mathbb{Z}$ then $j(E)$ is an algebraic integer. In this case we have that $j(E) = \frac{1728 \cdot 4}{31} \in \mathbb{Q} \setminus \mathbb{Z}$ which is not integral over $\mathbb{Z}$, therefore $\mathrm{End}(E) = \mathbb{Z}$.

2. Consider the elliptic curve $E/\mathbb{C}$ given by the affine equation

$$E : y^2 = x^3 + x.$$

The map $[i] : E \longrightarrow E$ given by $(x, y) \mapsto (-x, iy)$ is clearly an endomorphism and $[i]^2 = [-1] = -id$. Hence $\mathbb{Z} \subsetneq \mathrm{End}(E)$, in fact $\mathrm{End}(E) = \mathbb{Z}[i]$.

3. The same elliptic curve over $\mathbb{F}_p$ with $p \equiv 3 \pmod 4$ has a different behavior. The Frobenius endomorphism

$$\begin{array}{rccc} F_p : & E & \longrightarrow & E \\ & (x, y) & \mapsto & (x^p, x^p) \\ & \infty & \mapsto & \infty \end{array}$$

does not commute with $[i]$ (we mean that $i \in \mathbb{F}_{p^2}$ is a square root of -1):

$$([i] \circ F_p)(x, y) = (-x^p, iy^p) \neq (-x^p, -iy^p) = (F_p \circ [i])(x, y).$$

In this case $\mathrm{End}(E)$ is an order in a quaternion algebra.

We will see that the three examples above cover all the possible cases. It is useful to introduce the **endomorphism algebra**

$$\mathrm{End}^0(E) = \mathrm{End}(E) \otimes_{\mathbb{Z}} \mathbb{Q},$$

which is nontrivial since $\mathrm{End}(E)$ is a torsion free additive group. It is in fact a $\mathbb{Q}$-algebra, regarding $\mathbb{Q} \subseteq \mathrm{End}^0(E)$ through $r/s \hookrightarrow id \otimes r/s$.

Remark 3.1.13. If $S = \mathbb{Z} \setminus \{0\}$ then $\mathrm{End}(E) \otimes_{\mathbb{Z}} \mathbb{Q} \cong S^{-1} \mathrm{End}(E)$, thus we can think of the endomorphism algebra as

$$\mathrm{End}^0(E) = \left\{ \frac{\varphi}{n} \,\middle|\, \phi \in \mathrm{End}(E),\ n \in S \right\}.$$

Theorem 3.1.4 ([Sil09], Chapter 3, Theorem 9.3)**.** *Let E/K be an elliptic curve. Then* $\mathrm{End}(E)$ *is one of the following:*

1. *the ring $\mathbb{Z}$;*
2. *an order $\mathcal{O}$ in an imaginary quadratic field K;*
3. *an order R in a quaternion algebra H.*

3.2 Complex Multiplication

3.2.1 Elliptic curves with complex multiplication

Let $E/\mathbb{C}$ be an elliptic curve. By the correspondence established in Chapter 2 we can write $E \cong E_\Lambda$ for a lattice Λ that we can assume (up to homothety) to be of the form $\Lambda = \mathbb{Z} \oplus \mathbb{Z}\tau$ for some $\tau \in \mathcal{H}$. This correspondence is an equivalence of categories, in particular,

$$\mathrm{End}(E) \cong \mathrm{End}(\Lambda) = \{\alpha \in \mathbb{C} \mid \alpha\Lambda \subseteq \Lambda\}.$$

Remark 3.2.1. If $\Lambda' = \lambda\Lambda$ for $\lambda \in \mathbb{C}^\times$ then $\mathrm{End}(\Lambda') = \mathrm{End}(\Lambda)$, thus homothetic lattices give the same endomorphism ring.

Thanks to this identification we will consider $\mathrm{End}(E) \subseteq \mathbb{C}$. We have this first result.

Proposition 3.2.1. *Let E and Λ be as before. Then*

1. *if $\mathbb{Q}(\tau)$ is not an imaginary quadratic field then* $\mathrm{End}(E) = \mathbb{Z}$*;*
2. *if $\mathbb{Q}(\tau)$ is an imaginary quadratic field then* $\mathrm{End}(E) \subseteq \mathbb{Q}(\tau)$ *is an order and*
$$\mathrm{End}^0(E) = \mathrm{End}(E) \otimes_{\mathbb{Z}} \mathbb{Q} = \mathbb{Q}(\tau).$$

Proof. If $\alpha \in \mathrm{End}(\Lambda)$ then

- $\alpha \cdot 1 \in \Lambda \implies \alpha = m + n\tau,\ n, m \in \mathbb{Z}$;
- $\alpha \cdot \tau \in \Lambda \implies \alpha \cdot \tau = p + q\tau,\ p, q \in \mathbb{Z}$.

From the first condition we derive

$$\tau = \frac{\alpha - m}{n} \implies \tau \in \mathbb{Q}(\alpha)$$

which implies, by replacing it in the second condition,

$$\alpha\left(\frac{\alpha - m}{n}\right) = p + \frac{q}{n}(\alpha - m)$$

Therefore α satisfies a quadratic relation. If $\mathbb{Q}(\tau)$ is not an imaginary quadratic field then $\alpha \in \mathbb{Z}$ and $\mathrm{End}(E) = \mathbb{Z}$. Otherwise there exist $r, s \in \mathbb{Q}$ such that

$$\tau^2 + r\tau + s = 0,$$

therefore $\exists\ N \in \mathbb{Z}$ such that $N\tau, N\tau^2 \in \Lambda$, hence $N\tau \in \mathrm{End}(E)$, which is then strictly bigger than $\mathbb{Z}$. By theorem 3.1.4, $\mathrm{End}(E)$ is an order in $\mathbb{Q}(\tau)$ and

$$\mathrm{End}^0(E) = \mathrm{End}(E) \otimes_{\mathbb{Z}} \mathbb{Q} = \mathbb{Q}(\tau).$$

□

Definition 3.2.1. We say that an elliptic curve $E/\mathbb{C}$ has **complex multiplication** (briefly CM) by $\mathcal{O}$ if $\mathrm{End}(E) = \mathcal{O}$ is an order in an imaginary quadratic field.

Remark 3.2.2. From proposition 3.2.1 it follows that only a *countable* number of elliptic curves over $\mathbb{C}$ has complex multiplication. In fact, the lattice $\Lambda = \mathbb{Z} \oplus \mathbb{Z}\xi$, $\xi \in \mathcal{H}$ transcendental, defines an elliptic curve E_Λ with $\mathrm{End}(E_\Lambda) \neq \mathbb{Z}$.

For every order $\mathcal{O}$ in an imaginary quadratic field we set

$$\mathcal{ELL}(\mathcal{O}) = \frac{\{E/\mathbb{C} \mid \mathrm{End}(E) = \mathcal{O}\}}{\text{isomorphism}} = \frac{\{\Lambda \subseteq \mathbb{C} \text{ lattices} \mid \mathrm{End}(\Lambda) = \mathcal{O}\}}{\text{homothety}}$$

where the second "equality" follows from the correspondences established in chapter 2.

For every negative integer $D \equiv 0, 1 \pmod 4$ we will denote by $\mathcal{O}_D$ the order of discriminant D in $K = \mathbb{Q}(\sqrt{D})$ and by $\mathcal{O}_K$ the maximal order (i.e. the ring of integers) of K. We will focus on the set $\mathcal{ELL}(\mathcal{O}_D)$.

Let E_Λ be an elliptic curve with CM by $\mathcal{O}_D$, that is

$$\mathrm{End}(\Lambda) = \{\alpha \in \mathbb{C} \mid \alpha\Lambda \subseteq \Lambda\} = \mathcal{O}_D.$$

Let's say $\mathcal{O}_D = \mathbb{Z} \oplus \mathbb{Z}\omega$ and, up to homothety, $\Lambda = \mathbb{Z} \oplus \mathbb{Z}\tau$, $\tau \in \mathcal{H}$. It follows that $\omega \in \Lambda$, in particular $\omega = m + n\tau$ and hence $n\tau = \omega - m$, therefore

$$n\Lambda = \mathbb{Z}n \oplus \mathbb{Z}(\omega - m) \subseteq \mathcal{O}_D.$$

Because $\mathcal{O}_D\Lambda \subseteq \Lambda$, it follows that Λ is a fractional ideal of $\mathcal{O}_D$. Is the converse true? If Λ is a fractional ideal of $\mathcal{O}_D$, is $\mathrm{End}(\Lambda) = \mathcal{O}_D$? Of course Λ is a lattice in $\mathbb{C}$ and we have

$$\mathrm{End}(\Lambda) = \{\alpha \in \mathbb{C} \mid \alpha\Lambda \subseteq \Lambda\} = \{\alpha \in K \mid \alpha\Lambda \subseteq \Lambda\}$$

therefore

$$\mathrm{End}(\Lambda) = \mathcal{O}_D \iff \Lambda \text{ is a proper fractional ideal of } \mathcal{O}_D.$$

Remark 3.2.3. If $\mathcal{O}_D = \mathcal{O}_K$ is the maximal order, all nonzero fractional ideals are invertible and thus proper.

We have seen in Section 3.1 that a nonzero fractional ideal of $\mathcal{O}_D$ is invertible if and only if it is proper. If I, J are proper fractional ideals of $\mathcal{O}_D$ in the same class in $\mathrm{Cl}(\mathcal{O}_D)$ then there exists $\alpha \in K^\times$ such that $IJ^{-1} = (\alpha)$, therefore $I = \alpha J$, that is, I and J are homothetic lattices and $E_I \cong E_J$. Thus there is a well-defined map

$$\begin{array}{ccc} \mathrm{Cl}(\mathcal{O}_D) & \longrightarrow & \mathcal{ELL}(\mathcal{O}_D) \\ [I] & \mapsto & [E_I] \end{array}.$$

We are going to prove that it is a bijection, hence in particular that we have the equality

$$\#\,\mathrm{Cl}(\mathcal{O}_D) = \#\mathcal{ELL}(\mathcal{O}_D).$$

Proposition 3.2.2. *Let Λ be a lattice such that E_Λ has complex multiplication by $\mathcal{O}_D$, and I, J be non-zero proper fractional ideals of $\mathcal{O}_D$.*

1. *$I\Lambda$ is a lattice in $\mathbb{C}$.*
2. *The elliptic curve $E_{I\Lambda}$ has CM by $\mathcal{O}_D$, that is $\mathrm{End}(E_{I\Lambda}) \cong \mathcal{O}_D$.*
3. *$E_{I\Lambda} \cong E_{J\Lambda}$ if and only if $[I] = [J]$ in $\mathrm{Cl}(\mathcal{O}_D)$.*

Hence there is a well-defined action $\mathrm{Cl}(\mathcal{O}_D) \curvearrowright \mathcal{ELL}(\mathcal{O}_D)$ given by

$$\begin{array}{ccc} \mathrm{Cl}(\mathcal{O}_D) \times \mathcal{ELL}(\mathcal{O}_D) & \longrightarrow & \mathcal{ELL}(\mathcal{O}_D) \\ ([I], [E_\Lambda]) & \mapsto & [I] * [E_\Lambda] := [E_{I^{-1}\Lambda}] \end{array}.$$

Proof. We can suppose, up to homothety, $\Lambda = \mathbb{Z} \oplus \tau\mathbb{Z}$. Since $\mathrm{End}(\Lambda) = \mathcal{O}_D$, Λ is a proper fractional ideal, therefore $I\Lambda$ also is; in particular $I\Lambda$ is a lattice in $\mathbb{C}$ and the elliptic curve $E_{I\Lambda}$ has CM by $\mathcal{O}_D$.

$E_{I\Lambda} \cong E_{J\Lambda}$ if and only if $I\Lambda$ and $J\Lambda$ are homotethic lattices, that is, there exists $\beta \in \mathbb{C}^\times$ such that $I\Lambda = cJ\Lambda$. Actually $c \in K^\times$: $I\Lambda = cJ\Lambda$ implies $\Lambda = cI^{-1}J\Lambda$, hence $cI^{-1}J \subseteq \mathcal{O}_D$ because by assumption $\mathrm{End}(\Lambda) = \mathcal{O}_D$. It follows that $c \in IJ^{-1} \subseteq K$. As Λ is a proper fractional ideal (hence invertible) and $c \in K^\times$, we have

$$I\Lambda = cJ\Lambda \iff I = cJ \iff [I] = [J].$$

□

Corollary 3.2.1. *The action of $\mathrm{Cl}(\mathcal{O}_D)$ on $\mathcal{ELL}(\mathcal{O}_D)$ is simply transitive*[1]*, hence*

$$\#\,\mathrm{Cl}(\mathcal{O}_D) = \#\mathcal{ELL}(\mathcal{O}_D).$$

Proof. It is transitive because $[I] * E_{\mathcal{O}_D} = E_{I^{-1}}$ for all $[I] \in \mathrm{Cl}(\mathcal{O}_D)$. It is simply transitive because of part (3) of proposition 3.2.2. □

[1] By simply transitive we mean that $\forall\ x, y \in \mathcal{ELL}(\mathcal{O}_D)\ \exists!\ [I] \in \mathrm{Cl}(\mathcal{O}_D)$ such that $[I] * x = y$.

3.2.2 The Hilbert Polynomial

Let's progress in the study of $\mathcal{ELL}(\mathcal{O}_D)$. Since the j-invariant determines the isomorphism/homothety class of elliptic curves/lattices over $\mathbb{C}$ we have canonical identifications

$$\mathcal{ELL}(\mathcal{O}_D) = \big\{j(E) \in \mathbb{C} \mid \mathrm{End}(E) = \mathcal{O}_D\big\} = \big\{j(\Lambda) \in \mathbb{C} \mid \mathrm{End}(\Lambda) = \mathcal{O}_D\big\}.$$

In the last section we proved that $\mathcal{ELL}(\mathcal{O}_D)$ is finite, now we will move to the field of definition of (the isomorphism class of) an elliptic curve $E \in \mathcal{ELL}(\mathcal{O}_D)$.

Consider $E \in \mathcal{ELL}(\mathcal{O}_D)$ defined, up to isomorphism, by an affine Weierstrass equation

$$E : y^2 = x^3 + Ax + B, \quad A, B \in \mathbb{C}$$

with j-invariant

$$j(E) = 1728\frac{4A^3}{4A^3 + 27B^2}.$$

For all field automorphisms $\sigma \in \mathrm{Aut}(\mathbb{C}/\mathbb{Q})$ define the curve

$$E^\sigma : y^2 = x^3 + \sigma(A)x + \sigma(B)$$

which is in fact an elliptic curve because $\Delta(E^\sigma) = \sigma(\Delta(E)) \neq 0$ since σ is injective, and

$$j(E^\sigma) = 1728\frac{4\sigma(A)^3}{4\sigma(A)^3 + 27\sigma(B)^2} = \sigma(j(E)).$$

Proposition 3.2.3. *Let $E/\mathbb{C}$ be an elliptic curve and $\sigma \in \mathrm{Aut}(\mathbb{C}/\mathbb{Q})$, then $\mathrm{End}(E) \cong \mathrm{End}(E^\sigma)$.*

Proof. Let $\phi \in \mathrm{End}(E)$,

$$\begin{array}{rccc} \phi : & E & \longrightarrow & E \\ & (x,y) & \mapsto & (u(x,y), v(x,y)) \\ & \infty & \mapsto & \infty \end{array}$$

for some rational functions $u, v \in \mathbb{C}(E)$. We have

$$\begin{aligned} & v(x,y)^2 = u(x,y)^3 + Au(x,y) + B \\ \implies & \sigma(v)^2 = \sigma(u)^3 + \sigma(A)\sigma(u) + \sigma(B) \quad \forall \sigma \in \mathrm{Aut}(\mathbb{C}/\mathbb{Q}). \end{aligned}$$

Therefore $\forall \sigma \in \mathrm{Aut}(\mathbb{C}/\mathbb{Q})$ there is a well-defined rational map $\phi^\sigma \in \mathrm{End}(E^\sigma)$ given by

$$\begin{array}{rccc} \phi^\sigma : & E^\sigma & \longrightarrow & E^\sigma \\ & (x,y) & \mapsto & (\sigma(u(x,y)), \sigma(v(x,y))) \\ & \infty & \mapsto & \infty \end{array}$$

which induces the isomorphism $\sigma : \mathrm{End}(E) \longrightarrow \mathrm{End}(E^\sigma)$. $\square$

Corollary 3.2.2. *If $E \in \mathcal{ELL}(\mathcal{O}_D)$ then $E^\sigma \in \mathcal{ELL}(\mathcal{O}_D)$ $\forall \sigma \in \mathrm{Aut}(\mathbb{C}/\mathbb{Q})$. It follows that $[\mathbb{Q}(j(E)) : \mathbb{Q}] \leq h(D)$ is finite, hence $j(E)$ is algebraic over $\mathbb{Q}$.*

Proof. By the previous proposition $\sigma(j(E)) \in \mathcal{ELL}(\mathcal{O}_D)$ for all σ, therefore $j(E)$ has only finitely many conjugates and therefore it is algebraic over $\mathbb{Q}$ and

$$[\mathbb{Q}(j(E)) : \mathbb{Q}] \leq \# \operatorname{Cl}(\mathcal{O}_D) = h(D).$$

□

Corollary 3.2.3. *We have*

$$\mathcal{ELL}(\mathcal{O}_D) = \frac{\left\{E/\overline{\mathbb{Q}} \mid \operatorname{End}(E) = \mathcal{O}_D\right\}}{\text{isomorphism over } \overline{\mathbb{Q}}}.$$

In other words every elliptic curve $E/\mathbb{C}$ with complex multiplication is isomorphic to an elliptic curve $E'/\overline{\mathbb{Q}}$.

So far we have established that there exist only finitely many (isomorphism classes of) elliptic curves with complex multiplication by $\mathcal{O}_D$, they are exactly $h(D)$, and their j-invariants are algebraic numbers of degree at most $h(D)$.

Definition 3.2.2. For a discriminant $D < 0$, the **Hilbert polynomial** $H_D(x)$ is defined by

$$H_D(x) = \prod_{E \in \mathcal{ELL}(\mathcal{O}_D)} (x - j(E)).$$

Remark 3.2.4. As demonstrated above, $H_D(x)$ is a monic polynomial of degree $h(D) = \# \operatorname{Cl}(\mathcal{O}_D)$ with rational coefficients. It can be shown[2] that $[\mathbb{Q}(j(E)) : \mathbb{Q}] = h(D)$ for all $E \in \mathcal{ELL}(\mathcal{O}_D)$, hence $H_D(x)$ is irreducible and it is, in fact, the minimal polynomial of $j(E)$ over $\mathbb{Q}$.

Example 13. Let $K = \mathbb{Q}[i]$ and $\mathcal{O}_K = \mathbb{Z}[i]$ be its ring of integers. We have $D = \operatorname{disc}(K) = -4$. Since $\mathbb{Z}[i]$ is PID, $\# \operatorname{Cl}(\mathcal{O}_K) = 1$ and, up to isomorphism, there is exactly one curve with complex multiplication by $\mathcal{O}_K$. We already know that $E : y^2 = x^3 + x$ over $\mathbb{C}$ has endomorphism ring $\operatorname{End}(E) = \mathbb{Z}[i]$, therefore

$$H_{-4}(x) = x - j(E) = x - 1728.$$

[2]See chapter 4, theorem 4.1.1.

Example 14. We list a few polynomials $H_D(x)$ for some small values of D:

$$\begin{aligned}
H_{-3}(x) &= x \\
H_{-7}(x) &= x + 3375 \\
H_{-8}(x) &= x - 8000 \\
H_{-4\cdot 5}(x) &= x^2 - 1264000x - 681472000 \\
H_{-4\cdot 6}(x) &= x^2 - 4834944x + 14670139392 \\
H_{-23}(x) &= x^3 + 3491750x^2 - 5151296875x + 12771880859375.
\end{aligned}$$

In all the examples above $H_D(x) \in \mathbb{Z}[x]$. This result is not specific but it holds in general.

Theorem 3.2.1. *Let $E/\mathbb{C}$ be an elliptic curve with complex multiplication. Then $j(E)$ is an algebraic integer, therefore the Hilbert polynomial has integer coefficients.*

Proof. We will prove this theorem in the last section, theorem 3.3.2. □

One of our tasks consists in providing an explicit procedure to compute the Hilbert polynomial $H_D(x)$. In order to do this two questions arise:

- How can we parameterize $\mathcal{ELL}(\mathcal{O}_D)$? In other words, how can we choose representatives of isomorphism classes of elliptic curves with CM by $\mathcal{O}_D$? What we would like to have is a set of $\tau \in \mathcal{H}$, which correspond to elliptic curves E_{Λ_τ} with CM by $\mathcal{O}_D$.
- How can we efficiently compute $j(E_{\Lambda_\tau}) = j(\tau)$?

We will answer the first question in the next section using *binary quadratic forms.*

3.2.3 Binary Quadratic Forms

Definition 3.2.3. A **binary quadratic form** Q is a primitive homogeneous polynomial of degree two in x, y with integer coefficients:

$$Q(x, y) = ax^2 + bxy + cy^2,\ a, b, c \in \mathbb{Z},\ \gcd(a, b, c) = 1.$$

The **discriminant** of Q is $\operatorname{disc}(Q) = b^2 - 4ac$. For every $D \in \mathbb{Z}$, $D \equiv 0, 1 \pmod 4$ we denote by $\mathrm{QF}(D)$ the set of all binary quadratic forms of discriminant D.

Example 15. $Q(x, y) = x^2 - xy + 3y^2$ is a binary quadratic form and its discriminant is $\operatorname{disc}(Q) = 1 - 4 \cdot 3 = -11$.

We will henceforth assume $a \geq 0$ and that the discriminant D is negative. For all $\left(\begin{smallmatrix} \alpha & \beta \\ \gamma & \delta \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$ and $Q \in \mathrm{QF}(D)$ we define

$$\left(\begin{smallmatrix} \alpha & \beta \\ \gamma & \delta \end{smallmatrix}\right) \cdot Q(x, y) = Q(\alpha x + \beta y, \gamma x + \delta y).$$

Remark 3.2.5. This operation preserves the discriminant: if $\mathrm{disc}(Q) = D$ and $\left(\begin{smallmatrix} \alpha & \beta \\ \gamma & \delta \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$ then

$$\mathrm{disc}(\left(\begin{smallmatrix} \alpha & \beta \\ \gamma & \delta \end{smallmatrix}\right) \cdot Q) = \mathrm{disc}(Q(\alpha x + \beta y, \gamma x + \delta y)) = (\alpha\delta - \beta\gamma)^2 \,\mathrm{disc}(Q) = \mathrm{disc}(Q)$$

since $\alpha\delta - \beta\gamma = 1$. Therefore this defines an action of $\mathrm{SL}_2(\mathbb{Z})$ on $\mathrm{QF}(D)$.

Definition 3.2.4. We say that two binary quadratic forms $Q_1, Q_2 \in \mathrm{QF}(D)$ are *equivalent*, and write $Q_1 \sim Q_2$, if there exists $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ such that $\gamma \cdot Q_1 = Q_2$. We denote the quotient[3] by $\mathcal{Q}(D) := \mathrm{QF}(D)/\sim$.

Remark 3.2.6. If $Q(x, y) = ax^2 + bxy + cy^2 \in \mathrm{QF}(D)$, then we can write

$$Q(x, y) = a(x - zy)(x - \overline{z}y)$$

where $z = \frac{-b+\sqrt{D}}{2a} \in \mathcal{H}$. This is true because of our assumptions $a \geq 0$ and $D < 0$.

Thanks to the remark we have an injective map

$$\begin{array}{rccc} F: & \mathrm{QF}(D) & \longrightarrow & \mathcal{H} \\ & Q & \mapsto & z \end{array}$$

which commutes with the action on $\mathrm{SL}_2(\mathbb{Z})$, that is

$$F(\gamma \cdot Q) = \gamma \cdot F(Q) \quad \forall \gamma \in \mathrm{SL}_2(\mathbb{Z}).$$

Therefore F induces an injection on the quotients $F : \mathcal{Q}(D) \longrightarrow \mathcal{H}/\,\mathrm{SL}_2(\mathbb{Z})$. In Chapter 2 we determined a fundamental domain for the action of $\mathrm{SL}_2(\mathbb{Z})$ on $\mathcal{H}$, the set

$$\overline{\mathcal{F}} = \left\{ z \in \mathcal{H} \,\middle|\, |z| \geq 1 \wedge -\frac{1}{2} \leq \mathrm{Re}\, z \leq \frac{1}{2} \right\}.$$

We are thus allowed to identify the elements of $\mathcal{Q}(D)$ with their corresponding representatives[4] in $\overline{\mathcal{F}}$. We have the following result:

Proposition 3.2.4. *For every discriminant $D < 0$ the set $\mathcal{Q}(D)$ is finite.*

[3]Since $\mathrm{SL}_2(\mathbb{Z}) \curvearrowright \mathrm{QF}(D)$, the relation $\sim$ just defined is in fact an equivalence relation.
[4]More precisely, a set of representatives for the action of $\mathrm{SL}_2(\mathbb{Z})$ on $\mathcal{H}$ is the set $\mathcal{F}_0$.

Proof. Since $\overline{\mathcal{F}}$ contains a set of representatives for the action of $\mathrm{SL}_2(\mathbb{Z})$ on $\mathcal{H}$, it is enough to show that there is only a finite number of elements $z = \frac{-b+\sqrt{D}}{2a} \in \overline{\mathcal{F}}$.

1. From $|\operatorname{Re} z| \leq \frac{1}{2}$ we obtain $\left|\frac{b}{2a}\right| \leq \frac{1}{2} \implies |b| \leq a$ since $a \geq 0$;
2. The condition $|z| \geq 1$ implies that $z\overline{z} \geq 1$, therefore

$$z\overline{z} = \frac{b^2 - D}{4a^2} \geq 1 \implies 4a^2 \leq b^2 - D$$

which implies, using 1, that $3a^2 \leq |D|$.

By 2 it is clear that there are only finitely many values for a, and since $|b| \leq a$ the same holds for b. Since c is determined once a and b are given, there are only finitely many triples (a, b, c). Hence $\mathcal{Q}(D)$ is finite. □

Corollary 3.2.4. *There is a one-to-one correspondence between $\mathcal{Q}(D)$ and the set*

$$\left\{ \frac{-b+\sqrt{D}}{2a} \;\middle|\; \gcd(a,b,c) = 1,\ D = b^2 - 4ac,\ |b| \leq a \leq c,\ 3a^2 \leq |D|,\ (\star) \right\}.$$

where

$$(\star):\ b \geq 0 \text{ if } a = |b| \ \vee\ a = c.$$

Proof. $\mathcal{Q}(D)$ is in bijection with the set

$$\left\{ \frac{-b+\sqrt{D}}{2a} \in \mathcal{F}_0 \mid D = b^2 - 4ac,\ \gcd(a,b,c) = 1 \right\}$$

where $\mathcal{F}_0 \subseteq \overline{\mathcal{F}}$ is the set

$$\mathcal{F}_0 = \left\{ z \in \mathcal{H} \;\middle|\; \left(|z| \geq 1 \ \wedge\ -\frac{1}{2} \leq \operatorname{Re} z \leq 0\right) \ \vee\ \left(|z| > 1 \ \wedge\ 0 < \operatorname{Re} z < \frac{1}{2}\right) \right\}.$$

Retracing the proof of the last proposition we obtain $|b| \leq a \leq c$ and $3a^2 \leq |D|$. Condition $(\star)$ follows from the fact that we want elements in $\mathcal{F}_0$. □

3.2.4 Relation with the ideal class group

The reason why we introduced binary quadratic forms and the action of the modular group is that these give lots of information about the ideal class group of $\mathcal{O}_D$, the order of discriminant $D < 0$ in the imaginary quadratic field $K = \mathbb{Q}(\sqrt{D})$. We will establish an explicit correspondence between $\mathrm{Cl}(\mathcal{O}_D)$

and $\mathcal{Q}(D)$, which in turn provides us with an explicit set of representative for the ideal class group.

Let $I \subseteq K$ be a proper fractional ideal of $\mathcal{O}_D$. We can write $I = \mathbb{Z}\omega_1 \oplus \mathbb{Z}\omega_2$ where $\mathcal{B} = \{\omega_1, \omega_2\}$ is a positive basis. Define the binary quadratic form

$$Q_I(x, y) = \frac{\mathrm{N}_{K/\mathbb{Q}}(x\omega_1 - y\omega_2)}{\mathrm{N}(I)}$$

where, in the computation of the numerator, x and y must be considered as integers.

Remark 3.2.7.

1. If we compute Q_I in a different basis $\mathcal{B}' = \{\widetilde{\omega}_1, \widetilde{\omega_2}\}$, then we have $\omega_1 = a\widetilde{\omega}_1 + b\widetilde{\omega}_2$ and $\omega_2 = c\widetilde{\omega}_1 + d\widetilde{\omega}_2$, where $\left(\begin{smallmatrix} a & c \\ b & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$ (because we considered positive bases), and the norm does not change. It can be easily checked that

$$Q_{I^{\mathcal{B}}}(x, y) = Q_{I^{\mathcal{B}'}}(ax - cy, -bx + dy).$$

   Therefore Q_I is well defined up to equivalence of quadratic forms.

2. $\forall \alpha \in K^\times,\ Q_I = Q_{\alpha I}$. This is a consequence of the multiplicativity of the norm: indeed we have

$$\mathrm{N}(\alpha I) = |\,\mathrm{N}_{K/\mathbb{Q}}(\alpha)|\,\mathrm{N}(I)$$

   and

$$\mathrm{N}_{K/\mathbb{Q}}(\alpha\beta) = \mathrm{N}_{K/\mathbb{Q}}(\alpha)\,\mathrm{N}_{K/\mathbb{Q}}(\beta) \quad \forall \alpha, \beta \in K.$$

3. It is clear from the definition that $Q_I \in \mathbb{Q}[x, y]$. By the definition of fractional ideal there exists $d \in \mathcal{O}_D \setminus \{0\}$ such that $dI \subseteq \mathcal{O}_D$. Therefore we can suppose ω_1, ω_2 algebraic integers. It follows that the coefficients of Q_I are integers: for example, the coefficient of x^2 is

$$\frac{\omega_1\overline{\omega_1}}{\mathrm{N}(I)} = \frac{\mathrm{N}_{K/\mathbb{Q}}(\omega_1)}{\mathrm{N}(I)} \in \mathbb{Z}$$

   because $(\omega_1) \subseteq I$ and hence $\mathrm{N}(I) \mid \mathrm{N}_{K/\mathbb{Q}}(\omega_1)$. Analogous arguments show that the other coefficients are also integers. Therefore we have $Q_I \in \mathbb{Z}[x, y]$.

4. $\mathrm{disc}(Q_I) = \mathrm{disc}(\mathcal{O}_D) = D$. We can suppose that I is an $\mathcal{O}$-ideal, therefore

$$\mathrm{disc}(Q_I) = \frac{1}{\mathrm{N}(I)^2}(\overline{\omega}_1\omega_2 - \omega_1\overline{\omega}_2)^2 = \frac{1}{\mathrm{N}(I)^2}\det\begin{pmatrix} \omega_1 & \omega_2 \\ \overline{\omega}_1 & \overline{\omega}_2 \end{pmatrix}^2 = \frac{\mathrm{N}(I)^2}{\mathrm{N}(I)^2}D = D$$

   in fact the above determinant is equal to $\mathrm{disc}(I)$ which is equal to $[\mathcal{O}_D : I]^2 D = \mathrm{N}(I)^2 D$.

It follows that there is a well defined map

$$\begin{array}{ccc} \mathrm{Cl}(\mathcal{O}_D) & \xrightarrow{\Phi} & \mathcal{Q}(D) \\ [I] & \mapsto & [Q_I] \end{array}.$$

To go in the other direction, for any binary quadratic form $Q(x,y) = ax^2 + bxy + cy^2 = a(x - zy)(x - \overline{z}y)$ with $z \in \mathcal{H}$, $\mathrm{disc}(Q) = D$, we define

$$I_Q = \mathbb{Z} \oplus \mathbb{Z}z.$$

Remark 3.2.8. 1. I_Q is a proper fractional ideal of $\mathcal{O}_D$. The proof of this fact is similar to part of the proof of theorem 3.1.1 and hence it will be omitted.

2. If $Q' = \gamma \cdot Q$ for $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ then $I_Q \sim I_{Q'}$. Since $\mathrm{SL}_2(\mathbb{Z}) = \langle \pm I, S, T \rangle$ it is enough to show it for those generators. The nontrivial cases occur for $\gamma = S, T$. We have

$$\begin{aligned} S \cdot Q(x,y) &= \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \cdot Q(x,y) \\ &= Q(y,-x) = cx^2 - bxy + ay^2 \\ &= c(x - z'y)(x - \overline{z'}y), \end{aligned}$$

where $z' = \frac{b+\sqrt{D}}{2c}$. It is easy to show that $z' = -1/z$, therefore $\frac{-1}{z} \cdot I_Q = \mathbb{Z}z' \oplus \mathbb{Z}(-1) = \mathbb{Z} \oplus \mathbb{Z}z' = I_{S \cdot Q}$ and therefore $I_Q \sim I_{S \cdot Q}$. For T we have

$$\begin{aligned} T \cdot Q(x,y) &= \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \cdot Q(x,y) = Q(x+y,y) \\ &= ax^2 + (2a+b)xy + (a+b+c)y^2 \\ &= c(x - z'y)(x - \overline{z'}y), \end{aligned}$$

with $z' = \frac{-(2a+b)+\sqrt{D}}{2a} = z - 1$, hence we have $I_Q = I_{T \cdot Q}$.

Therefore we have a map in the other direction:

$$\begin{array}{ccc} \mathcal{Q}(D) & \xrightarrow{\Psi} & \mathrm{Cl}(\mathcal{O}_D) \\ [Q] & \mapsto & [I_Q] \end{array}.$$

It is quite easy to check that $\Phi \circ \Psi = id_{\mathcal{Q}(D)}$ and $\Psi \circ \Phi = id_{\mathrm{Cl}(\mathcal{O}_D)}$. We have established

Theorem 3.2.2. *There is a one to one correspondence*

$$\mathcal{Q}(D) \quad \longleftrightarrow \quad \mathrm{Cl}(\mathcal{O}_D).$$

In particular we can take as representatives for the elements of $\mathrm{Cl}(\mathcal{O}_D)$ *the ideals of the form* $I = \mathbb{Z} \oplus \mathbb{Z}\tau$ *as* τ *varies in the set*

$$\left\{ \frac{-b+\sqrt{D}}{2a} \;\middle|\; \gcd(a,b,c) = 1,\ D = b^2 - 4ac,\ |b| \leq a \leq c,\ 3a^2 \leq |D|,\ (\star) \right\}.$$

Remark 3.2.9. Summarizing what we have done so far, we have established correspondences

$$\mathcal{Q}(D) \ \longleftrightarrow\ \mathrm{Cl}(\mathcal{O}_D) \ \longleftrightarrow\ \mathcal{ELL}(\mathcal{O}_D)$$

in a quite explicit way. In this section we have constructed a set T of representatives for ideal classes of $\mathrm{Cl}(\mathcal{O}_D)$. Combining this with the results on elliptic curves with complex multiplication established at the beginning of this section we have

$$\mathcal{ELL}(\mathcal{O}_D) = \big\{ [E_\Lambda] \mid \Lambda = \mathbb{Z} \oplus \mathbb{Z}\tau,\ \tau \in T \big\}.$$

This is the first ingredient we will use for the computation of the Hilbert polynomial, which will be the subject of part of the next section.

3.3 The integrality of j

Let $E/\mathbb{C}$ be an elliptic curve. If there exists an isogeny $\phi : E \longrightarrow E$ of degree $\deg \phi = n$ a non-square, E has complex multiplication. If that were not the case, we would have $\mathrm{End}(E) = \mathbb{Z}$, therefore if $\phi \in \mathrm{End}(E)$, $\phi = [n]$ and $\deg[n] = n^2$ for all $n \in \mathbb{Z}$. Remarkably the converse is also true. If E has CM by $\mathcal{O}_D \subseteq K = \mathbb{Q}(\sqrt{D})$, then $D = f^2 d_K$ and $\mathcal{O}_D = \mathbb{Z}[f\alpha] \subseteq \mathbb{Z}[\alpha] = \mathcal{O}_K$ where $\alpha = \sqrt{d} \ \vee\ \frac{1+\sqrt{d}}{2}$ for some square-free negative integer d. There exists $m \in \mathbb{Z}$ such that $m\sqrt{d} \in \mathcal{O}_D = \mathrm{End}(E)$, therefore

$$\deg(m\sqrt{d}) = |\,\mathrm{N}_{K/\mathbb{Q}}(m\sqrt{d})| = m^2|d|$$

which is not a perfect square (at least for $d < -1$). The first equality in the formula above is a nontrivial fact, but it is a consequence of what we will establish below (remark 3.3.1). Because of the above, we are interested in degree n-isogenies $\phi : E \longrightarrow E$.

3.3.1 Isogenies of degree n

Let E, E' be two elliptic curves over $\mathbb{C}$ and let $\phi : E \longrightarrow E'$ be a degree n isogeny. In terms of lattices it corresponds to an analytic map

$$\begin{array}{rccc} \alpha : & \mathbb{C}/\Lambda & \longrightarrow & \mathbb{C}/\Lambda' \\ & z & \mapsto & \alpha z \end{array}$$

induced by the inclusion $\alpha\Lambda \subseteq \Lambda',\ \alpha \in \mathbb{C}^\times$. Further, we have

$$\deg(\alpha) = |\Lambda'/\alpha\Lambda| = n.$$

Remark 3.3.1. Let $E = E_\Lambda$ be an elliptic curve with CM by $\mathcal{O}_D \subseteq K$ and let $\alpha \in \mathrm{End}(\Lambda)$, then

$$\begin{aligned} \deg(\alpha) &= \#\ker(\alpha) = \#\{z \in \mathbb{C}/\Lambda \mid \alpha z \in \Lambda\} \\ &= |\Lambda/\alpha\Lambda| = |\mathcal{O}_D/\alpha\mathcal{O}_D| = \mathrm{N}(\alpha) = |\,\mathrm{N}_{K/\mathbb{Q}}(\alpha)|. \end{aligned}$$

Up to homothety, $\Lambda = \mathbb{Z} \oplus \mathbb{Z}\tau$ and $\Lambda' = \beta \cdot (\mathbb{Z} \oplus \mathbb{Z}\tau')$ for $\tau, \tau' \in \mathcal{H}$ and $\beta \in \mathbb{C}^\times$. We have

$$\begin{aligned} \alpha\tau \in \Lambda' &\Longrightarrow \alpha\tau = \beta(a\tau' + b),\ a, b \in \mathbb{Z} \\ \alpha \in \Lambda' &\Longrightarrow \alpha = \beta(c\tau' + d),\ c, d \in \mathbb{Z} \end{aligned}$$

with $ad - bc = n$. Therefore

$$\tau = \frac{\alpha\tau}{\alpha} = \frac{a\tau' + b}{c\tau' + d} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot \tau',\ ad - bc = n.$$

It follows that

$$j(\tau) = j(\gamma \cdot \tau')$$

for some matrix $\gamma \in M_2(\mathbb{Z})$ of determinant n. Let's introduce two sets of matrices:

$$\mathcal{D}_n = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_2(\mathbb{Z}) \,\middle|\, ad - bc = n \right\}$$

and

$$\mathcal{S}_n = \left\{ \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \in M_2(\mathbb{Z}) \,\middle|\, ad = n,\ d > 0,\ 0 \le b < d \right\}$$

Remark 3.3.2. $\mathrm{SL}_2(\mathbb{Z})$ acts on $\mathcal{D}_n$ by matrix multiplication on the left and $\mathcal{S}_n$ is a set of representatives for the quotient $\mathrm{SL}_2(\mathbb{Z})\backslash\mathcal{D}_n$.

Recall from Chapter 2 the j-function

$$\begin{array}{rccc} j: & \mathcal{H} & \longrightarrow & \mathbb{C} \\ & \tau & \mapsto & j(\tau) = j(\Lambda_\tau) \end{array}$$

which is a holomorphic $\mathrm{SL}_2(\mathbb{Z})$-invariant function, with a simple pole at ∞. For all $\alpha = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in M_2(\mathbb{Z})$, $\det \alpha > 0$, we define the function $j \circ \alpha$ by the formula

$$j \circ \alpha(\tau) = j(\alpha \cdot \tau) = j\left(\frac{a\tau + b}{c\tau + d}\right), \quad \forall \tau \in \mathcal{H}.$$

It is again a holomorphic function on $\mathcal{H}$.

Remark 3.3.3. Since j is $\mathrm{SL}_2(\mathbb{Z})$-invariant, $\forall \gamma \in \mathrm{SL}_2(\mathbb{Z})$, $\forall \alpha \in \mathcal{D}_n$ we have $j \circ \gamma\alpha = j \circ \alpha$.

We are going to study the polynomial

$$F_n(X) = \prod_{\alpha \in \mathcal{S}_n} (X - j \circ \alpha).$$

Theorem 3.3.1. *There is a polynomial $F_n(Y, X) \in \mathbb{Z}[Y, X]$ such that*

$$\prod_{\alpha \in \mathcal{S}_n} (X - j \circ \alpha) = F_n(j, X).$$

Further, if n is a non-square integer, then the polynomial $F_n(X, X) \in \mathbb{Z}[X]$ has leading coefficient equal to ± 1.

Proof (Sketch). $F_n(X) = \prod_{\alpha \in \mathcal{S}_n} (X - j \circ \alpha) = \sum_m s_m X^m$, where $s_m = s_m(\tau)$ is the mth symmetric function in the $j \circ \alpha$'s and hence is a holomorphic function on $\mathcal{H}$. First we are going to prove that $s_m \in \mathbb{C}[j]$. We will use that the s_m are holomorphic $\mathrm{SL}_2(\mathbb{Z})$-invariant functions[5]:

$$s_m(\gamma \cdot \tau) = s_m(\tau) \ \forall \gamma \in \mathrm{SL}_2(\mathbb{Z}) \text{ and } \tau \in \mathcal{H}.$$

It follows that $s_m(\tau + 1) = s_m(\tau)$ and so it has a Fourier expansion in $q = e^{2\pi i \tau}$. What happens to s_m for $q \to 0$? Recall that j has a Fourier expansion $j(\tau) = q^{-1} + \sum_{k \geq 0} c_k q^k$, so it has a simple pole at $q = 0$. For all $\alpha = \left(\begin{smallmatrix} a & b \\ 0 & d \end{smallmatrix}\right) \in \mathcal{S}_n$ we have

$$j \circ \alpha(\tau) = e^{-2\pi i \frac{a\tau + b}{d}} + \sum_{k=0}^{\infty} c_k e^{2k\pi i \frac{a\tau + b}{d}}$$

[5]See [Sil94], Chapter 2, Section 6.

therefore $q^{n+1} \cdot j \circ \alpha(\tau) \to 0$ for $q \to 0$. Since s_m is the mth symmetric function of the $j \circ \alpha$'s it follows that $\exists N$ such that $q^N s_n(\tau) \to 0$ for $q \to 0$. Therefore $s_m(\tau)$ is an $\mathrm{SL}_2(\mathbb{Z})$-invariant holomorphic function which is meromorphic at ∞, hence $s_m \in \mathbb{C}[j]$. We will use another fact: the Fourier expansion of s_m has integer coefficients, that is

$$s_m(\tau) \in \mathbb{Z}[[q, q^{-1}]].$$

Now we prove that $s_m \in \mathbb{Z}[j]$, showing that, in fact, $\mathbb{C}[j] \cap \mathbb{Z}[[q, q^{-1}]] = \mathbb{Z}[j]$. The inclusion $\supseteq$ is clear. Conversely, given $f \in \mathbb{C}[j] \cap \mathbb{Z}[[q, q^{-1}]]$, we can write

$$f = a_0 j^d + a_1 j^{d-1} + \cdots + a_d$$

and replace j with its Fourier expansion, yielding

$$f = \frac{a_0}{q^d} + \frac{a_1 + 744 d a_0}{q^{d-1}} + \dots$$

By assumption, $f \in \mathbb{Z}[[q, q^{-1}]]$, thus $a_0 \in \mathbb{Z}$. Therefore $f - a_0 j^d = a_1 j^{d-1} + \cdots + a_d \in \mathbb{C}[j] \cap \mathbb{Z}[[q, q^{-1}]] = \mathbb{Z}[j]$ and hence we can repeat the same argument and eventually obtain $a_i \in \mathbb{Z}\ \forall i \implies f \in \mathbb{Z}[j]$.

It follows that $s_m \in \mathbb{Z}[j]\ \forall m$, therefore $F_n(X) = F_n(j, X) \in \mathbb{Z}[j, X]$ and the first part of the theorem is proven.

We set $Q = q^{1/n}$ and $\zeta = e^{2\pi i/n}$. For all $\alpha = \left(\begin{smallmatrix} a & b \\ 0 & d \end{smallmatrix}\right) \in \mathcal{S}_n$ consider the Q-expansion of $j - j \circ \alpha$:

$$j - j \circ \alpha = \left(\frac{1}{Q^n} + \sum_{k=0}^{\infty} c_k Q^{nk} \right) - \left(\frac{1}{\zeta^{ab} Q^{a^2}} + \sum_{k=0}^{\infty} c_k \zeta^{abk} Q^{a^2 k} \right).$$

Since n is not a square, the leading terms cannot cancel out, so $j - j \circ \alpha$ has a pole as $Q \to 0$ and the leading coefficient (that can be 1 or $-\zeta^{-ab}$) is a root of unity. Therefore the same holds for $F_n(j, j)$, whose Q-expansion has integer coefficients, therefore its leading coefficient is ± 1. Turning to the q-expansion, we have

$$F_n(j, j) = \pm \frac{1}{q^m} + \cdots \in q^{-m} \mathbb{Z}[[q]]$$

for some $m > 0$. On the other hand we know that $F_n(j, j) \in \mathbb{Z}[j]$ and j has a simple pole at $q = 0$, therefore

$$F_n(j, j) = \pm j^m + \cdots \in \mathbb{Z}[j]$$

that is $F_n(X, X) \in \mathbb{Z}[X]$ is non-constant with leading coefficient equal to ± 1. $\square$

Corollary 3.3.1. *$F_n(j, j') = 0$ if and only if $j = j(\tau)$, $j' = j(\tau')$ are the j-invariants of $E_\Lambda, E_{\Lambda'}$ with an isogeny $\phi : E_\Lambda \longrightarrow E_{\Lambda'}$ of degree n.*

Corollary 3.3.2 (Theorem 3.2.1)**.** *Let $E/\mathbb{C}$ be an elliptic curve with complex multiplication. Then $j(E)$ is an algebraic integer.*

Proof. Let $E = E_\Lambda$ be an elliptic curve with CM by $\mathcal{O}_D$, $j(E) = j(\tau)$, $\tau \in \mathcal{H}$. As we noted at the beginning of this section, there is a non-square negative integer d and $m \in \mathbb{Z}$ such that $\alpha = m\sqrt{d} \in \mathrm{End}(E)$ and $\deg(\alpha) = n$ is not a square. The isogeny $\alpha : E \longrightarrow E$ corresponds to the inclusion of lattices $\alpha\Lambda \subseteq \Lambda$ and in particular there exists $\beta \in \mathcal{D}_n$ such that

$$j(E) = j(\tau) = j \circ \beta(\tau).$$

By theorem 3.3.1 it follows that $j(E)$ is a root of $F_n(X, X) \in \mathbb{Z}[x]$ and, since n is a non-square, this polynomial has leading coefficient equal to ± 1, so $j(E)$ is an algebraic integer. □

Corollary 3.3.3. *For every (negative) discriminant D, the Hilbert polynomial*

$$H_D(x) = \prod_{j(E) \in \mathcal{ELL}(\mathcal{O}_D)} (x - j(E))$$

has integer coefficients.

Remark 3.3.4. We have just proven that $j(E) \in \mathbb{A}$ if $E/\mathbb{C}$ has complex multiplication. The converse is not true: if $j(E)$ is an algebraic integer it could happen that $\mathrm{End}(E) = \mathbb{Z}$. There are indeed finitely many orders of discriminant D with class number $h(D) = 1$, and for these the Hilbert polynomial has degree one. It follows that there are only finitely many $j_0 \in \mathbb{Z}$ that are j-invariants of elliptic curves with complex multiplication. It can be shown, for example in [Cox13], Chapter 12, Section C, that the only 13 discriminants D such that $h(D) = 1$ are

$$D = -3, -4, -7, -8, -11, -12, -16, -19, -27, -28, -43, -67, -163.$$

3.3.2 Computing the Hilbert polynomial

In this section we provide an explicit algorithm for the computation of the Hilbert class polynomial, see algorithms 1 and 2 at the end. Given a discriminant D, we can compute $H_D(x)$ as follows:

(1) compute the set $T = \{\tau_1, \ldots, \tau_{h(D)}\} \subseteq \mathcal{H}$ of representatives for ideal classes of $\mathrm{Cl}(\mathcal{O}_D)$;

(2) for each $\tau \in T$ compute $\widetilde{j}(\tau)$, evaluating $j(\tau)$ with the appropriate precision, and then compute the polynomial

$$\widetilde{H}_D(x) = \prod_{\tau \in T} (x - \widetilde{j}(\tau_i));$$

(3) recover $H_D(x)$ by rounding the coefficients of $\widetilde{H}_D(x)$ to the nearest integer.

According to theorem 3.2.2 the set T consists of the elements

$$\tau = \frac{-b + \sqrt{b^2 - 4ac}}{2a}$$

where a, b, c are integers such that $\gcd(a, b, c) = 1,\ b^2 - 4ac = D,\ |b| \leq a \leq c,\ 3a^2 \leq |D|$ and $b \geq 0$ if $a = |b|\ \vee\ a = c$.

Remark 3.3.5. We can recover the set T using algorithm 1 which requires $O(|D|)$ arithmetic operations.

Next we have to evaluate $j(\tau)$ for $\tau \in T$. Recall that, by lemma 2.3.2, the q-expansion of $j(\tau)$ is

$$j(\tau) = \frac{1}{q} + 744 + \sum_{n \geq 1} c_n q^n \tag{3.1}$$

where $q = e^{2\pi i \tau}$, and c_n are positive integers. Evaluating $j(\tau)$ using (3.1) is not the best strategy. Let us introduce the **Dedekind η-function**

$$\eta(\tau) = q^{1/24} \prod_{n=1}^{\infty} (1 - q^n),$$

where $q^{1/24} = e^{2\pi i \tau/24}$. This function is related to $j(\tau)$ by the following theorem.

Theorem 3.3.2 ([Cox13], Chapter 12, Theorem 12.17)**.** *The Dedekind η-function is a non-zero holomorphic function on $\mathcal{H}$ and*

$$j(\tau) = \left(\left(\frac{\eta(\tau)}{\eta(2\tau)} \right)^8 + 2^8 \left(\frac{\eta(2\tau)}{\eta(\tau)} \right)^{16} \right)^3. \tag{3.2}$$

We can derive the q-expansion series of $\eta(\tau)$ using the **pentagonal number theorem**.

Theorem 3.3.3 ([Apo76], Chapter 14, Theorem 14.3)**.** *If $|x| < 1$ we have*

$$\prod_{n=1}^{\infty}(1-x^n) = 1 + \sum_{k=1}^{\infty}(-1)^k\left(x^{k(3k-1)/2} + x^{k(3k+1)/2}\right)$$
$$= 1 - x - x^2 + x^5 + x^7 - x^{12} - x^{15} + \cdots .$$

as convergent power series.

Remark 3.3.6. The theorem is due to Euler. The exponents of the summands $g_k = k(3k-1)/2$ are called pentagonal numbers and they are related to partitions.

Therefore, in order to compute the value of $j(\tau)$ we evaluate

$$\eta(\tau) = q^{1/24}\left(1 + \sum_{k=1}^{\infty}(-1)^k\left(q^{k(3k-1)/2} + q^{k(3k+1)/2}\right)\right) \tag{3.3}$$

and then use equation (3.2). The polynomial

$$\widetilde{H}_D(x) = \prod_{\tau\in T}(x - \widetilde{j}(\tau_i))$$

can be then computed with complex floating point arithmetic. To recover the exact value of $H_D(x)$ we round the coefficients of $\widetilde{H}_D(x)$ to the nearest integer, hence the need of high precision is crucial.

Remark 3.3.7. For $\tau = (-b + \sqrt{D})/2a \in T$ we have

$$\log|j(\tau)| \approx \log(|q^{-1}|) \approx \pi\sqrt{|D|}/a.$$

With this estimate, we obtain a heuristic bound for the height of the polynomial $H_D(x)$ of

$$\pi\sqrt{|D|}\sum_{\tau\in T}\frac{1}{a}.$$

It can be shown that the sum $\sum_{\tau\in T} 1/a$ is $O(\log^2|D|)$ and thus the precision needed $n(D)$ will be of the magnitude of

$$n(D) = O(\sqrt{|D|}\log^2|D|).$$

The heuristic can be made rigorous and in fact the above bound is essentially optimal, as shown in [Eng09]. Note that the coefficients of $H_D(x)$ can be very large, they are indeed of the magnitude of $O(\exp(\sqrt{|D|}\log^2|D|))$. On

the other hand, the degree of the class polynomial $H_D(x)$ is equal to the class number $h(D)$. By Siegel's theorem we have

$$h(D) = O(\sqrt{|D|}\log|D|),$$

thus the complexity of the computation of $H_D(x)$ via floating point approximation becomes

$$O(|D|\log^{\alpha+\epsilon}|D|) = O(|D|^{1+\epsilon})$$

for any $\epsilon > 0$, and where α is a positive integer.

Remark 3.3.8. Faster algorithms, which use the Chinese remainder theorem, are laid out in [BBEL08], [Sut11]. Under heuristic assumptions, the complexity of the computation of $H_D(x)$ with these algorithms becomes

$$O(|D|\log^{3+\epsilon}|D|)$$

for any $\epsilon > 0$, which is not so different asymptotically but is much better in practice.

Remark 3.3.9. An estimate of the number of terms of series (3.3) that we have to calculate is given in [AM93]. In practice, for the discriminant D, we compute $\eta(\tau)$ to the order

$$r(D) = \sqrt{S(D)\ a}$$

where

$$S(D) = \frac{2}{3}\ \frac{\log 6 + n(D)\log 10}{\pi\sqrt{D}}.$$

Algorithm 1

Require: A discriminant $D < 0$
Ensure: The set of representatives T for ideal classes of $\mathcal{O}_D$.

```
if D = −3 then
    return T = {(√(−3) − 1)/2}
end if
T ← ∅
a ← 1
while 3a^2 ≤ |D| do
    for b = −a, ..., a do
        if b^2 − D ≡ 0 (mod 4a) then
            c ← (b^2 − D)/(4a)
            if a ≤ c ∧ gcd(a, b, c) = 1 then
                if a = |b| ∨ a = c then
                    if b ≥ 0 then
                        τ ← {(−b + √D)/(2a)}
                        T ← T ∪ {τ}
                    end if
                else
                    τ ← {(−b + √D)/(2a)}
                    T ← T ∪ {τ}
                end if
            end if
        end if
    end for
    a ← a + 1
end while
return T
```

Algorithm 2

Require: A discriminant $D < 0$
Ensure: The Hilbert polynomial $H_D(x)$

```
if D = −3 then
    return H_D(x) = x
end if
prec ← precision(D)
T ← representatives(D)
H̃_D(x) ← 1
for τ ∈ T do
    j ← compute_jinv(τ, prec)
    H̃(x) ← H̃_D(x) · (x − j)
end for
H_D(x) ← round_coefficients(H̃_D(x))
return H_D(x)
```

Chapter 4

Class Invariants

At the end of Chapter 3 we have seen how large the coefficients of the Hilbert class polynomial can be. In order to overcome this inconvenient, we can replace $H_D(x)$ with other **class polynomials** with smaller height. Such polynomials come from **class invariants**, which were first introduced by Weber in [Web13]. It is a well known fact that the **ring class field** F of the order $\mathcal{O}_D$ in the imaginary quadratic field $K = \mathbb{Q}(\sqrt{D})$ is the splitting field of the Hilbert class polynomial $H_D(x)$ over K, and is generated over K by any $j(\tau) \in \mathcal{ELL}(\mathcal{O}_D)$:

$$F = K(j(\tau)),\ j(\tau) \in \mathcal{ELL}(\mathcal{O}_D).$$

A class invariant is a modular function w which plays the same role as j in the previous formula. More precisely we will say that w is a class invariant at $\tau \in \mathcal{H}$ if

$$F = K(w(\tau)).$$

Two examples of functions that were considered by Weber are

$$\mathfrak{f}(\tau) = \zeta_{48}^{-1}\ \frac{\eta(\frac{\tau+1}{2})}{\eta(\tau)} \text{ and } \gamma_2(\tau) = j(\tau)^{1/3}.$$

In this chapter we will focus on the latter $\gamma(\tau) := \gamma_2(\tau)$, one of the three cube roots of $j(\tau)$.

Most of the literature approaches class invariants with advanced mathematical tools, such as *Shimura's reciprocity law* (see, for example, [Sch02], [SE21] or [Gee99]). We are going to take an elementary point of view and we will show that γ is a class invariant for all imaginary discriminants relatively prime to 3 (that is to say, for all $\tau \in \mathcal{H}$ such that $\mathbb{C}/\langle 1, \tau\rangle \in \mathcal{ELL}(\mathcal{O}_D)$).

4.1 Introduction

4.1.1 The Hilbert class field

Let $\mathcal{O}_D$ be the order of discriminant D in the imaginary quadratic field $K = \mathbb{Q}(\sqrt{D})$. Define F as the splitting field of $H_D(x)$ over K. Therefore F is generated over K by the j-invariants of the elliptic curves with complex multiplication by $\mathcal{O}_D$:

$$F = K(j(E) \mid j(E) \in \mathcal{ELL}(\mathcal{O}_D)).$$

We can describe F much more precisely, in fact we have the following result, a proof of which can be found in [Sil94], chapter II, theorem 4.1.

Theorem 4.1.1. *Let $\mathcal{O}_D$ be the order of discriminant D in the imaginary quadratic field $K = \mathbb{Q}(\sqrt{D})$, and let F be the splitting field of $H_D(x)$ over K. There exists a group isomorphism*

$$\Phi : \mathrm{Gal}(F/K) \longrightarrow \mathrm{Cl}(\mathcal{O}_D)$$

which commutes with the action of $\mathrm{Gal}(F/K)$ *and* $\mathrm{Cl}(\mathcal{O}_D)$ *on* $\mathcal{ELL}(\mathcal{O}_D)$. *In addition, the Hilbert class polynomial $H_D(x)$ is irreducible over K and $F = K(j(E))$ for any $j(E) \in \mathcal{ELL}(\mathcal{O}_D)$.*

The splitting field of $H_D(x)$ over K is known as the **ring class field** of $\mathcal{O}_D$ and, if $\mathcal{O}_D = \mathcal{O}_K$ is the maximal order, it is called the **Hilbert class field**.

Remark 4.1.1. The Hilbert class field of a generic number field K is defined to be the maximal unramified abelian extension of K. It can be shown that, in the case of imaginary quadratic fields, it coincides with the splitting field over K of $H_{d_K}(x)$. Similarly, the ring class field of $\mathcal{O}_D$ is the maximal abelian extension of K unramified at any primes which are relatively prime to the conductor of D. We will not deal with these issues.

If $\tau \in \mathcal{H}$ is an imaginary quadratic number of discriminant D and w is a class invariant at τ, then, by definition of class invariant, the ring class field of $\mathcal{O}_D$ is equal to

$$F = K(w(\tau)).$$

The minimal polynomial of $w(\tau)$ over K is called the **class polynomial** of w and can be computed in a similar way to that of the Hilbert class polynomial.

4.1.2 The curve $X_{\mathrm{ns}}^{+}(3)$

Let $\Gamma(1) = \mathrm{SL}_2(\mathbb{Z})$ be the modular group. Consider the subgroup $\Gamma < \Gamma(1)$ defined by

$$\Gamma = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}) \;\middle|\; b \equiv c \pmod 3 \;\vee\; a \equiv d \equiv 0 \pmod 3 \right\}.$$

Remark 4.1.2. It is clear from the definition that $\Gamma(3) \subseteq \Gamma$, therefore Γ is a congruence subgroup.

Lemma 4.1.1. *The index* $[\Gamma(1) : \Gamma]$ *is* 3 *and a set of representatives for the (right) cosets is given by* $\{I,\ T,\ T^2\}$*, in other words*

$$\Gamma \backslash \Gamma(1) = \{\Gamma,\ \Gamma T,\ \Gamma T^2\}.$$

Proof. We want to prove that

$$\Gamma(1) = \bigsqcup_{i=0}^{2} \Gamma T^i.$$

It is clear that $\Gamma T^i \cap \Gamma T^j = \emptyset$ for $i \neq j$. Let $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma(1)$, we want to show that $M \in \Gamma T^i$ for some $i = 0, 1, 2$. If $M \in \Gamma$ we are done. Suppose instead $M \notin \Gamma$. Note that

$$MT^{-1} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} a & b-a \\ c & d-c \end{pmatrix}, \text{ and}$$
$$MT^{-2} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} a & b-2a \\ c & d-2c \end{pmatrix}.$$

We distinguish two cases. If $a \equiv 0 \pmod 3$ it follows that $d \not\equiv 0 \pmod 3$ and $c \not\equiv 0 \pmod 3$, thus if $c \equiv d \pmod 3$ then $M \in \Gamma T$, if $c \not\equiv d \pmod 3$ then $M \in \Gamma T^2$. The case $a \not\equiv 0 \pmod 3$ is completely similar and will be omitted. □

Corollary 4.1.1. *A fundamental domain for the action of* Γ *on* $\mathcal{H}$ *is given by*

$$\mathcal{F}_\Gamma = \mathcal{F} \cup T \cdot \mathcal{F} \cup T^2 \cdot \mathcal{F}$$

where $\mathcal{F}$ *is the usual fundamental domain for* $\Gamma(1)$.

Proof. It is a consequence of

$$\Gamma(1) = \Gamma \sqcup T^{-1}\Gamma \sqcup T^{-2}\Gamma$$

which follows by taking the inverse on both sides of $\Gamma(1) = \Gamma \sqcup \Gamma T \sqcup \Gamma T^2$. □

We define the curve $X^+_{\mathrm{ns}}(3) := \Gamma\backslash\mathcal{H}^*$ and we denote by $Y^+_{\mathrm{ns}}(3) := \Gamma\backslash\mathcal{H}$ the curve obtained by removing the cusps.

Proposition 4.1.1. *The natural map $X^+_{\mathrm{ns}}(3) \longrightarrow X(1)$ has degree 3 and the other parameters are $\epsilon_2 = 3, \epsilon_3 = 0$ and $\epsilon_\infty = 1$. It follows that $X^+_{\mathrm{ns}}(3)$ has genus 0, in particular*

$$X^+_{\mathrm{ns}}(3) \cong \mathbb{P}^1, \textit{ and } Y^+_{\mathrm{ns}}(3) \cong \mathbb{A}^1.$$

Proof. We will use the same notation of [DS05], chapter 3. Since $-I \in \Gamma$, it follows that our map has degree $d_\Gamma = [\Gamma(1) : \Gamma] = 3$. For $j = 2, 3$ we have the relation

$$d_\Gamma = j\nu_j + \epsilon_j$$

between d_Γ and the number of the elliptic points of $X^+_{\mathrm{ns}}(3)$. It follows that there are three two-elliptic points

$$\Gamma \cdot i,\ \Gamma T \cdot i,\ \Gamma T^2 \cdot i$$

and no three-elliptic points. Therefore $\epsilon_2 = 3$ and $\epsilon_3 = 0$. Now we shall prove that there is only one cusp. We write

$$\mathbb{P}^1(\mathbb{Q}) = \big\{[x : 1] \mid x \in \mathbb{Q}\big\} \cup \big\{[1 : 0]\big\}$$

and we will prove that $[x : 1] \sim_\Gamma [1 : 0]$ for all $x \in \mathbb{Q}$. Recall that

$$[x : 1] \sim_\Gamma [1 : 0] \iff \exists A \in \Gamma \text{ s.t. } A \cdot [x : 1] = [1 : 0].$$

Let $x = \frac{s}{t}$ for some coprime integers s, t and let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, then

$$A \cdot [x : 1] = [ax + b : cx + d] = [1 : 0]$$

if and only if $cx + d = 0$, that is $cs + td = 0$. We set $c = -t$ and $d = s$ and we will find a, b such that $A \in \Gamma$. If $s = 0$ we can take $A = S \in \Gamma$, so we can assume $s \neq 0$. Consider the linear Diophantine equation (in a, b)

$$sa + tb = 1$$

which has integer solutions because by assumption $(s, t) = 1$. Let (a_0, b_0) be a particular solution, then the general solution is

$$(a_0 + tk, b_0 - sk),\ k \in \mathbb{Z}.$$

If $3 \mid s$ we can find $k \in \mathbb{Z}$ such that $3 \mid a = a_0 + tk$, hence $A \in \Gamma$. It is enough to solve (for k) the congruence

$$a_0 + kt \equiv 0 \pmod 3 \iff kt \equiv -a_0 \pmod 3$$

which can be solved since $3 \nmid t$, because $(s,t) = 1$. Now suppose $3 \nmid s$. We want to find $k \in \mathbb{Z}$ such that $b = b_0 - sk \equiv -t \pmod 3$, i.e.

$$sk \equiv t + b_0 \pmod 3$$

which has a solution because $3 \nmid s$ by hypothesis. It follows that $\epsilon_\infty = 1$. The genus of $X^+_{\mathrm{ns}}(3)$ can be finally computed by the formula

$$g(X^+_{\mathrm{ns}}(3)) = 1 + \frac{d_\Gamma}{12} - \frac{\epsilon_2}{4} - \frac{\epsilon_3}{3} - \frac{\epsilon_\infty}{2} = 1 + \frac{1}{4} - \frac{3}{4} - \frac{1}{2} = 0.$$

Therefore $X^+_{\mathrm{ns}}(3) \cong \mathbb{P}^1(\mathbb{C})$ and, removing the cusp, $Y^+_{\mathrm{ns}}(3) \cong \mathbb{A}^1(\mathbb{C})$. $\square$

Now we are going to explicitly write an isomorphism $X^+_{\mathrm{ns}}(3) \xrightarrow{\sim} \mathbb{P}^1(\mathbb{C})$ and an (affine) equation for our curve.

Remark 4.1.3. The modular curve $X(1) := \Gamma(1)\backslash\mathcal{H}^*$ has also genus 0 and an isomorphism with $\mathbb{P}^1(\mathbb{C})$ is given by the j-function

$$j : X(1) \xrightarrow{\sim} \mathbb{P}^1(\mathbb{C}),$$

where

$$j(\tau) = 1728\frac{g_2^3(\tau)}{\Delta(\tau)}$$

and

$$\Delta(\tau) = g_2^3(\tau) - g_3^2(\tau) = (2\pi)^{12} q \prod_{n=1}^{\infty}(1-q^n)^{24}.$$

Definition 4.1.1. Define $\gamma : \mathcal{H} \longrightarrow \mathbb{C}$ by

$$\gamma(\tau) = 12q^{-1/3}\frac{g_2(\tau)}{(2\pi)^4\prod_{n=1}^{\infty}(1-q^n)^8},$$

where $q^{-1/3} := e^{-(2\pi i/3)\cdot\tau}$.

Remark 4.1.4. It follows from the definition that γ is holomorphic on $\mathcal{H}$ and that

$$j(\tau) = \gamma(\tau)^3 \quad \forall \tau \in \mathcal{H}$$

which means that $\gamma(\tau)$ is a cube root of $j(\tau)$; for this reason we will sometimes write $\gamma(\tau) = j(\tau)^{1/3}$.

Proposition 4.1.2. *$\gamma : \mathcal{H} \longrightarrow \mathbb{C}$ is a holomorphic Γ-invariant function, meromorphic at the cusp. More precisely*

1. *$\gamma(\tau+1) = \zeta_3^2\gamma(\tau)$ and $\gamma(-1/\tau) = \gamma(\tau)$;*
2. *for all $\alpha = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma(1)$*

$$\gamma(\alpha \cdot \tau) = \zeta_3^{ac-ab+a^2cd-cd}\gamma(\tau) \ \forall \tau \in \mathcal{H}.$$

In particular γ is Γ-invariant.

Proof. Since j is $\Gamma(1)$-invariant and $j(\tau) = \gamma(\tau)^3$, it follows that

$$\gamma(\alpha \cdot \tau) = \zeta_3^{i(\alpha)}\gamma(\tau) \quad \forall \alpha \in \Gamma(1)$$

where $i(\alpha) \in \mathbb{Z}$ is an exponent depending on α (a priori we have $i(\alpha, \tau)$, but it is a continuous function in τ taking values in the discrete set $\mathbb{Z}$, hence it is constant in τ). Note that, up to a factor $q^{-1/3}$, γ admits a Fourier q-expansion in $\mathbb{Q}[[q]]$. In fact we can write

$$\gamma(\tau) = 12q^{-1/3}\frac{E_4(\tau)}{\prod_{n=1}^{\infty}(1-q^n)^8} = q^{-1/3}f(q)$$

where $f(q)$ is an holomorphic function in $\mathbb{Z}[[q]]$ (because the numerator and the denominator are in $\mathbb{Z}[[q]]$ and $(1-q^n)^{-1} = 1 + q^n + q^{2n} + \dots$). From this it follows that $\gamma(\tau+1) = \zeta_3^2\gamma(\tau)$. Now, from the definition, we have that $\gamma(\tau)$ is real for τ in the positive imaginary axis. Therefore it follows that $\gamma(-1/\tau) = \gamma(\tau)$ for $\tau \in i\mathbb{R}^+$, since both $\gamma(-1/\tau)$ and $\gamma(\tau)$ are real. Because they are holomorphic functions and coincide on the positive imaginary axis, they coincide on $\mathcal{H}$.

The second part follows from 1. by induction on the length of the representation of α as a word in S, T. □

Corollary 4.1.2. *The field of meromorphic functions on $X_{\mathrm{ns}}^+(3)$ is generated, as a field, by γ:*

$$\mathbb{C}(X_{\mathrm{ns}}^+(3)) = \mathbb{C}(\gamma).$$

Proof. By the previous proposition we have the inclusions

$$\mathbb{C}(X(1)) = \mathbb{C}(j) \subsetneq \mathbb{C}(\gamma) \subseteq \mathbb{C}(X_{\mathrm{ns}}^+(3)).$$

On the other hand $[\mathbb{C}(\gamma) : \mathbb{C}(j)] = [\mathbb{C}(X_{\mathrm{ns}}^+(3)) : \mathbb{C}(X(1))] = d_\Gamma = 3$, hence $\mathbb{C}(X_{\mathrm{ns}}^+(3)) = \mathbb{C}(\gamma)$. □

The situation is nicely described by the following commutative diagram:

$$\begin{tikzcd}
\mathcal{H}^* \arrow[d] \arrow[dr, "\gamma"] \arrow[ddr, bend right, "\pi"'] \arrow[ddrr, bend left, "j"] & & \\
X_{\mathrm{ns}}^+(3) \arrow[r, dash, "\sim"] \arrow[d] & \mathbb{P}^1(\mathbb{C}) \arrow[d] & \\
X(1) \arrow[r, dash, "\sim"] & \mathbb{P}^1(\mathbb{C}) &
\end{tikzcd}$$

The projection between the curves $X_{\mathrm{ns}}^+(3) \longrightarrow X(1)$ corresponds to the inclusion $\mathbb{C}(j) \subseteq \mathbb{C}(j, \gamma)$ of their fields of functions, therefore $X_{\mathrm{ns}}^+(3)$ can be defined over $\mathbb{Q}$ by the equation

$$\gamma^3 = j,$$

and the projection can be identified with $(j, \gamma) \mapsto j$ (to be more precise, their projective closures). This means, in particular, that the quotient map $\pi : \mathcal{H}^* \longrightarrow X_{\mathrm{ns}}^+(3)$ can be identified with $\tau \mapsto (j(\tau), \gamma(\tau))$.

4.2 On the class invariant $\gamma(\tau) = j(\tau)^{1/3}$

In the last section we introduced the Γ-modular function $\gamma(\tau) = j(\tau)^{1/3}$, now we are going to show that this function is a class invariant for any discriminant relatively prime to 3.

4.2.1 The Galois representation

Let n be a positive integer and let F be a field of characteristic $\neq 2, 3$ and which does not divide n. Let E/F be an elliptic curve. The Galois group $G = \mathrm{Gal}(\overline{F}/F)$ acts on the group of n-torsion points $E[n]$ via

$$\sigma \cdot P = \begin{cases} (\sigma(x), \sigma(y)) & \text{if } P = (x, y) \\ \infty & \text{if } P = \infty \end{cases}$$

where $\sigma \in G$ and $P \in E[n]$. Since addition on E is given by rational functions, the action of G preserves the group structure of $E[n]$:

$$\sigma \cdot (P_1 + P_2) = \sigma \cdot P_1 + \sigma \cdot P_2.$$

This leads to the **Galois representation**

$$\rho_{n,E/F} : G \longrightarrow \mathrm{Aut}(E[n]) \cong \mathrm{GL}_2(\mathbb{Z}/n\mathbb{Z}).$$

Remark 4.2.1. The isomorphism $\mathrm{Aut}(E[n]) \cong \mathrm{GL}_2(\mathbb{Z}/n\mathbb{Z})$ depends on the choice of a $\mathbb{Z}/n\mathbb{Z}$-basis of $E[n] \cong (\mathbb{Z}/n\mathbb{Z})^2$.

Remark 4.2.2. For all $n \in \mathbb{N}$ there is also an action of $\mathrm{End}(E)$ on the n-torsion points of E given by

$$\varphi \cdot P := \varphi(P) \quad \forall P \in E[n],\ \forall \varphi \in \mathrm{End}(E).$$

From now on we will focus on the special case $n = 3$. The following two results are crucial for our purposes.

Theorem 4.2.1. *Let F be a field whose characteristic is $\neq 2, 3$ and let E be an elliptic curve defined over F. The image of the Galois representation ρ_3 is contained in the 2-Sylow of $\mathrm{GL}_2(\mathbb{F}_3)$ (up to conjugation) if and only if $j(E)$ is a cube in F.*

Proof. Let $E : y^2 = x^3 + Ax + B$ and $P_i = (x_i, y_i) \in E(\overline{F})$ for $i = 1, \ldots, 8$ the 3-torsion points different from the point at infinity. We claim that $\mathrm{Im}\,\rho_3$ can be identified with the Galois group of the extension $F_3 := F(x_i, y_i \mid i = 1, \ldots, 8)$ over F. In fact, $\sigma \in \mathrm{Gal}(\overline{F}/F)$ acts trivially if and only if σ fixes P_i for all i, that is fixes x_i and y_i for all i, if and only if σ fixes F_3. Hence $\ker \rho_3 = \mathrm{Gal}(\overline{F}/F_3)$ and by the *first isomorphism theorem* we have

$$\mathrm{Im}\,\rho_{3,E/F} \cong \frac{\mathrm{Gal}(\overline{F}/F)}{\ker \rho_3} = \frac{\mathrm{Gal}(\overline{F}/F)}{\mathrm{Gal}(\overline{F}/F_3)} \cong \mathrm{Gal}(F_3/F).$$

Thus $\mathrm{Im}\,\rho_3$ is contained in a 2-Sylow if and only if $[F_3 : F]$ is a power of 2. Now we note that, letting $F_{3,x} := F(x_i \mid i = 1, \ldots, 8)$, we have $F \subseteq F_{3,x} \subseteq F_3$. The second extension has degree a power of two because it is obtained by adjoining square roots (in fact $y_i^2 = x_i^3 + Ax_i + B \in F_{3,x}$ for all i). Hence $[F_3 : F]$ is a power of two if and only if $[F_{3,x} : F]$ is. $F_{3,x}$ is the splitting field of the 3-division polynomial of E, that is

$$\psi_3(x) = 3x^4 + 6Ax^2 + 12Bx - A^2.$$

It is a classical fact that a degree 4 polynomial has Galois group over F of order not divisible by 3 if and only if its *resolvent cubic* has a root in F, for this see [Con]. The resolvent cubic of $\frac{1}{3}\psi_3(x)$ is

$$R(y) = 8y^3 + 16Ay^2 + \frac{32}{3}Ay^2 - 16B^2.$$

In order to test whether $R(y)$ has a root in F or not we can consider

$$\begin{aligned}\frac{1}{8}R(y) &= y^3 + 2Ay^2 + \frac{4}{3}Ay^2 - 2B^2 \\ &= \left(y + \frac{2}{3}A\right)^3 - \frac{8}{27}A^3 - 2B^2,\end{aligned}$$

which has a root in F if and only if $\frac{8}{27}A^3 + 2B^2 = \left(\frac{2}{3}\right)^3\left(A^3 + \frac{27}{4}B^2\right)$ is a cube in F. Rearranging yields

$$\left(\frac{2}{3}\right)^3\left(A^3 + \frac{27}{4}B^2\right) = (8A)^3\frac{A^3 + 27B^2}{(12A)^3} = \frac{(8A)^3}{j(E)},$$

therefore it is a cube in F if and only if $j(E)$ is, as desired. □

Remark 4.2.3. The relation $j =$ "a cube" is actually the equation of $Y_{\mathrm{ns}}^{+}(3)$. In fact, under the hypothesis of the theorem, the F-rational points of $Y_{\mathrm{ns}}^{+}(3)$ parameterize ($\overline{F}$-isomorphism classes of) elliptic curves E/F such that $\operatorname{Im}\rho_{3,E/F}$ is a 2-group.

Proposition 4.2.1. *Let F be a number field and E/F be an elliptic curve with complex multiplication by $\mathcal{O}_D$. Suppose that $3 \nmid D$ and $\operatorname{End}_F(E) = \mathcal{O}_D$, that is, all the endomorphisms of E are defined over F. The image of the Galois representation $\operatorname{Im}\rho_{3,E/F}$ is contained in a 2-Sylow of $\mathrm{GL}_2(\mathbb{F}_3)$.*

Proof. Since $\operatorname{End}(E) = \operatorname{End}_F(E)$, the actions of $\operatorname{Gal}(\overline{F}/F)$ and $\operatorname{End}(E)$ on $E[3]$ commute. In fact, for all $\varphi \in \operatorname{End}(E)$, $\sigma \in \operatorname{Gal}(\overline{F}/F)$, $P \in E[3]$ we have $\varphi = (\varphi_1, \varphi_2)$ with $\varphi_i \in F(E)$ and $P = (x, y)$, therefore

$$\begin{aligned}
\sigma \cdot (\varphi \cdot P) &= \sigma \cdot (\varphi(P)) \\
&= \sigma(\varphi_1(x,y), \varphi_2(x,y)) \\
&= (\sigma(\varphi_1(x,y)), \sigma(\varphi_2(x,y))) \\
&= (\varphi_1(\sigma(x), \sigma(y)), \varphi_2(\sigma(x), \sigma(y))) \\
&= \varphi \cdot (\sigma \cdot P).
\end{aligned}$$

By contradiction, suppose that $\operatorname{Im}\rho_{3,E/F} \subsetneq$ 2-Sylow. There exists $\sigma \in \operatorname{Gal}(\overline{F}/F)$ such that $M = \rho_3(\sigma) \in \mathrm{GL}_2(\mathbb{F}_3)$ has order three. Choosing an appropriate basis $\mathcal{B}$, we can suppose that

$$M = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Since the actions of $\operatorname{Gal}(\overline{F}/F)$ and $\operatorname{End}(E)$ on $E[3]$ commute, for all $\varphi \in \operatorname{End}(E)$ the matrix M commutes with

$$[\varphi|_{E[3]}]_{\mathcal{B}} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

Imposing the condition $M \cdot [\varphi|_{E[3]}]_{\mathcal{B}} = [\varphi|_{E[3]}]_{\mathcal{B}} \cdot M$, we obtain the equality

$$\begin{pmatrix} a+c & b+d \\ c & d \end{pmatrix} = \begin{pmatrix} a & a+b \\ c & c+d \end{pmatrix}$$

which implies $c \equiv 0 \pmod 3$ and $a \equiv d \pmod 3$. This shows that in the base $\mathcal{B}$ every $\varphi \in \mathrm{End}(E)$ acts on $E[3] \cong (\mathbb{Z}/3\mathbb{Z})^2$ via a matrix of the form

$$[\varphi|_{E[3]}]_{\mathcal{B}} = \begin{pmatrix} a & b \\ 0 & a \end{pmatrix}. \tag{4.1}$$

Now we distinguish two cases. If $|D| \equiv -1 \pmod 3$, then we consider $\alpha = r\sqrt{D} \in \mathcal{O}_D \cong \mathrm{End}(E)$ for some integer $r \not\equiv 0 \pmod 3$, hence

$$\deg(\alpha) = |\,\mathrm{N}_{K/\mathbb{Q}}(\alpha)| = r^2|D| \equiv -1 \pmod 3.$$

On the other hand by equation (4.1) we have $\deg(\alpha) = \det([\alpha|_{E[3]}]_{\mathcal{B}}) \equiv 0, 1 \pmod 3$, which leads to a contradiction. If $|D| \equiv 1 \pmod 3$ we consider $\alpha = s + r\sqrt{D} \in \mathcal{O}_D$ with $s, r \not\equiv 0 \pmod 3$. We have

$$\det(\alpha) = |\,\mathrm{N}_{K/\mathbb{Q}}(\alpha)| = s^2 + r^2|D| \equiv -1 \pmod 3$$

which leads to a contradiction as before. □

Remark 4.2.4. The result of the last proposition remains true without the assumption that the endomorphisms of E are defined over F. Let $K = \mathbb{Q}(\sqrt{D})$, since $\mathrm{End}(E) = \mathcal{O}_D$ it follows that the endomorphisms of E are defined over the product FK (this fact is not trivial, see [Sil94], Chapter 2, Theorem 2.2). Consider the following diagram

$$\begin{tikzcd}
 & \overline{F} \arrow[d, dash] & \\
 & FK \arrow[dl, dash, "\leq 2"'] \arrow[dr, dash] & \\
F \arrow[dr, dash] & & K \arrow[dl, dash, "2"] \\
 & \mathbb{Q} &
\end{tikzcd}$$

where $\overline{F}$ is a common algebraic closure of K and F. By the proposition just proved we have that the image of

$$\mathrm{Gal}(\overline{F}/FK) \longrightarrow \mathrm{GL}_2(\mathbb{F}_3)$$

is a 2-group. On the other hand $[FK : F]$ is at most two, therefore it follows that also $\mathrm{Im}\,\rho_{3,E/F}$ is a 2-group.

Combining the results proved above, we have

Proposition 4.2.2. *Let $\tau \in \mathcal{H}$ be an imaginary quadratic number. Let $E : y^2 = x^3 + Ax + B$ be the elliptic curve over $\mathbb{C}$ given by $E = \mathbb{C}/\langle 1, \tau\rangle$ and let $\mathcal{O}_D$ be its endomorphism ring, the order of discriminant D in the imaginary quadratic field $K = \mathbb{Q}(\sqrt{D})$. Suppose that $3 \nmid D$, then*

1. *$j(\tau)$ generates F, the ring class field of $\mathcal{O}_D$;*
2. *$\gamma^3(\tau) = j(\tau)$;*
3. *$j(\tau)$ has a cube root in the field F, hence at least one among*
$$\gamma(\tau), \gamma(\tau)\zeta_3, \gamma(\tau)\zeta_3^2$$
is in F.

Proof. The first two assertions are clear. By theorem 4.2.1 $j(\tau)$ is a cube in F if and only if $\operatorname{Im}\rho_{3,E/F}$ is a 2-group. Since $3 \nmid D$, by the previous proposition and remark, $\operatorname{Im}\rho_{3,E/F}$ is in fact a 2-group and hence $j(\tau)$ has a cube root in the field F. □

It remains to prove that in fact $\gamma(\tau) \in F$, that is, the cube root of $j(\tau)$ in F is $\gamma(\tau)$ and not $\gamma(\tau)\zeta_3$ or $\gamma(\tau)\zeta_3^2$. We are also interested in the computation of the conjugates of $\gamma(\tau)$. These questions are discussed in the next section.

4.2.2 3-torsion points

As in the previous sections, let $E = \mathbb{C}/\langle 1, \tau\rangle$ be an elliptic curve with CM by $\mathcal{O}_D$, $3 \nmid D$. Let $F = K(j(E)) = K(j(\tau))$ and let $P_i = (x_i, y_i)$ be the 3-torsion points of E. We denote by $F(E[3])$ the smallest extension of F which contains the coordinates of all the 3-torsion points,

$$F(E[3]) = F(x_i, y_i \mid i = 1, \ldots, 8).$$

Remark 4.2.5. $F(E[3]) = \overline{F}^{\ker \rho_{3,E/F}}$ is the fixed field of $\ker \rho_{3,E/F}$.

Lemma 4.2.1. *The field $F(E[3])$ contains all the three cube roots of $j(\tau)$.*

Proof. Because of proposition 4.2.2, $\gamma(\tau)\zeta_3^i \in F \subseteq F(E[3])$ for some $i = 0, 1, 2$ hence it is enough to prove that $\zeta_3 \in F(E[3])$. This is a consequence of the Galois-invariance of the Weil paring

$$e_3 : E[3] \times E[3] \longrightarrow \langle \zeta_3 \rangle.$$

In fact, if $P, Q \in E[3]$ are such that $e_3(P, Q) = \zeta_3$ then for all $\sigma \in \operatorname{Gal}(\overline{F}/F(E[3]))$

$$\begin{aligned}\sigma(\zeta_3) &= \sigma(e_3(P, Q)) = e_3(\sigma \cdot P, \sigma \cdot Q)\\ &= e_3(P, Q) = \zeta_3\end{aligned}$$

therefore $\zeta_3 \in F(E[3])$. □

We are actually able to compute these cube roots.

Proposition 4.2.3. *Let $E : y^2 = x^3 + Ax + B$ be an elliptic curve defined over a number field F with CM by $\mathcal{O}_D$, $3 \nmid D$. There exists a numbering of the four roots r_1, r_2, r_3, r_4 of the 3-division polynomial $\psi_3(x)$ such that*

$$\frac{-48A}{2A - 3(r_1r_2 + r_3r_4)}$$

is a cube root of $j(E)$ in $F(E[3])$.

Proof. By definition we have

$$j(\tau) = j(E) = 1728\frac{4A^3}{4A^3 + 27B^2} = 1728\frac{A^3}{A^3 + \frac{27}{4}B^2}.$$

Set $\alpha = A^3 + \frac{27}{4}B^2$ and let r_1, r_2, r_3, r_4 be the four roots of $\frac{1}{3}\psi_3(x) = x^4 + 2Ax^2 + 4Bx - \frac{A^2}{3}$. From to the proof of theorem 4.2.1 we have

$$\frac{1}{8}R(y) = \left(y + \frac{2}{3}A\right)^3 - \left(\frac{2}{3}\right)^3\alpha$$

where $R(y)$ is the cubic resolvent of $\frac{1}{3}\psi_3(x)$. We know that it has a solution in F because $j(\tau)$ is a cube in F. If $\beta \in F$ is such that $\beta^3 = \alpha$ then

$$y_0 = \frac{2}{3}(\beta - A)$$

is a solution. By the properties of the resolvent, the roots of $\frac{1}{3}\psi_3(x)$ correspond to the roots of the quadratic polynomials

$$x^2 - \sqrt{2y_0}x + A + y_0 + \frac{2B}{\sqrt{2y_0}}$$

and

$$x^2 - \sqrt{2y_0}x + A + y_0 - \frac{2B}{\sqrt{2y_0}}.$$

Up to renaming the roots of $\frac{1}{3}\psi_3(x)$, we can suppose r_1, r_2 are the solutions of the first polynomial and r_3, r_4 are the solutions of the second. It follows that

$$r_1r_2 + r_3r_4 = 2A + 2y_0$$

and substituting $y_0 = \frac{2}{3}(\beta - A)$ leads to

$$\beta = -\frac{1}{4}(2A - 3(r_1r_2 + r_3r_4)).$$

It follows that $\frac{-48A}{2A-3(r_1r_2+r_3r_4)}$ is a cube root of $j(\tau)$, as required. $\square$

Remark 4.2.6. We note that there are exactly three ways of divide the four roots r_1, r_2, r_3, r_4 in two pairs: these three ways lead to the three cube roots of $j(\tau)$, namely

$$\frac{-48A}{2A - 3(r_1r_2 + r_3r_4)}, \quad \frac{-48A}{2A - 3(r_1r_3 + r_2r_4)} \quad \text{and} \quad \frac{-48A}{2A - 3(r_1r_4 + r_3r_2)}.$$

If we treat the coefficients A, B of the equation defining E as variables, then we can consider $\psi_3(x)$ as a polynomial in $\mathbb{Q}(A, B)[x]$. The extension

$$\mathbb{Q}(A, B) \subseteq \mathbb{Q}(A, B, r_1, r_2, r_3, r_4),$$

where now the roots r_i depend on A, B, C, is a Galois extension (it is in fact the splitting field over $\mathbb{Q}(A, B)$ of $\psi_3(x)$) with Galois group isomorphic to the symmetric group on four elements S_4[1]. It follows that the Galois group acts transitively on the four roots r_1, r_2, r_3, r_4. For any $\sigma \in S_4$, since $\sigma(A) = A$ and $\sigma(B) = B$, we have that

$$\begin{aligned}
\sigma\left(\frac{-48A}{2A - 3(r_1r_2 + r_3r_4)}\right)^3 &= \sigma\left(\left(\frac{-48A}{2A - 3(r_1r_2 + r_3r_4)}\right)^3\right) \\
&= \sigma\left(1728\frac{4A^3}{4A^3 + 27B^2}\right) \\
&= 1728\frac{4A^3}{4A^3 + 27B^2} \\
&= j(E)
\end{aligned}$$

is still a cube root. On the other hand

$$\sigma\left(\frac{-48A}{2A - 3(r_1r_2 + r_3r_4)}\right) = \frac{-48A}{2A - 3(r_{\sigma(1)}r_{\sigma(2)} + r_{\sigma(3)}r_{\sigma(4)})},$$

therefore the other two cube roots of $j(E)$ correspond to the other ways of dividing r_1, r_2, r_3, r_4 in two pairs.

There is no algebraic way to distinguish the four roots r_i, and thus we cannot distinguish the three cube roots of $j(\tau)$. However, such task can be accomplished analytically. By fixing τ, E has a *canonical* equation given by

$$Y^2 = 4X^3 - g_2X - g_3$$

[1]This is true because when we specialize the polynomial $f = \psi_3(A, B, C, x) \in \mathbb{Q}[A, B, C, x]$ in some $A_0, B_0, C_0 \in \mathbb{Q}$ such that the resulting polynomial $g = \psi_3(A_0, B_0, C_0, x) \in \mathbb{Q}[x]$ is a separable polynomial of the same degree, then the Galois group of g over $\mathbb{Q}$ is a subgroup of that of f over $\mathbb{Q}(A, B)$. We can conclude observing that the 3-division polynomial of $E : y^2 = x^3 + x + 1$ has Galois group over $\mathbb{Q}$ isomorphic to S_4.

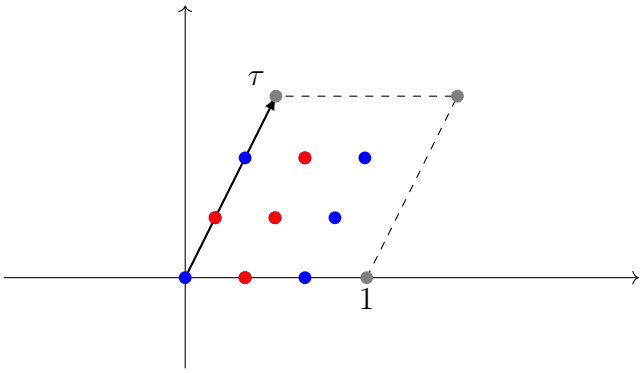


Figure 4.1: A fundamental parallelogram for $\Lambda = \langle 1, \tau \rangle$. The blue and red dots corresponds to the 3-torsion points of E_Λ, the red dots are those we have chosen to determine the X-coordinates of the 3-torsion points.

where $g_2 = 60G_4(\tau)$ and $g_3 = 140G_6(\tau)$. In these coordinates the 3-torsion points have X-coordinates given by

$$\wp\left(\frac{1}{3}\right), \wp\left(\frac{\tau}{3}\right), \wp\left(\frac{1+\tau}{3}\right), \wp\left(\frac{1+2\tau}{3}\right).$$

Multiplying by 16 on both sides we have

$$(4Y)^2 = (4X)^3 - 4g_2(4X) - 16g_3$$

and we can set $x = 4X$ and $y = 4Y$. In these coordinates $A = -4g_2$ and the 3-torsion points have x-coordinates given by

$$4\wp\left(\frac{1}{3}\right), 4\wp\left(\frac{\tau}{3}\right), 4\wp\left(\frac{1+\tau}{3}\right), 4\wp\left(\frac{1+2\tau}{3}\right).$$

Remark 4.2.7. Note that the function $\wp$ also depends on the lattice Λ. In our case we have, for example,

$$\wp\left(\frac{\tau}{3}\right) = \wp\left(\frac{\tau}{3}, \langle 1, \tau \rangle\right).$$

Proposition 4.2.4. *Let $E = \mathbb{C}/\langle 1, \tau \rangle$ be an elliptic curve with CM by $\mathcal{O}_D$, $3 \nmid D$, then*

$$\gamma(\tau) = \frac{-48A}{2A - 3 \cdot 16(\wp(\frac{1}{3})\wp(\frac{\tau}{3}) + \wp(\frac{1+\tau}{3})\wp(\frac{1+2\tau}{3}))}$$

where $A = -240G_4(\tau)$ and $\wp$ is the Weierstrass function of $\Lambda = \langle 1, \tau \rangle$.

Proof. From what we have discussed above, it follows that the three cube roots of $j(\tau)$ are

$$\begin{aligned}
\gamma_1(\tau) &= \frac{-48A}{2A-3\cdot 16(\wp\left(\frac{1}{3}\right)\wp\left(\frac{\tau}{3}\right)+\wp\left(\frac{1+\tau}{3}\right)\wp\left(\frac{1+2\tau}{3}\right))},\\
\gamma_2(\tau) &= \frac{-48A}{2A-3\cdot 16(\wp\left(\frac{1}{3}\right)\wp\left(\frac{1+\tau}{3}\right)+\wp\left(\frac{\tau}{3}\right)\wp\left(\frac{1+2\tau}{3}\right))},\\
\gamma_3(\tau) &= \frac{-48A}{2A-3\cdot 16(\wp\left(\frac{1}{3}\right)\wp\left(\frac{1+2\tau}{3}\right)+\wp\left(\frac{1+\tau}{3}\right)\wp\left(\frac{\tau}{3}\right))}.
\end{aligned}$$

We prove that $\gamma(\tau) = \gamma_1(\tau)\ \forall \tau \in \mathcal{H}$. We know that $\gamma(\tau)$ is real on the positive imaginary axis, on the other hand $\gamma_1(\tau)$ is the only one of the functions above that is real for $\tau \in i\mathbb{R}^+$, in fact we have

$$\overline{\gamma_1(\tau)} = \gamma_1(\tau),\ \overline{\gamma_2(\tau)} = \gamma_3(\tau)$$

for those values of τ. Therefore we have $\gamma(\tau) = \gamma_1(\tau)\ \forall \tau \in i\mathbb{R}^+$ and since they are holomorphic functions on $\mathcal{H}$ it follows that $\gamma(\tau) = \gamma_1(\tau)\ \forall \tau \in \mathcal{H}$. $\square$

Remark 4.2.8. Note that the *canonical* equation $y^2 = x^3 - 4g_2x - 16g_3$, that we used to make the computations in the above proof, may not have coefficients in F. We have to appropriately scale the coordinates x, y in order to get an equation defined over F. Fortunately, this inconvenient does not affect our argument: the expression $\frac{-48A}{2A-3(r_1r_2+r_3r_4)}$ is in fact invariant under such scaling (both A and r_ir_j are multiplied by λ^4, where λ is the rescaling factor).

Given $\tau \in \mathcal{H}$, we consider its minimal polynomial

$$ax^2 + bx + c \in \mathbb{Z}[x].$$

By replacing τ with $-1/\tau$ or $\tau/(1+\tau)$ (which are $\Gamma(1)$-equivalent to τ), respectively in the case $a \equiv 0 \pmod 3$, $c \not\equiv 0 \pmod 3$ and $a, c \equiv 0 \pmod 3$, we can assume $a \not\equiv 0 \pmod 3$. Now, exactly one among

$$\tau,\ \tau+1,\ \tau+2$$

has minimal polynomial

$$Ax^2 + Bx + C, \quad A, B, C \in \mathbb{Z}, \text{ with } (A, B, C) = 1 \text{ and } 3 \mid B.$$

We work with this representative and we will call it τ_0. We will come back to this in the next section.

Note that, by $\Gamma(1)$-equivalence, the discriminant of τ_0 is also equal to D.

Remark 4.2.9. Because, by assumption, $3 \nmid D = B^2 - 4AC$ and $B \equiv 0 \pmod 3$, it follows that $3 \nmid A$ and $3 \nmid C$.

Proposition 4.2.5. *Let $E = \mathbb{C}/\langle 1, \tau_0 \rangle$, with τ_0 as above and let $P = [\frac{1}{3}]$ and $Q = [\frac{\tau_0}{3}]$.*

1. *The image of $\rho_{3,E/F}$, in the basis $\mathcal{B} = \{P, Q\}$, is contained either in the group*
$$\left\{ M \in \mathrm{GL}_2(\mathbb{F}_3) \mid M = \begin{pmatrix} a & b \\ b & a \end{pmatrix} \right\}$$
or in the group
$$\left\{ M \in \mathrm{GL}_2(\mathbb{F}_3) \mid M = \begin{pmatrix} a & -b \\ b & a \end{pmatrix} \right\}.$$

2. *$\gamma(\tau_0)$ is invariant under the Galois group $\mathrm{Gal}(\overline{F}/F)$ and hence $\gamma(\tau_0) \in F$.*

Proof. The minimal polynomial of τ_0 is $Ax^2 + Bx + C$ with $3 \mid B$. The action of $\mathrm{Gal}(\overline{F}/F)$ commutes with the action of $\mathrm{End}_F(E) = \mathrm{End}_{\overline{K}}(E) = \mathbb{Z} \oplus \mathbb{Z}A\tau_0$, hence it must commute with the action of $A\tau_0$. In the basis $\mathcal{B} = \{P, Q\}$ we can explicitly write this action:

$$P = \left[\frac{1}{3}\right] \mapsto \left[\frac{A\tau}{3}\right],$$
$$Q = \left[\frac{\tau_0}{3}\right] \mapsto \left[\frac{A\tau_0^2}{3}\right] = \left[-\frac{B\tau_0}{3} - \frac{C}{3}\right]$$

hence it is represented by the matrix

$$M = \begin{pmatrix} 0 & -C \\ A & -B \end{pmatrix} = \begin{pmatrix} 0 & -C \\ A & 0 \end{pmatrix}$$

since $B \equiv 0 \pmod 3$. Because we are working modulo 3 and A and C are not zero modulo 3 it follows that, up to a sign,

$$M = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \vee M = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Therefore $\mathrm{Im}\, \rho_{3,E/F}$ is contained in the subgroup

$$\{g \in \mathrm{GL}_2(\mathbb{F}_3) \mid gM = Mg\} = \mathrm{Span}_{\mathbb{F}_3}(\mathrm{Id}, M) \cap \mathrm{GL}_2(\mathbb{F}_3)$$

which coincides with one of the subgroups claimed.

By proposition 4.2.4 we have

$$\gamma(\tau_0) = \frac{-48A}{2A - 3(r_1 r_2 + r_3 r_4)}$$

where $A = -240G_4(\tau_0)$ and r_1, r_2, r_3, r_4 are the images through $4\wp$ of the numbers $1/3, \tau_0/3, (1+\tau_0)/3$ and $(1+2\tau_0)/3$. In particular, r_1 and r_2 are the x-coordinates of the two 3-torsion points we have chosen as basis of $E[3]$. By the description of $\operatorname{Im}\rho_{3,E/F}$ just given, it follows that the action of $\operatorname{Gal}(\overline{F}/F)$ on $\gamma(\tau_0)$ is trivial, thus $\gamma(\tau_0) \in F$. This can be checked case by case: for example, suppose $\sigma \in \operatorname{Gal}(\overline{F}/F)$ acts as the matrix

$$\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$$

then $P \mapsto P+Q,\ Q \mapsto -P+Q,\ P+Q \mapsto Q$ and $P+2Q \mapsto -P$, therefore

$$\sigma(\wp(1/3)) = \sigma(x(P)) = x(P+Q) = \wp(\tau/3 + 1/3)$$

and similarly $\sigma(\wp(\tau_0/3)) = \wp((1+2\tau_0)/3)$, $\sigma(\wp(\tau_0/3+1/3)) = \wp(\tau_0/3)$ and $\sigma(\wp((1+2\tau_0)/3)) = \wp(1/3)$. Then

$$\sigma\big(\wp\big(\tfrac{1}{3}\big)\wp\big(\tfrac{\tau}{3}\big) + \wp\big(\tfrac{1+\tau}{3}\big)\wp\big(\tfrac{1+2\tau}{3}\big)\big) = \wp\big(\tfrac{1}{3}\big)\wp\big(\tfrac{\tau}{3}\big) + \wp\big(\tfrac{1+\tau}{3}\big)\wp\big(\tfrac{1+2\tau}{3}\big).$$

□

We can finally prove that γ is a class invariant.

Theorem 4.2.2. *Let $\tau \in \mathcal{H}$ be a quadratic number with discriminant $D \not\equiv 0 \pmod 3$, and let $\tau_0 \in \mathcal{H}$ as above. Let $F = K(j(\tau))$ be the Hilbert class field of $K = \mathbb{Q}(\sqrt{D})$, then*

$$F = K(\gamma(\tau_0)).$$

Proof. We have $F = K(j(\tau)) = K(j(\tau_0)) \subseteq K(\gamma(\tau_0))$, since j is $\Gamma(1)$-invariant and $j(\tau_0) = \gamma(\tau_0)^3$. On the other hand, by the last proposition, $\gamma(\tau_0) \in F$ and then $K(\gamma(\tau_0)) \subseteq F$. □

The only thing left to answer is how to compute the conjugates of $\gamma(\tau_0)$ and consequently the class polynomial. We are going to provide an answer to this question in the next section.

4.2.3 Computing the conjugates of $\gamma(\tau_0)$

We have seen in chapter 2 that

$$\mathcal{ELL}(\mathcal{O}_D) = \big\{ j(\tau_i) \mid \tau_i \in T \big\}$$

where $T = \{\tau_1, \ldots, \tau_{h(D)}\} \subseteq \mathcal{F}$ is a set of representatives for isomorphism classes of elliptic curves with complex multiplication by $\mathcal{O}_D$. As usual, we suppose that $3 \nmid D$.

For all $\tau_i \in T$ we can construct, as at the end of the last section, the representative $\tau_{i,0} \in \mathcal{H}$ determined by the condition that its minimal polynomial

$$A_i x^2 + B_i x + C_i \in \mathbb{Z}[x]$$

has trace $B_i \equiv 0 \pmod 3$, and therefore $3 \nmid A_i, C_i$. By theorem 4.2.2 we also have

$$F = K(j(\tau_i)) = K(\gamma(\tau_{i,0})).$$

We will call L the set of these representatives:

$$L = \{\tau_{1,0}, \ldots, \tau_{h(D),0}\}.$$

We will need the following lemma.

Lemma 4.2.2. *Let $f(x) \in K[x]$ and let $\alpha_1, \ldots, \alpha_n$ be its roots in an algebraic closure $\overline{K}$. Let F be the splitting field of $f(x)$ over K, that is $F = K(\alpha_1, \ldots, \alpha_n)$. Then $f(x)$ is irreducible over K if and only if* $\mathrm{Gal}(F/K)$ *acts transitively on $\{\alpha_1, \ldots, \alpha_n\}$.*

We conclude with the following theorem

Theorem 4.2.3. *Let $\mathcal{O}_D$ be the order of discriminant D in the field $K = \mathbb{Q}(\sqrt{D})$ and let F be the corresponding ring class field. Suppose $3 \nmid D$, then for all $\tau \in L$ we have $F = K(\gamma(\tau))$, and the polynomial*

$$G_D(x) = \prod_{\tau \in L} (x - \gamma(\tau))$$

is the minimal polynomial over K of $\gamma(\tau)$ for any $\tau \in L$. Moreover $G_D(x) \in \mathbb{Z}[x]$.

Proof. The first part has been proved at the end of the previous section. From the theory developed in chapter 2, we know that $\mathrm{Cl}(\mathcal{O}_D)$ acts transitively on $\mathcal{ELL}(\mathcal{O}_D) = \{j(\tau_i) \mid \tau_i \in T\}$. On the other hand by theorem 4.1.1 also $\mathrm{Gal}(F/K)$ acts transitively on $\mathcal{ELL}(\mathcal{O}_D)$, therefore

$$\forall h, k \ \exists \sigma \in \mathrm{Gal}(F/K) \text{ such that } \sigma(j(\tau_h)) = j(\tau_k).$$

For all h, k we have

$$j(\tau_h) = j(\tau_{h,0}) = \gamma(\tau_{h,0})^3, \text{ and } j(\tau_k) = j(\tau_{k,0}) = \gamma(\tau_{k,0})^3$$

and there exists $\sigma \in \mathrm{Gal}(F/K)$ such that

$$(\sigma\gamma(\tau_{h,0}))^3 = \sigma j(\tau_h) = j(\tau_k) = \gamma(\tau_{k,0})^3.$$

It follows that

$$\sigma\gamma(\tau_{h,0}) = \zeta_3^s \gamma(\tau_{k,0})$$

for some $s \in \mathbb{Z}$, which implies $\zeta_3^s = \sigma\gamma(\tau_{h,0})/\gamma(\tau_{k,0}) \in F$. We claim that $s \equiv 0 \pmod 3$ and therefore

$$\sigma\gamma(\tau_{h,0}) = \gamma(\tau_{k,0}).$$

Let's prove the claim. We have $\zeta_3 \in F$ if and only if $\sqrt{-3} \in F$. Then suppose, by contradiction, that $\sqrt{-3} \in F$. We have, thus, that the ideal

$$3\mathcal{O}_F = (\sqrt{-3})^2$$

is ramified. On the other hand $3 \nmid D = f^2 d_K \implies 3 \nmid f$ and $3 \nmid d_K$, hence 3 does not ramify in $\mathcal{O}_K$ ($3 \nmid d_K$) and, also, all primes $\mathfrak{p} \subseteq \mathcal{O}_K$ above 3 are unramified in $\mathcal{O}_F$ ($3 \nmid f$ and F is the ring class field of $\mathcal{O}_D$). It follows that 3 does not ramify in $\mathcal{O}_F$, a contradiction.

We have proved the following

$$\forall h, k \ \exists \sigma \in \mathrm{Gal}(F/K) \text{ s.t. } \sigma(\gamma(\tau_{h,0})) = \gamma(\tau_{k,0}),$$

therefore $\mathrm{Gal}(F/K)$ acts transitively on L, hence, by lemma 4.2.2, the polynomial

$$G_D(x) = \prod_{\tau \in L} (x - \gamma(\tau))$$

is irreducible over K and it is the minimal polynomial of any of its roots.

So far we have proved that $G_D(x) \in K[x]$. In order to show that its coefficients are integers it is enough to show that they are rational: in fact $\gamma(\tau_{i,0})^3 = j(\tau_{i,0})$ and $j(\tau_{i,0}) \in \mathbb{A}$, thus the coefficients of $G_D(x)$ are rational algebraic integers and hence integers.

Let $G_D(x) = x^{h(D)} + b_{h(D)-1}x^{h(D)-1} + \cdots + b_1 x + b_0$, we will prove that $b_i \in K \cap \mathbb{R} = \mathbb{Q}(\sqrt{D}) \cap \mathbb{R} = \mathbb{Q}$ for all i. Since the b_i's are the elementary symmetric functions in the $\gamma(\tau_{i,0})$'s, to show that $b_i \in \mathbb{R}$ it is enough to prove that $\{\gamma(\tau_{1,0}), \ldots, \gamma(\tau_{h(D),0})\}$ is stable under complex conjugation. We note that for any $\tau = a + ib \in \mathcal{H}$ we have

$$\overline{q(\tau)} = q(\tau'), \ \overline{q^{1/3}(\tau)} = q^{1/3}(\tau')$$

where $\tau' = -a + ib$. Therefore looking at the q-expansion series of j and γ, it follows that

$$\overline{j(\tau)} = j(\tau'), \ \overline{\gamma(\tau)} = \gamma(\tau') \ \ \forall \tau \in \mathcal{H}.$$

Since $H_D(x)$ has integer coefficients the set

$$\{j(\tau_i) \mid \tau_i \in T\} = \{j(\tau_{1,0}), \ldots, j(\tau_{h(D),0})\}$$

is stable under complex conjugation. For all i there exists j such that $\overline{j(\tau_{i,0})} = j(\tau_{j,0})$, then, since the trace $\operatorname{tr}(\tau'_{i,0}) = -\operatorname{tr}(\tau_{i,0})$ is $0 \pmod 3$, we have

$$j(\tau'_{0,i}) = \overline{j(\tau_{0,i})} = j(\tau_{j,0}) \implies \gamma(\tau'_{0,i}) = \overline{\gamma(\tau_{0,i})} = \gamma(\tau_{j,0}),$$

and hence $\{\gamma(\tau_{1,0}), \ldots, \gamma(\tau_{h(D),0})\}$ is stable under complex conjugation. □

Corollary 4.2.1. *Let $\mathcal{O}_D$ be the order of discriminant D in the field $K = \mathbb{Q}(\sqrt{D})$, $3 \nmid D$, and let $H_D(x)$ be the Hilbert class polynomial. The constant term $H_D(0)$ is a cube in $\mathbb{Z}$.*

We can easily recover the set of representative L from T, and then the computation of $G_D(x)$ is analogue to that of $H_D(x)$.

Remark 4.2.10. We note that

$$O(|j(\tau)|) = O(|q|^{-1}),\ O(|\gamma(\tau)|) = O(|q|^{-1/3})$$

therefore

$$\log(|\gamma(\tau)|) \approx \frac{1}{3}\log(|j(\tau)|).$$

It follows that the precision needed to compute $G_D(x)$ and the size of its coefficients are approximately one third of that of $H_D(x)$. Compare, for example, the following:

$$\begin{aligned} H_{-83}(x) = x^3 &+ 2691907584000x^2 - 41490055168000000x + \\ &+ 549755813888000000000 \end{aligned}$$

and

$$G_{-83}(x) = x^3 + 13920x^2 + 128000x + 8192000.$$

Chapter 5

The CM method

The theory of complex multiplication developed in the previous chapters enables us to design an efficient algorithm to construct elliptic curves, defined over some finite fields, with a prescribed number of rational points.

In order to better understand the method, in the first section we will briefly review some basic facts of **elliptic curves over finite fields** and some notation about **ramification of primes** in extensions of number fields. We will then state the **Deuring lifting theorem** which establishes the connection between elliptic curves with complex multiplication over $\mathbb{C}$ and elliptic curves over finite fields.

5.1 Preliminary

5.1.1 Elliptic curves over finite fields

Let E be an elliptic curve defined over a finite field $\mathbb{F}_q$. We will assume that $\operatorname{Char}\mathbb{F}_q \neq 2, 3$, so that E can be defined by an (affine) short Weierstrass equation $y^2 = x^3 + Ax + B$ with $A, B \in \mathbb{F}_q$. Since we are working over a finite field, the set $E(\mathbb{F}_q)$ is finite and its cardinality is controlled by the **Hasse-Weil bound**. For any integer n we define the Hasse-Weil interval

$$\mathcal{H}(n) := [n + 1 - 2\sqrt{n}, n + 1 + 2\sqrt{n}] \cap \mathbb{Z}.$$

Theorem 5.1.1 ([Sil09] Chapter V, Theorem 1.1)**.** *Let E be an elliptic curve defined over the finite field $\mathbb{F}_q$. Then*

$$q + 1 - 2\sqrt{q} \leq |E(\mathbb{F}_q)| \leq q + 1 + 2\sqrt{q}.$$

In other words, $|E(\mathbb{F}_q)| \in \mathcal{H}(q)$.

A significant difference with elliptic curves over $\mathbb{C}$ is that the endomorphism ring of E is always strictly bigger than $\mathbb{Z}$. In fact, we have the so-called q-**Frobenius** $F_q \in \mathrm{End}(E)$ given by

$$\begin{array}{rccc} F_q: & E & \longrightarrow & E \\ & (x,y) & \mapsto & (x^q, y^q). \\ & \infty & \mapsto & \infty \end{array}$$

Therefore $\mathbb{Z}[F_q] \subseteq \mathrm{End}(E)$ and, if equality holds, $\mathrm{End}(E)$ is an order in a quadratic imaginary field. In this case we say that E is *ordinary*; otherwise, $\mathrm{End}(E)$ is an order in a quaternion algebra and we say that E is *supersingular*. In any case, F_q satisfies a quadratic equation

$$F_q^2 - [t]F_q + [q] = 0$$

where $t \in \mathbb{Z}$ is called the **trace** of q-Frobenius and sometimes denoted by $\mathrm{tr}(F_q)$. The trace plays a significant role, in fact we have

$$|E(\mathbb{F}_q)| = q + 1 - t. \tag{5.1}$$

Remark 5.1.1. Equation (5.1) and theorem 5.1.1 imply that $|t| \leq 2\sqrt{q}$.

Remark 5.1.2. If $E : y^2 = x^3 + Ax + B$ is an elliptic curve over $\mathbb{F}_q$ with $\mathrm{tr}(F_{q,E}) = t$ then the curve defined by $E' : y^2 = x^3 + Ad^2x + Bd^3$ where $d \in \mathbb{F}_q^\times$ is any quadratic non-residue is such that $\mathrm{tr}(F_{q,E'}) = -t$. The curve E' is called the **quadratic twist** of E, and $j(E) = j(E')$.

5.1.2 Ramification of primes

In this section we will briefly recall the usual notation in the theory of ramification of primes in extensions of number fields. Given a number field K, we will denote by $\mathcal{O}_K$ its maximal order, also called the ring of integers of K.

Remark 5.1.3. $\mathcal{O}_K$ is a Dedekind domain. In particular, every non-zero fractional ideal I is invertible and can be decomposed into the product of prime ideals:

$$I = \mathfrak{p}_1^{e_1} \dots \mathfrak{p}_r^{e_r},$$

where $\mathfrak{p}_1, \dots, \mathfrak{p}_r \subseteq \mathcal{O}_K$ are prime ideals and $e_1, \dots, e_r \in \mathbb{Z}$. Moreover, the primes $\mathfrak{p}_i$ and the exponents e_i are uniquely determined by I.

Let F/K be an extension of number fields and let $0 \neq \mathfrak{p} \subseteq \mathcal{O}_K$ be a prime ideal. The extension $\mathfrak{p}\mathcal{O}_F$ is a fractional $\mathcal{O}_F$-ideal and hence can be factored as

$$\mathfrak{p}\mathcal{O}_F = \mathfrak{q}_1^{e_1} \dots \mathfrak{q}_r^{e_r}.$$

We will say that $\mathfrak{q}_i$ **lies above** $\mathfrak{p}$ and we will write $\mathfrak{q}_i \mid \mathfrak{p}$. The exponent e_i is called the **ramification index** of $\mathfrak{q}_i$ over $\mathfrak{p}$.

Remark 5.1.4. A prime $\mathfrak{q} \subseteq \mathcal{O}_F$ lies above $\mathfrak{p} \subseteq \mathcal{O}_K$ if and only if $\mathfrak{p} = \mathfrak{q} \cap \mathcal{O}_K$.

Because $\mathcal{O}_K$ is a Dedekind domain, $\mathfrak{p}$ is a maximal ideal and therefore the quotient $\mathcal{O}_K/\mathfrak{p}$ is a finite field of characteristic p, where $\mathfrak{p} \cap \mathbb{Z} = (p)$. It is called the **residue field** at $\mathfrak{p}$ and will be denoted by $\mathbb{F}_\mathfrak{p}$. The inclusion $\mathcal{O}_K \subseteq \mathcal{O}_F$ induces an inclusion $\mathbb{F}_\mathfrak{p} \subseteq \mathbb{F}_{\mathfrak{q}_i}$ between the residue fields for any prime $\mathfrak{q}_i \mid \mathfrak{p}$. The degree $f_i := [\mathbb{F}_{\mathfrak{q}_i} : \mathbb{F}_\mathfrak{p}]$ is called the **inertia degree** of $\mathfrak{q}_i$ over $\mathfrak{p}$.

Remark 5.1.5. If $\mathcal{O}_K/\mathfrak{p} \cong \mathbb{F}_q$ then $\mathcal{O}_F/\mathfrak{q}_i \cong \mathbb{F}_{q^{f_i}}$.

We will say that $\mathfrak{p}$ is **unramified** in $\mathcal{O}_F$ if $e_i = 1\ \forall i$, and we say that $\mathfrak{p}$ **splits completely** if, in addition, $f_i = 1\ \forall i$. If $\mathfrak{p}\mathcal{O}_F = \mathfrak{q}$ is still prime we will say that $\mathfrak{p}$ is **inert**.

5.1.3 Good reduction

So far we have only considered (except for section 2.1) elliptic curves over $\mathbb{C}$ or over number fields. In this section we will see how a suitable equation defining an elliptic curve over a number field can be *reduced* to an elliptic curve over a finite field.

Let L be a number field and $E : y^2 = x^3 + Ax + B$ an elliptic curve defined over L. Up to isomorphism over L, E can be defined by the equation $y^2 = x^3 + d^4Ax + d^6B$ where $d \in L^\times$. Because $L = \mathrm{Frac}(\mathcal{O}_L)$ we can appropriately choose $d \in \mathcal{O}_F$ clearing both the denominators of A and B, yielding an equation with integral coefficients. Therefore we can suppose that E is defined by an equation

$$E : y^2 = x^3 + Ax + B, \text{ with } A, B \in \mathcal{O}_L. \tag{5.2}$$

Let $\mathfrak{p} \subseteq \mathcal{O}_L$ be a prime ideal. The projection map

$$\pi : \mathcal{O}_L \longrightarrow \mathcal{O}_L/\mathfrak{p} = \mathbb{F}_\mathfrak{p}$$

allows us to reduce the equation (5.2) of E, leading to the new curve

$$E_\mathfrak{p} : y^2 = x^3 + \overline{A}x + \overline{B}$$

defined over the finite field $\mathbb{F}_\mathfrak{p}$. To avoid troubles we will always assume that $\mathrm{Char}(\mathbb{F}_\mathfrak{p}) \neq 2, 3$, namely that $\mathfrak{p} \cap \mathbb{Z} \neq (2), (3)$.

Definition 5.1.1. Let $E : y^2 = x^3 + Ax + B$ be an elliptic curve defined over a number field L and let $\mathfrak{p} \subseteq \mathcal{O}_L$ be a prime ideal. We say that E has **good reduction** modulo $\mathfrak{p}$ if there exists a Weierstrass equation $y^2 = x^3 + Ax + B$ for E/L with $A, B \in \mathcal{O}_L$ such that the reduced curve $E_{\mathfrak{p}}/\mathbb{F}_{\mathfrak{p}}$ is still an elliptic curve. If E does not have good reduction modulo $\mathfrak{p}$ we say that E has **bad reduction** modulo $\mathfrak{p}$.

Remark 5.1.6. Recall that a Weierstrass equation defines an elliptic curve E if and only if the discriminant $\Delta(E)$ is nonzero. Therefore $E_{\mathfrak{p}}$ as above defines an elliptic curve if and only if

$$\Delta(E_{\mathfrak{p}}) \not\equiv 0 \pmod{\mathfrak{p}} \iff \mathfrak{p} \nmid (\Delta(E)).$$

It follows that there exists only a finite number of primes $\mathfrak{p}$ of $\mathcal{O}_L$ for which E has bed reduction.

Example 16. Consider the elliptic curve $E : y^2 = x^3 + 5^6$ defined over $\mathbb{Q}$ and the prime $p = 5$. If we reduce the equation defining E modulo p we get the singular curve $E_p : y^2 = x^3$. On the other hand, through the isomorphism $(x, y) \mapsto (5^2x, 5^3y)$, E can be defined by the equation $y^2 = x^3 + 1$ whose reduction modulo p yields to an elliptic curve E_p. Therefore, according to our definition, E has good reduction modulo p.

Example 17. Consider the curves $E : y^2 = x^3 + 1$ and $E' : y^2 = x^3 + 5$ over $\mathbb{Q}$. Of course $j(E) = j(E') = 0$, on the other hand the first has good reduction modulo $p = 5$ but E' does not. To see this, we note that the discriminant of $y^2 = x^3 + d^6 5$ is $\Delta = d^{12}\Delta(E') = 27d^{12}5^2$ where $d \in \mathbb{Q}^\times$, therefore

$$v_5(\Delta') = 12v_5(d) + 2 = 0 \implies v_5(d) = \frac{1}{6}$$

but this can never be for $d \in \mathbb{Q}^\times$. Note that both E and E' have good reduction over $\mathbb{Q}(\sqrt[6]{5})$.

Remark 5.1.7. The example above generalizes as follows. Let $E : y^2 = x^3 + Ax + B$ defined over L and $\mathfrak{p} \subseteq \mathcal{O}_L$ a prime ideal. E is isomorphic over L to $E' : y^2 = x^3 + d^4A + d^6B$ for any $d \in L^\times$ and $\Delta(E') = d^{12}\Delta(E)$. Therefore

$$0 < v_{\mathfrak{p}}(\Delta(E)) < 12 \implies E \text{ has bad reduction modulo } \mathfrak{p}.$$

Definition 5.1.2. Let E/L be an elliptic curve defined over a number field L and let $\mathfrak{p} \subseteq \mathcal{O}_L$ be a prime ideal. We say that E/L has **potential good reduction** modulo $\mathfrak{p}$ if there exists a finite extension M/L such that E/M has good reduction modulo every prime $\mathfrak{q} \subseteq \mathcal{O}_M$ over $\mathfrak{p}$.

Example 18. The curve $E : y^2 = x^3 + 5$ over $\mathbb{Q}$ has potential good reduction modulo $p = 5$. Consider the field $K = \mathbb{Q}(\sqrt[6]{5})$, we have that p is totally ramified in $\mathcal{O}_K$

$$p\mathcal{O}_K = (5) = (\sqrt[6]{5})^6 = \mathfrak{q}^6.$$

Let $d = \sqrt[6]{5} \in \mathcal{O}_K$, then E is isomorphic over K to

$$E' : y^2 = x^3 + d^{-6}5 = x^3 + 1$$

which has good reduction modulo $\mathfrak{q}$.

Theorem 5.1.2. *The elliptic curve E/L has potential good reduction modulo $\mathfrak{p}$ if and only if $v_{\mathfrak{p}}(j(E)) \geq 0$.*

Proof. We will prove the theorem only for those primes $\mathfrak{p}$ with $\mathfrak{p} \cap \mathbb{Z} \neq (2), (3)$. For the general case we refer to [Tat75].

($\Longrightarrow$) Let M/L where E has good reduction. For each prime $\mathfrak{q} \mid \mathfrak{p}$ we fix an equation $E(\mathfrak{q}) : y^2 = x^3 + A_{\mathfrak{q}}x + B_{\mathfrak{q}}$ with good reduction modulo $\mathfrak{q}$, therefore, for every prime $\mathfrak{q} \mid \mathfrak{p}$

$$j(E) = j(E(\mathfrak{q})) = 1728\frac{4A_{\mathfrak{q}}^3}{\Delta(E(\mathfrak{q}))}$$

with $v_{\mathfrak{q}}(\Delta(E(\mathfrak{q}))) = 0,\ A_{\mathfrak{q}} \in \mathcal{O}_M$. It follows that

$$v_{\mathfrak{q}}(j(E)) = 3v_{\mathfrak{q}}(A_{\mathfrak{q}}) - v_{\mathfrak{q}}(\Delta(E(\mathfrak{q}))) = 3v_{\mathfrak{q}}(A_{\mathfrak{q}}) \geq 0.$$

Since this holds for any prime $\mathfrak{q} \mid \mathfrak{p}$ it follows that also $v_{\mathfrak{p}}(j(E)) \geq 0$.

($\Longleftarrow$) Suppose E has bad reduction modulo $\mathfrak{p}$ (otherwise the claim is trivial). Let us fix an equation $E : y^2 = x^2 + Ax + B$ over L with $A, B \in \mathcal{O}_L$. Let $e = v_{\mathfrak{p}}(\Delta(E)) > 0$ and $\pi \in \mathfrak{p} \setminus \mathfrak{p}^2$ an element of valuation $v_{\mathfrak{p}}(\pi) = 1$. Define $M = L(\pi^{1/r})$, where $r = 12/\gcd(e, 12)$. Then $[M : L] = r$ and, by the theorem of Ore, $\mathfrak{p}$ is totally ramified in $\mathcal{O}_M$:

$$\mathfrak{p}\mathcal{O}_M = \mathfrak{q}^r.$$

Let $\mu \in \mathcal{O}_M$ be such that

$$v_{\mathfrak{q}}(\mu) = \frac{1}{12}v_{\mathfrak{q}}(\Delta(E)),$$

for example we can take[1] $\mu = \pi^{a/r}$ where $a = e/\gcd(e, 12)$. Consider the curve

$$E' : y^2 = x^3 + \mu^{-4}Ax + \mu^{-6}B, \tag{5.3}$$

[1] In fact

$$v_{\mathfrak{q}}(\mu) = \frac{a}{r}v_{\mathfrak{q}}(\pi) = \frac{v_{\mathfrak{p}}(\Delta(E))}{12}v_{\mathfrak{q}}(\pi) = \frac{1}{12}v_{\mathfrak{q}}(\Delta(E)).$$

then

$$\Delta(E') = \frac{1}{\mu^{12}}\Delta(E) \implies v_{\mathfrak{q}}(\Delta(E')) = 0.$$

The coefficients of equation (5.3) have zero $\mathfrak{q}$-adic valuation. By assumption $v_{\mathfrak{p}}(j(E)) \geq 0$, hence

$$v_{\mathfrak{q}}(j(E)) = v_{\mathfrak{q}}\left(1728\frac{4A^3}{\Delta(E)}\right) \geq 0$$

and then, since the characteristic is not 2 or 3, $v_{\mathfrak{q}}(A^3) \geq v_{\mathfrak{q}}(\Delta(E))$. Therefore

$$\begin{aligned} v_{\mathfrak{q}}(\mu^{-4}A) \geq 0 &\iff v_{\mathfrak{q}}((\mu^{-4}A)^3) \geq 0 \\ &\iff v_{\mathfrak{q}}(A^3) - v_{\mathfrak{q}}(\mu^{12}) = v_{\mathfrak{q}}(A^3) - v_{\mathfrak{q}}(\Delta(E)) \geq 0 \\ &\iff v_{\mathfrak{q}}(A^3) \geq v_{\mathfrak{q}}(\Delta(E)). \end{aligned}$$

For the coefficient $\mu^{-6}B$ we distinguish two cases. If $v_{\mathfrak{q}}(B^2) \geq v_{\mathfrak{q}}(A^3)$ we can do the same as before because $v_{\mathfrak{q}}((\mu^{-6}B)^2) \geq v_{\mathfrak{q}}(A^3) - v_{\mathfrak{q}}(\mu^{12}) \geq 0$. Otherwise $v_{\mathfrak{q}}(B^2) < v_{\mathfrak{q}}(A^3)$ and in this case

$$v_{\mathfrak{q}}(\lambda^{12}) = v_{\mathfrak{q}}(\Delta(E)) = v_{\mathfrak{q}}(4A^3 + 27B^2) = \min(v_{\mathfrak{q}}(A^3), v_{\mathfrak{q}}(B^2)) = v_{\mathfrak{q}}(B^2)$$

and hence $v_{\mathfrak{q}}(\mu^{-6}B) = 0$.

Now, if some denominators arise in the equation (5.3), they would have zero $\mathfrak{q}$-adic valuation and hence we can clear them with a change of variables without changing the $\mathfrak{q}$-adic valuation of the discriminant. Therefore E' can be defined over M by an equation with integral coefficients with good reduction modulo $\mathfrak{q}$, and then E/L has potential good reduction modulo $\mathfrak{p}$. □

Corollary 5.1.1. *Let E/F be an elliptic curve with complex multiplication defined over a number field F. Then E has potential good reduction modulo any prime $\mathfrak{p}$ of $\mathcal{O}_F$.*

Proof. By theorem 3.3.2 $j(E)$ is an algebraic integer, hence $j(E) \in \mathcal{O}_F$ and therefore $v_{\mathfrak{p}}(j(E)) \geq 0$ for any prime $\mathfrak{p} \subseteq \mathcal{O}_F$. □

We conclude with an important result due to Deuring.

Theorem 5.1.3 (Deuring)**.** *Let E/F be an elliptic curve defined over a number field with complex multiplication by the order $\mathcal{O}_D$ in the imaginary quadratic field $K = \mathbb{Q}(\sqrt{D})$. Let $p \nmid D$ be a prime number which splits completely in F and $\mathfrak{p}$ be a prime of $\mathcal{O}_F$ above p where E/F has good reduction. Then there exists an isomorphism*

$$\mathrm{End}(E) \xrightarrow{\sim} \mathrm{End}(E_{\mathfrak{p}})$$

and every elliptic curve over $\mathbb{F}_p$ with endomorphism ring isomorphic to $\mathcal{O}_D$ arises in this way. If $\pi \in \mathcal{O}_D \cong \mathrm{End}(E)$ is the element which corresponds to the Frobenius $F_p \in \mathrm{End}(E_\mathfrak{p})$ through the isomorphism above then

$$|E_\mathfrak{p}(\mathbb{F}_p)| = p + 1 - (\pi + \overline{\pi}).$$

Proof. For the first part of the theorem see [Lan87], Chapter 13, Theorems 12, 13 and 14.

Since p splits completely in F, $\mathbb{F}_\mathfrak{p} \cong \mathbb{F}_p$ and thus $E_\mathfrak{p}$ is defined over $\mathbb{F}_p$. Let $\pi \in \mathcal{O}_D \cong \mathrm{End}(E)$ be the element which corresponds, through the isomorphism $\mathrm{End}(E) \xrightarrow{\sim} \mathrm{End}(E_\mathfrak{p})$, to the Frobenius endomorphism F_p. Because the isomorphism preserves the degree and, by remark 3.3.1, for any $\alpha \in \mathcal{O}_D \cong \mathrm{End}(E)$ we have $\deg(\alpha) = |\,\mathrm{N}_{K/\mathbb{Q}}(\alpha)|$, we have

$$p = \deg(F_p) = \deg(\pi) = \mathrm{N}_{K/\mathbb{Q}}(\pi) = \pi\overline{\pi}.$$

On the other hand we have

$$\begin{aligned} |E_\mathfrak{q}(\mathbb{F}_p)| &= \deg(1 - F_p) = \deg(1 - \pi) = \mathrm{N}_{K/\mathbb{Q}}(1 - \pi) \\ &= (1 - \pi)(1 - \overline{p}) = 1 - \overline{\pi} - \pi + \pi\overline{\pi} \\ &= p + 1 - (\pi + \overline{\pi}). \end{aligned}$$

□

Remark 5.1.8. By the proof of the theorem $\pi \in \mathcal{O}_D$ has norm equal to p and, since p is prime, it is uniquely determined up to complex conjugation and multiplication by elements of $\mathcal{O}_D^\times$. The equation

$$p = \mathrm{N}_{K/\mathbb{Q}}(\pi),$$

plays a crucial role, and will be studied in the next section.

5.1.4 The norm equation

Let $\mathcal{O}_D$ be the order of discriminant D in the the imaginary quadratic field $K = \mathbb{Q}(\sqrt{D})$ and let $p \in \mathbb{Z}$ be a prime. Suppose that there exists $\pi \in \mathcal{O}_D$ of norm p, that is

$$\mathrm{N}_{K/\mathbb{Q}}(\pi) = \pi\overline{\pi} = p.$$

Then π satisfies the quadratic equation

$$x^2 - tx + p = 0 \tag{5.4}$$

where $t = \pi + \overline{\pi} \in \mathbb{Z}$ is the trace. The solutions of (5.4) are

$$x_{1,2} = \frac{t \pm \sqrt{t^2 - 4p}}{2} \implies 2x_{1,2} = t \pm \sqrt{t^2 - 4p}.$$

Taking the norm on both sides we get

$$4p = 4\,\mathrm{N}_{K/\mathbb{Q}}(\pi) = \mathrm{N}_{K/\mathbb{Q}}(2\pi) = (t + \sqrt{t^2 - 4p})(t - \sqrt{t^2 - 4p}),$$

but $t^2 - 4p = \mathrm{disc}_{K/\mathbb{Q}}(\pi) = v^2 D$ for some positive integer v, therefore

$$\begin{aligned} 4p &= (t + v\sqrt{D})(t - v\sqrt{D}) \\ &= t^2 - v^2 D. \end{aligned}$$

Definition 5.1.3. The equation

$$4p = t^2 - v^2 D \tag{5.5}$$

is called the **norm equation**.

Let F be the ring class field of $\mathcal{O}_D$. As we mentioned in chapter 3, the ring class field of $\mathcal{O}_D$ is the maximal abelian extension of K unramified at any prime $p \nmid D$.

The following theorem is crucial for our algorithm.

Theorem 5.1.4. *Let $\mathcal{O}_D$ be the order of discriminant D in the imaginary quadratic field $K = \mathbb{Q}(\sqrt{D})$, let $p \nmid D$ be an odd prime and let F be the ring class field of $\mathcal{O}_D$. The following are equivalent:*

(i) p is the norm of an element $\pi \in \mathcal{O}_D$;

(ii) $4p = t^2 - v^2 D$ for some integers t and v, and $t \not\equiv 0 \pmod{p}$;

(iii) p splits completely in F.

Proof. $(i) \implies (ii)$. If $\pi \in \mathcal{O}_D$ is an element of norm p then, by the above discussion, there exists integers t, v such that $4p = t^2 - v^2 D$. Now suppose, by contradiction, that $t \equiv 0 \pmod{p}$, then equation $4p = t^2 - v^2 D$ implies $p \mid v^2 D$. By assumption we have $p \nmid D$, hence $p \mid v$ and then $p^2 \mid t^2 - v^2 D = 4p$, which is a contradiction because p is an odd prime.

$(ii) \implies (i)$. Let $\pi \in \mathcal{O}_K$ be a solution of the equation $x^2 - tx + p = 0$. Therefore we have $\mathrm{N}_{K/\mathbb{Q}}(\pi) = \pi\overline{\pi} = p$ and $\mathrm{disc}_{K/\mathbb{Q}}(\pi) = t^2 - 4p = v^2 D$, hence $\pi \in \mathcal{O}_{v^2 D} \subseteq \mathcal{O}_D$ (see remark 3.1.4).

For equivalence $(ii) \iff (iii)$ we refer to the proof of Theorems 9.4 in [Cox13], Chapter two. $\square$

Theorems 5.1.4 and 5.1.3 lead to the following corollary.

Corollary 5.1.2. *Let $D < -4$ be a discriminant and let $p > 3$ be a prime number such that $p \nmid D$ and*

$$4p = t^2 - v^2 D$$

for some integers t and v. Let E be an elliptic curve with complex multiplication by $\mathcal{O}_D$ defined over the ring class field F and let $\mathfrak{p}$ be a prime ideal of $\mathcal{O}_F$ over p. Then there exists an elliptic curve $\overline{E}$ defined over $\mathbb{F}_p \cong \mathbb{F}_{\mathfrak{p}}$ such that

1. $j(\overline{E}) \equiv j(E) \pmod{\mathfrak{p}}$;
2. $\mathrm{End}(\overline{E}) \cong \mathcal{O}_D$;
3. $|\overline{E}(\mathbb{F}_p)| = p + 1 \pm t$.

Proof. Let E/F be an elliptic curve with complex multiplication by $\mathcal{O}_D$. If E has good reduction modulo modulo $\mathfrak{p}$ we take $\overline{E} = E_{\mathfrak{p}}$. Otherwise E has bad reduction modulo $\mathfrak{p}$, and we let $L = F(\pi^{1/r})$ where $\pi \in \mathfrak{p} \setminus \mathfrak{p}^2$. According to the proof of theorem 5.1.2 the prime $\mathfrak{p}$ is totally ramified in L, $\mathfrak{p}\mathcal{O}_L = \mathfrak{q}^r$, and E has good reduction modulo $\mathfrak{q}$. We take $\overline{E} = E_{\mathfrak{q}}$ which is defined over $\mathbb{F}_{\mathfrak{q}} \cong \mathbb{F}_{\mathfrak{p}}$ since $\mathfrak{p}$ is totally ramified. By equivalence $(ii) \iff (iii)$ the prime p splits completely in F, therefore, $\mathbb{F}_{\mathfrak{p}} \cong \mathbb{F}_p$ and $\overline{E}$ is defined over $\mathbb{F}_p$. Therefore we have $j(\overline{E}) \equiv j(E) \pmod{\mathfrak{p}}$ and, by theorem 5.1.3, $\mathrm{End}(\overline{E}) \cong \mathrm{End}(E) \cong \mathcal{O}_D$ and $|\overline{E}(\mathbb{F}_p)| = p+1-(\pi+\overline{\pi})$ where $\pi \in \mathcal{O}_D$ corresponds to the Frobenius F_p. Because $D < -4$ it follows that $\mathcal{O}_D^\times = \{\pm 1\}$, therefore by the equivalence $(i) \iff (ii)$ and remark 5.1.8 we have $\pi + \overline{\pi} = \pm t$ and

$$|\overline{E}(\mathbb{F}_p)| = p + 1 - (\pi + \overline{\pi}) = p + 1 \pm t.$$

□

Corollary 5.1.3. *Let $p \geq 5$ be a prime number. For all $N \in \mathcal{H}(p)$ there exists an elliptic curve E defined over $\mathbb{F}_p$ with $|E(\mathbb{F}_p)| = N$.*

Proof. Let $E/\mathbb{F}_p$ be an elliptic curve, by theorem 5.1.1 we have $|E(\mathbb{F}_p)| = p + 1 - t$ where $t = \mathrm{tr}(F_q)$ satisfies $|t| \leq 2\sqrt{p}$. We want to prove that for all $t \in [-2\sqrt{p}, 2\sqrt{p}] \cap \mathbb{Z}$ there exists an elliptic curve $E/\mathbb{F}_p$ such that $\mathrm{tr}(F_p) = t$. We distinguish two cases. If $t \neq 0$, then $t \not\equiv 0 \pmod{p}$. We define $D = t^2 - 4p$, therefore $p \nmid D$ and, by construction, $4p = t^2 - D$. By corollary 5.1.2 any $j(E) \in \mathcal{ELL}(\mathcal{O}_D)$ reduces modulo p to the j-invariant of an elliptic curve $E'/\mathbb{F}_p$ with $\mathrm{tr}(E') = \pm t$. Therefore, up to replacing E' with its quadratic twist, we get the desired curve.

If $t = 0$ then $|E(\mathbb{F}_p)| = p + 1$ is supersingular, hence we cannot construct it using the corollary. It can be shown that if $p \equiv 3 \pmod 4$ the curve $E : y^2 = x^3 + x$ over $\mathbb{F}_p$ is supersingular, and the same holds for the curve $E : y^2 = x^3 + 1$ when $p \equiv 2 \pmod 3$. For the remaining cases we refer to [Br09]. □

Remark 5.1.9. For $p = 2, 3$ corollary 5.1.3 is still true and examples can be constructed by hand.

5.2 The method

Corollary 5.1.2 allows us to design an efficient algorithm to construct elliptic curves with a desired number of points over some finite field. Before delving into the details, let us present the main steps of the algorithm. Given a positive integer N in input, together with its factorization into primes, the following procedure returns a prime number p and an elliptic curve $E/\mathbb{F}_p$ with $|E(\mathbb{F}_p)| = N$.

(1) Find a discriminant D and a solution (x, v) to $4N = x^2 - v^2 D$ such that $p = N - 1 + t$ is prime, where $t = x + 2$.

(2) Compute the Hilbert class polynomial $H_D(x)$, or a more convenient Weber class polynomial, and find a root j_0 modulo p.

(3) Write down (an equation of) an elliptic curve $E/\mathbb{F}_p$ with $j(E) = j_0$. Then $|E(\mathbb{F}_p)| = p + 1 \pm t$, and therefore E or its quadratic twist E' has the desired cardinality.

5.2.1 Cornacchia's algorithm

The first step of the algorithm is to find the smallest (in absolute value) discriminant $D < -4$ such that a solution $(x, v) \in \mathbb{Z}^2$ to the equation

$$4N = x^2 - v^2 D \tag{5.6}$$

exists. We can efficiently compute a solution of (5.6) (if one exists) using the method of Cornacchia, which essentially computes the continued fraction expansion of the square root of a given rational. The algorithm is given below (see [BW05] for more details).

Remark 5.2.1. In order to solve the congruence $a^2 \equiv D \pmod N$ in algorithm 3, that is, to find a square root of D modulo N, we need to know the factorization of N. In fact for every prime $q \mid N$ we can efficiently find a

Algorithm 3 Cornacchia's algorithm

Require: A positive integer N together with its factorization and a discriminant $D < -4$
Ensure: A solution (x, v) to $4N = x^2 - v^2 D$, if one exists

```
r_0 ← 4N
r_1 ← a solution of a^2 ≡ D (mod 4N) with 0 ≤ r_1 ≤ 2N
s ← r_1^2
while s ≥ 4N do
    t ← r_1
    r_1 ← r_0 (mod r_1)
    r_0 ← t
    s ← r_1^2
end while
x ← r_1
v ← √((4N − s)/|D|)
if v ∈ Z then
    return (x, v)
else
    return No solutions
end if
```

square root of D modulo q using the Tonelli-Shanks algorithm (see [Coh93], Chapter 1), then we lift the solution modulo q^e, the maximal power of q dividing N, by Hensel's lemma, and finally we recover the solution modulo N with the Chinese remainder theorem.

Then, given a solution (x, y) to (5.6), we set $t := x + 2$ and $p := N - 1 + t$ and check whether p is prime or not (for example with some probabilistic primality test, such as the Miller-Rabin test, see [Kob94], Chapter 5). If p is not prime, or no solution to (5.6) exists, we try with another discriminant D.

Remark 5.2.2. Once we have $p = N - 1 + t$ prime, then

$$\begin{aligned} 4p &= 4N - 4 + 4t = x^2 - v^2 D - 4 + 4t \\ &= x^2 - v^2 D - 4 + 4x + 8 = (x + 2)^2 - v^2 D \\ &= t^2 - v^2 D, \end{aligned}$$

that is, the norm equation. Therefore, provided p is an odd prime not dividing D, we can use corollary 5.1.2 to obtain an elliptic curve over $\mathbb{F}_p$ with $p + 1 \pm t$ points.

5.2.2 Roots modulo p

The j-invariants of elliptic curves over $\mathbb{C}$ with complex multiplication by $\mathcal{O}_D$ are precisely the roots of the Hilbert class polynomial:

$$H_D(x) = \prod_{j(E) \in \mathcal{ELL}(\mathcal{O}_D)} (x - j(E)).$$

By corollary 5.1.2 every $j(E) \in \mathcal{ELL}(\mathcal{O}_D)$ is the j-invariant of an elliptic curve with good reduction modulo $\mathfrak{q}$ for any prime $\mathfrak{q}$ above p and $j(E_{\mathfrak{q}}) = \overline{j(E)}$.

Therefore the next step will be to recover a root of $H_D(x)$ modulo p. Once we have computed the polynomial, this can be efficiently done in at least two ways.

1. We can (completely) factor $H_D(x)$ in $\mathbb{F}_p[x]$ with Berlekamp's algorithm (see for example [Coh93], Chapter 3) and then get any root modulo p.

2. We can recover a root of $H_D(x)$ modulo p using a randomized root-finding algorithm (see for example [GvdHL15]).

The first method is more expensive than the second, but on the other hand it allows us to construct more than one curve at the same time.

Remark 5.2.3. This step is quite expensive since it requires the computation of the Hilbert class polynomial. The algorithms to compute $H_D(x)$ have an expected running time of $O(|D|^{1+\epsilon})$ for any $\epsilon > 0$. In order to reduce the computational cost we can use other class polynomials with smaller height. If $3 \nmid D$ we can use, for example, $G_D(x)$ which requires one third of the precision involved in the computation of $H_D(x)$. In the same way we can compute a root γ_0 of $G_D(x)$ modulo p and recover the j-invariant as $j_0 = \gamma_0^3$.

Remark 5.2.4. A little trick that could save more time is to consider the maximal order $\mathcal{O}_K$ instead of $\mathcal{O}_D$. If $D = f^2 d_K$, where f is the conductor and d_K is the fundamental discriminant, we have

$$4N = x^2 - v^2 D = x^2 - (vf)^2 d_K.$$

Thus working with $\mathcal{O}_K$ does not alter x, and therefore t. This is particularly useful when the conductor f is big. Note that we have assumed $D < -4$, hence if after the simplification we come up with the discriminants -3 and -4 we have to use the original D.

5.2.3 Recovering the correct curve

From the last step we have recovered a j-invariant $j_0 = j(E) \in \mathbb{F}_p$ of an elliptic curve $E/\mathbb{F}_p$ with $|E(\mathbb{F}_p)| = p + 1 \pm t$. Provided $j_0 \neq 0, 1728$, we can take $E : y^2 = x^3 + Ax + B$ with $A = 3j_0(1728 - j_0)$ and $B = 2j_0(1728 - j_0)^2$.

On the other hand we have $p = N - 1 + t \implies N = p + 1 - t$, therefore one among E and its quadratic twist E' has the desired number of rational points. Checking which one has the correct order is then done by choosing any random point $P \in E(\mathbb{F}_p)$ and checking whether $[p + 1 - t]P = \infty$ or $[p + 1 + t]P = \infty$ holds (at least one of these must hold). If only one of the equalities holds then the exact value of the trace is determined, otherwise we try with a different point. For much more efficient methods to detect the correct curve see [RS10].

Remark 5.2.5. If $j_0 = 0, 1728$ additional cubic, quartic or sextic twists have to be taken into account. Since we have considered only $D < -4$ these cases can be ignored.

Remark 5.2.6. The proof of corollary 5.1.3 can be turned into an algorithm to construct ordinary elliptic curves E over $\mathbb{F}_p$, where p and the cardinality of E are given. Unfortunately, this algorithm is far from being efficient, its complexity is, indeed, $O(N)$ since $D = t^2 - 4p = (1 + p - N)^2 - 4p$ has, typically, the same magnitude as N.

5.2.4 Analysis of the cost

We summarize what we have said so far in algorithm 4.

We note that if q is any number then

$$N \in \mathcal{H}(q) \iff q \in \mathcal{H}(N).$$

Therefore algorithm 4 will in fact return the desired elliptic curve if there exists a prime $p \in \mathcal{H}(N)$. We do not have a rigorous proof of this fact, but in practice there is always an abundance of primes p for which there exists an elliptic curve $E/\mathbb{F}_p$ of order N. The run time analysis is based on *heuristic assumptions*: a very detailed analysis can be found in [BS07] or in [BS08].

It can be shown that the discriminant D returned after the first while loop is of size

$$D = O((\log N)^2 + 2^{\omega(N)})$$

and it can be found, together with a solution (x, v) of equation (5.6), in an expected time of $O(2^{\omega(N)}(\log N)^{4+\epsilon})$. Here we denoted by $\omega(N)$ the number of distinct prime factors of N. Then the computation of $H_D(x)$ requires $O(|D|^{1+\epsilon}) = O(((\log N)^2 + 2^{\omega(N)})^{1+\epsilon})$ operations and finding a root of $H_D(x)$

Algorithm 4 CM method

Require: A positive integer N together with its factorization
Ensure: A prime number p and an elliptic curve $E/\mathbb{F}_p$ such that $|E(\mathbb{F}_p)| = N$

```
D ← −5
p ← 1
while p is not prime do
    D ← next_discriminant(D)
    if there exists a solution to 4N = x^2 − v^2 D then
        (x, v) ← cornacchia(D, 4 * N)
        t ← x + 2
        p ← N − 1 + t
    end if
end while
D ← fundamental_discriminant(D)
H(x) ← HilbertClassPolynomial(D)
H_p ← H(x) (mod p)
j_0 ← get_root(H_p)
A ← 3j_0(1728 − j_0) (mod p)
B ← 2j_0(1728 − j_0)^2 (mod p)
E ← EllipticCurve(A, B, p)
Q ← ∞, R ← ∞
while Q = ∞ and R = ∞ do
    P ← RandomPoint(E)
    Q ← [p + 1 − t]P, R ← [p + 1 + t]P
end while
if Q = ∞ then
    return E
else
    return QuadraticTwist(E)
end if
```

modulo p requires $O(\deg(H_D(x))(\log p)^2) = O((\log N)^3)$ operations. The rest of the algorithm requires very few operations, therefore the entire cost of algorithm 4 is

$$O(2^{\omega(N)}(\log N)^{4+\epsilon}).$$

Remark 5.2.7. The entire cost of the algorithm is not exactly polynomial in the number of digits of N. On the other hand, for the cryptographic application, N should be a huge prime number or at least a product of few prime numbers. Therefore, in practice, the algorithm is indeed polynomial in $\log N$.

5.2.5 Examples

We close with some explicit examples following algorithm 4 step-by-step.

Example 19. We want to find an elliptic curve of order 2022.

1. Starting with $D = -5$ we use Cornacchia's algorithm to find a solution to $4N = x^2 - v^2 D$ such that $p = N - 1 + t$ is prime (recall that $t = x + 2$). A solution is $D = -344,\ p = 2111$ and $t = x + 2 = 90$.

2. $D = -2^3 \cdot 43$ is a fundamental discriminant, then we compute $H_D(x)$ modulo p:

$$\begin{aligned} H_D(x) = {} & x^{10} + 524x^9 + 1498x^8 + 1431x^7 + 725x^6 + 1363x^5 + 1910x^4 + \\ & + 1858x^3 + 1568x^2 + 270x + 1249 \end{aligned}$$

   and we find that $j_0 = 2098$ is a root of $H_D(x)$ modulo p.

3. We define $E : y^2 = x^3 + Ax + B$, over $\mathbb{F}_p$, where $A = 3j_0(1728 - j_0) = 1764$ and $B = 2j_0(1728 - j_0)^2 = 1857$, therefore $|E(\mathbb{F}_p)| = p + 1 \pm t$. In order to determine the sign of the trace of p-Frobenius we determine a random point $P = (635, 219) \in E(\mathbb{F}_p)$ and compute

$$Q = [p + 1 - t]P = [2022]P = (843, 1480)$$

   and

$$R = [p + 1 + t]P = [2202]P = \infty.$$

   Since $Q \neq \infty$ and $R = \infty$ it follows that the quadratic twist $E' : y^2 = x^3 + 1202x + 1918$ is the desired curve.

We note that $3 \nmid D$, hence, in order to speed up the computation, we can replace step 2, with the following:

2bis. Compute the class polynomial $G_D(x)$ modulo p:

$$G_D(x) = x^{10} + 942x^9 + 1976x^8 + 664x^7 + 586x^6 + 1945x^5 + 1543x^4 + \\ + 1790x^3 + 1158x^2 + 1954x + 662$$

then find a root $\gamma_0 = 1805$ and set $j_0 = \gamma_0^3$. Comparing the polynomials $H_D(x)$ and $G_D(x)$, we have that the largest coefficient of the former has 80 decimal digits while for the latter it has $27 \approx 80/3$.

Example 20. Now we wish to construct an elliptic curve of order

$$N = \texttt{next_prime}(10^{30}) = 1000000000000000000000000000057.$$

1. A suitable solution to the norm equation is

   $D = -1243$, $p = 1000000000000000843585789526627$ and $t = 843585789526571$.

2. D is a fundamental discriminant and $3 \nmid D$, hence, after computing $H_D(x)$ or $G_D(x)$ (which we do not display because of their magnitude), we find

   $$j_0 = 227591831816459974952384260327.$$

3. We set $E : y^2 = x^3 + Ax + B$ where

   $$A = 896498963166364177596790241008$$

   and

   $$B = 651010137682304974324561040374.$$

   We then pick a random point $P = (x_P, y_P)$ where

   $$x_P = 383491826115538075114036307473$$

   and

   $$y_P = 895324831600199993461923726194$$

   and we find that $[p + 1 - t]P \neq \infty$ and $[p + 1 + t]P = \infty$. Therefore the quadratic twist $E' : y^2 = x^3 + A'x + B'$ where

   $$A' = 591632761323954608541573771316$$

   and

   $$B' = 654843311949341113186202658729$$

   has the desired cardinality.

Appendix A

Implementations

A working implementation of the CM method is given. The algorithms that follows were written in the SageMath language. They are intended to produce some explicit examples of what we established in the thesis and they do not claim to be efficient.

A.1 Class polynomials

The following functions returns the Hilbert class polynomials and the Weber polynomials for the class invariant γ.

A.1.1 Modular forms

```
# THESE FUNCTIONS EVALUATE SOME MODULAR FORMS
# prec denote the complex floating point precision
# r denote the truncation number of the series

# 1. eta function
def eta(tau, r, prec):
    R = RealField(prec)
    C = R.algebraic_closure()
    q = exp(2*pi*I*tau)
    e = 1
    for i in range(r):
        k = i + 1
        e += ((-1)**k)*(q**(k*(3*k+1)/2) + q**(k*(3*k-1)/2))
    return C(q**(1/24) * e)

# 2. j function
def j_inv(tau, r, prec):
    R = RealField(prec)
```

```
    C = R.algebraic_closure()
    q = exp(2 * pi * I * tau)
    a = eta(tau, r, prec)
    b = eta(2 * tau, r, prec)
    return ((a / b)**8 + (2**8) * (b / a)**16)**3

# 3. gamma function
def gamma(tau, r, prec):
    R = RealField(prec)
    C = R.algebraic_closure()
    q = exp(2 * pi * I * tau)
    a = eta(tau, r, prec)
    b = eta(2 * tau, r, prec)
    return (a / b)**8 + (2**8) * (b / a)**16
```

A.1.2 Representatives

```
# THESE FUNCTIONS PROVIDES VARIOUS SETS OF REPRESENTATIVES
#
# This function compute the set T of representatives for ideal classes for
# the order of discriminant D in K = Q(sqrt(D))
# INPUT: a discriminant D<0
# OUTPUT: the set of representatives T
def representatives(D):
    a = 1
    if D == -3:
        return []
    T = []
    while 3*(a**2) <= abs(D):
        for b in range(-a,a+1):
            if mod(b**2 - D, 4*a) == 0:
                c = (b**2 - D)/(4*a)
                if a <= c and gcd(a,gcd(b,c)) == 1:
                    if a == abs(b) or a == c:
                        if b >= 0:
                            T.append((-b + sqrt(D))/(2*a))
                    else:
                        T.append((-b + sqrt(D))/(2*a))
        a = a + 1
    return T

# This function computes the minimal polynomial of any imaginary quadratic tau
# INPUT: tau
# OUTPUT: [A, B, C] such that A*tau**2 + B*tau * C = 0, gcd(A, B, C) = 1
def min_poli(tau):
    t = 2*real(tau)
```

```
    N = real(tau)**2 + imag(tau)**2
    m = lcm(denominator(t),denominator(N))
    A = m
    B = t*m
    C = N*m
    return [A, -B, C]


# INPUT: tau in H
# OUTPUT: tau* in H such that j(tau) = gamma(tau*)
def find_rep(tau):
    list = min_poli(tau)
    A = ZZ(list[0])
    B = ZZ(list[1])
    C = ZZ(list[2])
    # Forcing A to be not divisible by 3
    if mod(A, 3) == 0:
        if mod(C, 3) == 0:
            A = A + B + C
            B = 2*C + B
        else:
            z = A
            A = C
            B = -B
            C = z
    # find the right tau: A != 0 (mod 3), B = 0 (mod 3), C !=
    0 (mod 3)
    while mod(B, 3) != 0:
        C = A - B + C
        B = B - 2*A
    return A, B, C

# This function computes the set L
# INPUT: a discriminant D<0
# OUTPUT: 'ERROR' if 3|D, L otherwise
def gamma_rep(D):
    L = []
    if mod(D, 3) == 0:
        return 'ERROR'
    list = representatives(D)
    for tau in list:
        A, B, C = find_rep(tau)
        tau = -B/(2*A) + sqrt(B**2 - 4*A*C)/(2*A)
        L.append(tau)
    return L
```

A.1.3 Hilbert and Weber class polynomials

```
# THE HILBERT AND WEBER-GAMMA POLYNOMIALS
```

```

R.<x> = ZZ[x]

# Parameters:
# INPUT: a discriminant D and a marker i
# OUTPUT: the required precision for computing H_D(x) if i = 0, or G_D(x) if i = 1
def precision(D, i):
    if i == 0:
        prec = ceil(pi*sqrt(-D)*log(-D)**2)
    else:
        prec = 100 + ceil((pi*sqrt(-D)*log(-D)**2)/3)
    return prec

# This function computes the Hilbert class polynomial H_D(x)
# INPUT: a discriminant D
# OUTPUT: H_D(x)
def hilbert_poly(D):
    if not is_a_discriminant(D):
        return 'Not a discriminant'
    if D == -4:
        return x - 1728
    if D == -3:
        return x
    prec = precision(D, 0)
    r = 10
    h = 1
    rap = representatives(D)
    for tau in rap:
        h = h*(x - j_inv(tau, r, prec))
    c = h.coefficients()
    h = 0
    for i in range(len(c)):
        val = round(real(c[i][0]))
        h = h + val*(x**i)
    return h

# This function compute the Weber polynomial G_D(x)
# INPUT: a discriminant D
# OUTPUT: If 3|D 'The discriminant must be relatively prime to 3', otherwise G_D(x)
def gamma_poly(D):
    if not is_a_discriminant(D):
        return 'Not a discriminant'
    if mod(D, 3) == 0:
        return 'The discriminant must be relatively prime to 3'
    if D == -4:
        return x - 12
```

```
    prec = precision(D, 1)
    r = 10
    g = 1
    rap = gamma_rep(D)
    for tau in rap:
        g = g*(x - gamma(tau, r, prec))
    c = g.coefficients()
    g = 0
    for i in range(len(c)):
        val = round(real(c[i][0]))
        g = g + val*(x**i)
    return g

#####################################

# OTHER FUNCTIONS

# Check if D is a discriminant
def is_a_discriminant(D):
    if mod(D, 4) == 0 or mod(D, 4) == 1:
        return true
    else:
        return false

# Next discriminant
def next_discriminant(D):
    D = D - 1
    while True:
        if mod(D, 4) == 0 or mod(D, 4) == 1:
            return D
        else:
            D = D - 1
```

A.2 The CM method

Two versions of the CM method are given. The first `CMM(N)` uses the SageMath built-in `hilbert_class_polynomial(D)` function, and is in fact faster than our `myCMM(N)` which uses our `hilbert_poly(D)` and `gamma_poly(D)` functions.

A.2.1 The Cornacchia algorithm

```
# THE CORNACCHIA ALGORITHM

#INPUT: integers m, d such that 1 <= d < m
#OUTPUT: (x, y) primitive solution of x^2 + d*y^2 = m

```

```
def cornacchia(d, m):
    if not mod(-d, m).is_square():
        return "No solutions"
    r_zero = m
    r_one = int(mod(-d, m).sqrt())
    if r_one < 0:
        r_one = -r_one
    while True:
        s = r_one^2
        t = r_one
        if s < m:
            break
        r_one = int(mod(r_zero, r_one))
        r_zero = t
    if sqrt((m - s)/d) == int(sqrt((m-s)/d)):
        return [r_one, sqrt((m - s)/d)]
    else:
        return "No solutions"
```

A.2.2 The CM method

```
# CM elliptic curve with a prescribed number of rational
    points
# INPUT: an integer N
# OUTPUT: a prime p and an elliptic curve defined over GF(p)
    with N F_p-rational points

def CMM(N):
    found = False
    D = -4
    while not found:
        D = next_discriminant(D)
        x = cornacchia(-D, 4 * N)
        if len(x) == 2:
            a = x[0]
            v = x[1]
            t = a + 2
            p = N - 1 + t
            if is_prime(p):
                found = True
    d = fundamental_discriminant(D)
    if d != -3 and d != -4:
        D = d
    if mod(D, 3) == 0:
        H = hilbert_poly(D)
    else:
        H = gamma_poly(D)
    R = PolynomialRing(GF(p),'x')
    H = R(H)
```

```
    if mod(D, 3) == 0:
        j = H.any_root()
    else:
        j = H.any_root()
        j = j**3
    A = mod( 3 * j * (1728 - j), p)
    B = mod( 2 * j * ((1728 - j)**2), p)
    E = EllipticCurve(GF(p), [A,B])
    if N < 65:
        if E.cardinality() == N:
            return E
        else:
            return E.quadratic_twist()
    infinity = E(0)
    P = infinity
    Q = infinity
    while P == infinity and Q == infinity:
        R = E.random_element()
        P = (p + 1 - t) * R
        Q = (p + 1 + t) * R
    if P == infinity:
        return E
    else:
        return E.quadratic_twist()
```

Bibliography

[AM93] A. O. L. Atkin and F. Morain. Elliptic curves and primality proving. *Math. Comp.*, 61(203):29–68, 1993.

[AM16] M. F. Atiyah and I. G. Macdonald. *Introduction to commutative algebra*. Addison-Wesley Series in Mathematics. Westview Press, Boulder, CO, economy edition, 2016. For the 1969 original see [MR0242802].

[Apo76] T. M. Apostol. *Introduction to analytic number theory*. Undergraduate Texts in Mathematics. Springer-Verlag, New York-Heidelberg, 1976.

[BBEL08] J. Belding, R. Bröker, A. Enge, and K. Lauter. Computing Hilbert class polynomials. In *Algorithmic number theory*, volume 5011 of *Lecture Notes in Comput. Sci.*, pages 282–295. Springer, Berlin, 2008.

[Br09] R. Bröker. Constructing supersingular elliptic curves. *J. Comb. Number Theory*, 1(3):269–273, 2009.

[BS07] R. Bröker and P. Stevenhagen. Efficient CM-constructions of elliptic curves over finite fields. *Math. Comp.*, 76(260):2161–2179, 2007.

[BS08] R. Bröker and P. Stevenhagen. Constructing elliptic curves of prime order. In *Computational arithmetic geometry*, volume 463 of *Contemp. Math.*, pages 17–28. Amer. Math. Soc., Providence, RI, 2008.

[BSS00] I. F. Blake, G. Seroussi, and N. P. Smart. *Elliptic curves in cryptography*, volume 265 of *London Mathematical Society Lecture Note Series*. Cambridge University Press, Cambridge, 2000. Reprint of the 1999 original.

[BSS05] I. F. Blake, G. Seroussi, and N. P. Smart, editors. *Advances in elliptic curve cryptography*, volume 317 of *London Mathematical Society Lecture Note Series*. Cambridge University Press, Cambridge, 2005.

[BV07] J. Buchmann and U. Vollmer. *Binary quadratic forms*, volume 20 of *Algorithms and Computation in Mathematics*. Springer, Berlin, 2007. An algorithmic approach.

[BW05] J. M. Basilla and H. Wada. On the solution of $x^2 - dy^2 = \pm m$. *Proc. Japan Acad. Ser. A Math. Sci.*, 81(8):137–140, 2005.

[Coh93] H. Cohen. *A course in computational algebraic number theory*, volume 138 of *Graduate Texts in Mathematics*. Springer-Verlag, Berlin, 1993.

[Con] K. Conrad. Galois Groups of Cubics and Quartics (Not in Characteristic 2). `https://kconrad.math.uconn.edu/blurbs/galoistheory/cubicquarticallchar.pdf`.

[Cox13] D. A. Cox. *Primes of the form $x^2 + ny^2$*. Pure and Applied Mathematics (Hoboken). John Wiley & Sons, Inc., Hoboken, NJ, second edition, 2013. Fermat, class field theory, and complex multiplication.

[DS05] F. Diamond and J. Shurman. *A first course in modular forms*, volume 228 of *Graduate Texts in Mathematics*. Springer-Verlag, New York, 2005.

[Eng09] A. Enge. The complexity of class polynomial computation via floating point approximations. *Math. Comp.*, 78(266):1089–1107, 2009.

[Fre11] E. Freitag. *Complex analysis. 2*. Universitext. Springer, Heidelberg, 2011. Riemann surfaces, several complex variables, abelian functions, higher modular functions.

[Gee99] A. Gee. Class invariants by Shimura's reciprocity law. volume 11, pages 45–72. 1999. Les XXèmes Journées Arithmétiques (Limoges, 1997).

[GvdHL15] B. Grenet, J. van der Hoeven, and G. Lecerf. Randomized root finding over finite FFT-fields using tangent Graeffe transforms.

In *ISSAC'15—Proceedings of the 2015 ACM International Symposium on Symbolic and Algebraic Computation*, pages 197–204. ACM, New York, 2015.

[Kob93] N. Koblitz. *Introduction to elliptic curves and modular forms*, volume 97 of *Graduate Texts in Mathematics*. Springer-Verlag, New York, second edition, 1993.

[Kob94] N. Koblitz. *A course in number theory and cryptography*, volume 114 of *Graduate Texts in Mathematics*. Springer-Verlag, New York, second edition, 1994.

[Lan87] S. Lang. *Elliptic functions*, volume 112 of *Graduate Texts in Mathematics*. Springer-Verlag, New York, second edition, 1987. With an appendix by J. Tate.

[Mar18] D. A. Marcus. *Number fields*. Universitext. Springer, Cham, 2018. Second edition of [MR0457396], With a foreword by Barry Mazur.

[Neu99] J. Neukirch. *Algebraic number theory*, volume 322 of *Grundlehren der mathematischen Wissenschaften [Fundamental Principles of Mathematical Sciences]*. Springer-Verlag, Berlin, 1999. Translated from the 1992 German original and with a note by Norbert Schappacher, With a foreword by G. Harder.

[Rei88] M. Reid. *Undergraduate algebraic geometry*, volume 12 of *London Mathematical Society Student Texts*. Cambridge University Press, Cambridge, 1988.

[RS10] K. Rubin and A. Silverberg. Choosing the correct elliptic curve in the CM method. *Math. Comp.*, 79(269):545–561, 2010.

[Sch02] R. Schertz. Weber's class invariants revisited. *J. Théor. Nombres Bordeaux*, 14(1):325–343, 2002.

[SE21] M. Streng and A. Enge. Schertz style class invariants for quartic CM fields. `https://www.math.leidenuniv.nl/~streng/agct2021.pdf`, 6 2021.

[Sil94] J. H. Silverman. *Advanced topics in the arithmetic of elliptic curves*, volume 151 of *Graduate Texts in Mathematics*. Springer-Verlag, New York, 1994.

[Sil09] J. H. Silverman. *The arithmetic of elliptic curves*, volume 106 of *Graduate Texts in Mathematics*. Springer, Dordrecht, second edition, 2009.

[Sut11] A. V. Sutherland. Computing Hilbert class polynomials with the Chinese remainder theorem. *Math. Comp.*, 80(273):501–538, 2011.

[Tat75] J. Tate. Algorithm for determining the type of a singular fiber in an elliptic pencil. In *Modular functions of one variable, IV (Proc. Internat. Summer School, Univ. Antwerp, Antwerp, 1972)*, pages 33–52. Lecture Notes in Math., Vol. 476, 1975.

[Was08] L. C. Washington. *Elliptic curves*. Discrete Mathematics and its Applications (Boca Raton). Chapman & Hall/CRC, Boca Raton, FL, second edition, 2008. Number theory and cryptography.

[Web13] H. Weber. *Lehrbuch der Algebra. Band III: Elliptische Funktionen und algebraische Zahlen.* Braunschweig, F. Vieweg und Sohn, 1842-1913.